

Affirming a Racial Divide?

The Political Consequences of Affirmative Action in Brazil

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Abstract

Affirmative action policies are designed to increase opportunities for underrepresented groups. This paper explores a potential unintended effect: whether by emphasizing group identity in resource allocation, these policies can polarize voters along identity lines. I investigate this by focusing on the implementation of the first nationwide racial affirmative action policy in Brazil, the Law of Quotas, which mandated that half of all admission seats in public federal universities be reserved for historically underrepresented groups, including non-White students. To assess the local effects of the policy, I construct a municipality-level exposure measure based on pre-policy enrolment patterns and the expansion of reserved seats across federal universities. Combining geographically granular voting data, electorates' demographics, and an automated machine-learning model trained to approximate candidates' racial identity based on official photographs, I find that the reform increased support for ideologically polarized Black candidates in Brazilian legislative elections. This effect emerged as greater support for left-wing Black candidates among the non-White electorate and a rise in support for Black conservative candidates among White voters. I argue that this polarization was driven by Black candidates becoming more vocal on racial issues, while adopting opposing positions on these topics across ideological lines.

Keywords: affirmative action, tertiary education, identity politics, polarization, racial voting

JEL Classification: D72, H7, J15, I28

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1 Introduction

Affirmative action policies are highly popular among governments, institutions, and corporations as means to redress systematic group-based inequalities in education, labour markets, and politics (Gisselquist et al., 2023). These policies provide preferential access to opportunities and resources for historically disadvantaged groups, commonly through the use of quotas. However, by placing group identity at the core of resource allocation, affirmative action may inadvertently shift the political conflict towards this now more salient identity (Bonomi et al., 2021). A common view is that voters may prefer candidates from their own group, believing that they are better positioned to advocate for their interests (Osborne and Slivinski, 1996; Besley and Coate, 1997). In this paper, I argue instead that affirmative action could lead both winners and losers from the policy to perceive minority candidates as uniquely suited to represent opposing sides of this political conflict.

Understanding this potential consequence of affirmative action policies is of economic interest. While voting along identity lines can enhance the representation of disadvantaged groups in policymaking (Pande, 2003; Chattopadhyay and Duflo, 2004; Aneja and Ritadhi, 2022), it may also lead voters to prioritize group identity over other critical dimensions of policymaking, such as the competence of policymakers (Banerjee and Pande, 2007). Furthermore, as political divisions deepen along identity lines, voters may become less inclined to support funding for public goods if these resources are perceived as disproportionately benefiting other groups (Alesina et al., 1999; Fong and Luttmer, 2009). Despite debates surrounding affirmative action, the political consequences of affirmative action policies remain unexplored.¹

In this paper, I investigate whether affirmative action, by increasing the salience of the targeted identity in policymaking, polarizes voters along identity lines. The implementation of the *Lei de Cotas* (*Lei nº 12.711/2012*, hereafter Law of Quotas) in Brazil, the country’s first nationwide affirmative action policy targeting disadvantaged racial groups within its regressive public tertiary education system, offers a natural setting for examining this question. The policy was implemented in 2013 and mandated the reservation of half of admission seats in every program offered in federal universities, among the most prestigious in the country, to a combination of public high-school, low-income, and non-White students. The inclusion of race as an eligibility criterion sparked considerable debate, given the fluidity of racial identity in Brazil and fears of reverse discrimination (Pimenta, 2022).

For identification, I exploit municipality-level exposure to non-White quotas within a difference-in-differences framework for the period spanning 2006 to 2020. In particular, I use municipality pre-policy

¹Examples include the banning of affirmative action policies in nine U.S. states and the high-profile lawsuit *Students for Fair Admissions, Inc. v. President and Fellows of Harvard College* (also known as “The Harvard Admissions Lawsuit”) against the use of racial preferences in admissions at Harvard University for alleged discrimination against Asian Americans. Similarly, India, a country with a long history of affirmative action, have recently experienced riots over which ethnic groups receive special tribal status and hence are eligible for these policies (NY Times, 2023). Additionally, commissions to verify the racial identity of students were introduced in Brazilian federal universities following the implementation of racial quotas in 2013 (UOL Educação, 2022).

enrolment patterns to federal universities, together with the expansion of racial quotas at the university level, in the spirit of a shift-share IV. The instrument aims to predict changes in local enrolment across racial groups as a result of the reform, which is the one that would result if university-level quotas were apportioned across municipalities precisely based on pre-reform enrolment patterns. Beyond changes in enrolment across racial groups, the measure aims to capture how these changes in access to resources among racial groups heightened the salience of racial political conflict in exposed municipalities.

I show that the municipality exposure measure effectively captures the distribution of non-White quotas across municipalities, resulting in an almost one-to-one increase in non-White enrolment, as intended by the policy. However, past enrolment patterns fail to predict the incidence of the policy on White students. One possibility is that, since students could apply to universities nationwide, the local increase in non-White enrolment did not necessarily have to reduce enrolment opportunities for White students within the same municipality. In other words, White students might have borne the costs of the policy in aggregate, regardless of the local enrolment gains experienced by non-Whites, as predicted by pre-policy enrolment patterns. This frames the interpretation of the policy’s political effects as being driven by the direct policy gains experienced by non-Whites, as well as broader attitudinal shifts across racial groups triggered by exposure to the policy, such as fear of displacement among Whites or feelings of empowerment among non-Whites.

Studying whether the introduction of racial affirmative action influenced voting patterns across racial groups in exposed municipalities is particularly challenging in the Brazilian context due to the lack of systematic records on voters’ racial identities. To overcome this, I assembled a novel dataset covering the period 2006 to 2020, which merges voting outcomes at the municipality and polling station levels with information on the electorate’s racial composition using Census data. In addition to measuring changes in ideological stances across racial groups, I also assess whether the policy led to shifts in support for candidates of different racial groups. This presents an additional challenge: data on candidates’ racial identities is not available prior to the policy’s implementation. To address this, I train an automated machine-learning model for image classification to approximate candidates’ racial group using more than 1.5 million candidates’ official pictures. Disentangling identity-based voting from ideology-based preferences is inherently complex, as vote choices often reflect both factors. To tackle this, I leverage the open-list proportional system used in Brazilian legislative races, which allows voters to directly select their preferred candidate from within a party list. This system makes it possible to compare vote shares between parties and candidates of different racial groups, enabling a quantification of the extent to which voting choices are driven by candidate racial identity versus party ideology.

I start by analysing the local effects of the policy in federal legislative elections, the government level responsible for implementing the reform and where racial considerations were explicitly included in policymaking. I conduct a reduced-form analysis to account for the possibility that voter realignment along racial lines may be influenced not only by the policy’s direct effects but also by shifts in attitudes

across racial groups. At the aggregate level, I find no statistically significant ideological shift in municipalities more exposed to the policy. However, the results reveal that the policy had a sizeable impact on the support for non-White federal candidates in exposed municipalities. Among non-White candidates, who include both mixed-race and Black individuals, the effect was primarily driven by increased support for Black candidates. Given that the reform led to a roughly one-to-one increase in non-White enrolment, the results imply that for every additional non-White student admitted through quotas per 100 eligible students, there were approximately 4 additional votes per 100 votes cast for Black candidates. This represents a 50% increase in their electoral success relative to their baseline support. Moreover, I find that this increase in support for Black candidates in exposed municipalities was evenly split between those running in left-wing parties and those in right-wing parties.

To gain additional insights into how different racial groups responded to the expansion of racial quotas in federal universities, I exploit variation in the racial composition of over 90,000 polling stations, each serving approximately a thousand voters. I begin by documenting that the policy led to an ideological polarization of the electorate along racial lines, consistent with literature on the political effects of targeted programs (Manacorda et al., 2011; Zucco, 2013; Caprettini et al., 2021). On the one hand, non-White voters in exposed municipalities increased support for left-wing candidates, who advocated for affirmative action in public universities and advancing the rights of non-White communities. On the other hand, predominantly White polling stations exhibited increased support for conservative candidates. This finding also aligns with existing literature documenting a backlash by racial social majorities in response to the socioeconomic advancement of minority groups (Trebbi et al., 2008; Kuziemko and Washington, 2018; Zonszein and Grossman, 2022; Bernini et al., 2023).

Within this ideological realignment, both racial groups disproportionately shifted their support toward Black candidates. I argue that this pattern can be explained by the stronger emphasis Black candidates placed on race-related issues, albeit holding opposing stances on racial policies, which likely resonated differently with the potential winners and losers of affirmative action. To support this argument, I conduct a text analysis of over 40,000 campaign programs from candidates running for president, state governors, and mayors to infer their policy stances. I find that left-wing Black candidates have traditionally, and increasingly, placed more emphasis on race-related issues in their electoral platforms compared to candidates from other racial groups and ideologies. If these patterns in executive elections extended to legislative elections, this would suggest that left-wing Black candidates' focus on racial issues signaled their commitment to addressing racial inequalities, likely attracting support from non-White communities benefiting from affirmative action. On the other hand, the surge in support for Black conservative candidates among the White electorate could have stemmed from these candidates becoming more vocal on racial issues after the policy was implemented, yet opposing racial affirmative action and distancing themselves from the left's racial justice agenda — stances that likely appealed those who viewed themselves as disadvantaged by affirmative action.

Results from city council elections reinforce my findings in federal elections, supporting the argument

that affirmative action policies can shift political conflict toward the policy’s targeted identity. By placing race at the centre of federal policy, local exposure to the Law of Quotas could have signalled to voters that racial considerations could extend to other national policies, as seen with subsequent federal initiatives on non-White quotas in public employment and campaign funding allocations that assign greater weight on votes for non-White candidates. Importantly, city council election results indicate that the Law of Quotas intensified identity-based political conflict even in the absence of explicit racial policies in local political offices. Finally, I provide evidence that these results are not driven by other demographic factors correlated with race, such as income and educational attainment.

The documented increase in the descriptive representation of non-White candidates in key Brazilian legislative offices could potentially strengthen the representation of the non-White electorate in policymaking (Pande, 2003; Chattopadhyay and Duflo, 2004; Bernini et al., 2023; Aneja and Ritadhi, 2022). However, exposure to affirmative action could have led voters to prioritize group identity over other critical dimensions of policymaking, such as candidates’ competence or political experience (Banerjee and Pande, 2007). While results are imprecise, they suggest a potential decline in support for competence in local but not federal elections, indicated by a reduced share of votes for candidates with tertiary education. Additionally, evidence points to an increase in support for less experienced candidates in municipalities with higher exposure to the reform. Given the historical over-representation of White individuals in political office, these outcomes suggest a short-term trade-off between advancing racial diversity and electing more experienced politicians. Next, I assess whether the racialization of politics influenced public goods provision, as voters may become less supportive of policies perceived to benefit other racial groups (Alesina et al., 1999; Luttmer, 2001). Using municipality spending data, I find no evidence that increased electoral polarization along racial lines diminished public goods provision or significantly altered its composition at the local level.

This paper contributes to several strands of literature. Firstly, it contributes to the well-established and growing literature exploring the impact of affirmative action policies in politics, the labour market, and particularly, education. Prior studies have documented the effectiveness of affirmative action policies in increasing representation of disadvantaged groups in the targeted settings (Pande, 2003; Chattopadhyay and Duflo, 2004; Beaman et al., 2009; Miller, 2017; Vieira and Arends-Kuenning, 2019; Estevan et al., 2019), and that race-targeted policies outperform race-neutral alternatives at boosting diversity (Bleemer, 2023). This paper advances the literature in three key ways. First, while earlier works have documented the success of Brazil’s Law of Quotas in increasing the enrolment of low-income and non-White students without harming the system’s efficiency at the university level (Mello, 2022; Otero et al., 2021), this paper provides an examination of the local effects of the policy for the population at risk. Second, this paper shows that affirmative action policies, even when not explicitly aimed at increasing political representation, can enhance the selection of minority candidates. Third, this paper is the first to rigorously investigate a crucial trade-off of affirmative action policies: whether accelerating representation of disadvantaged groups through resource reallocation fosters political divisions along identity lines.

Secondly, the paper advances the literature on identity politics by documenting how policies that emphasize group identities, such as affirmative action, can foster identity-based voting and providing new insights into the trade-offs this may create for policymaking. My paper builds on previous research in economics and social psychology that indicate that individuals' identity is shaped by the most salient policy conflicts (Tajfel et al., 1979; Depetris-Chauvin et al., 2020; Bonomi et al., 2021) and can be influenced by political actors (Blouin and Mukand, 2019; Esposito et al., 2023; Ochsner and Roesel, 2024). Although voting along identity lines, such as gender, race, or ethnicity, is a well-documented phenomenon that can be attributed to voters deriving utility from ascribing to their identity, or as an attempt to mitigate information asymmetries and candidate's commitment issues (Beaman et al., 2009; Ichino and Nathan, 2013; Casey, 2015; Aguilar et al., 2015), my paper expands on existing research that warns of its negative impact on the well-functioning of democracies, including a potential decrease in policymakers competence (Banerjee and Pande, 2007) and the reduction of public goods provision (Alesina et al., 1999; Miguel and Gugerty, 2005).

Finally, this paper contributes to the growing literature in economics and political science on racial politics, which documents how the de facto enfranchisement of Black voters during the U.S. Civil Rights movement in the 1950s and 1960s (Bernini et al., 2023; Calderon et al., 2023) was met with a backlash from the White majority, who sought to protect their historically dominant position (Trebbi et al., 2008; Kuziemko and Washington, 2018; Bernini et al., 2023). This paper highlights that similar backlash effects can emerge from policies aimed at promoting the socioeconomic advancement of disadvantaged groups.

The rest of the paper is structured as follows. Section 2 introduces the Brazilian context and provides a detailed overview of the Law of Quotas. Section 3 outlines the data used in the analysis and presents descriptive trends. In Section 4, I define the municipality-level exposure to non-White quotas, and in Section 5 I discuss the enrolment effects of the policy. Section 6 examines the impact of the policy on voting along racial lines, while Section 7 explores the potential trade-offs. Section 8 concludes.

2 Institutional Setting of Brazil

Brazil provides a unique setting to explore the political consequences of racial affirmative action due to its complex racial relations coupled with significant inequalities, which have historically coexisted with a relative lack of emphasis on race in politics. Additionally, the implementation of the Law of Quotas, the country's first nationwide affirmative action policy targeting disadvantaged racial groups within its regressive public tertiary education system, offers a natural case for examining its effects. In this section, I provide an overview of the country's racial landscape, explain how the policy works, and discuss its political context.

2.1 Racial and Political Context

Brazil was the largest recipient of African enslaved people during the Atlantic slave trade era, and was the last one to abolish this institution. However, unlike other nations with similar experiences, Brazil did not develop a caste-like system, neither strong ethnic alliances. Instead, historical racial mixing between colonizers and enslaved Africans, together with European migration in the 20th century, led to a highly diverse society (Harris, 1964; Dos Santos, 2002). According to the 2010 census, 49% of the population self-identified as White, and 51% as non-White, including 43% who self-identified as mixed-race and 8% as Black or Indigenous. This legacy of racial mixing gave rise to a widespread discourse of “racial democracy”, which affirmed that race did not determine economic opportunities neither divided society.² However, this narrative sharply contrasts the observed socioeconomic disparities across racial groups and the perceptions of the Brazilian population, which reflect a reality where race influences labour market outcomes, social interactions, experiences with justice, and education (IBGE, 2008). This influence also extends to the political sphere, historically dominated by White individuals (Johnson and Heringer, 2015).

Despite their under-representation in formal politics, the Black Brazilian movement has exerted considerable influence in promoting policies aimed at reducing racial inequality. Over the years, members of this movement put forward several proposals for racial quotas in both education and employment, although many faced resistance in Congress (Pimenta, 2022). Supporters of racial quotas framed them as compensatory measures for the legacy of slavery and as tools to reducing racial inequality and acknowledging structural racism. Opponents, however, highlighted the difficulties of implementing such policies in a country with fluid racial identities, and warned that they might institutionalize discrimination (Pimenta, 2022).

Among the many legislative proposals, it was a 1999 bill introduced by Deputy Nice Lobão (*Projeto de Lei n° 73/1999*), later amended and approved after years of political negotiation, that eventually led to the enactment of the current Law of Quotas in 2012. This reform was passed under the Partido dos Trabalhadores (PT) government, which took other significant steps to address racial inequality. These included the introduction of African history into the national school curriculum in 2003, the establishment of Black Consciousness Day in 2010, the appointment of non-White ministers, and the implementation of quotas for non-Whites in public sector jobs starting in 2014.

2.2 The Law of Quotas

The Law of Quotas (*Lei Ordinaria 12.711/2012*, commonly known as Lei de Cotas) was approved in August 2012 and implemented at the start of the 2013 academic year. Its introduction marked a significant reform within the Brazilian tertiary education system, which is characterized by a mixture

²The term “racial democracy” is often attributed to the Brazilian sociologist Gilberto Freyre, after the publication of the book *Casa Grande e Senzala* (translated, “The Master and the Slaves”) in 1933, which studied the Brazilian societal architecture based on racial mixing between Portuguese masters and African enslaved people in Brazil.

of public and private institutions. Private institutions account for 60% of total university enrolment, while public federal universities constitute 30% of total enrolment and are considered among the most prestigious in the country. The remaining 10% of students are enrolled in public state universities. Despite their public nature, federal universities are particularly regressive, with low-income students making up less than 20% of total admissions (Figure A.1).

The Law of Quotas mandated that all federal universities reserve 50% of admission seats within all majors for a combination of public high-school, low-income, and non-White students. While the reservation rule of public-school and low-income seats was homogeneous across universities, the share of seats reserved for non-White students had to be proportional to the state's share of non-White population, as determined by the last demographic census.³ To benefit from these reserved seats, eligible students typically took a university entry exam, applied directly to these reserved seats, and self-reported their non-White status. These reserved seats often had lower admission cut-offs, which meant that applicants competed primarily with peers from similar socioeconomic backgrounds.

Figure 1 exhibits the advertised share of racial quotas in federal universities in the period 2010-2018. Before the enactment of the Law of Quotas, some universities already allocated approximately 5% of admission seats to non-White students on average. In 2013, the first year of implementation, the proportion of seats reserved for non-White individuals quadrupled. Racial quotas continued increasing until 2016, with 30% of seats reserved to those students. Moreover, the timing of the policy's implementation varied across universities, with only a few achieving the targeted quota levels in the first year of enforcement.

In response to allegations of fraud regarding non-targeted White students enrolling through non-White quotas, universities established panels to verify that applicants for racial quotas met the socially prescribed phenotypical traits of non-White individuals, first voluntarily and later under government guidelines (Conselho Nacional do Ministério Público, 2016; UOL Educação, 2022). The debate over racial quotas extended beyond the educational sphere, attracting significant media attention and igniting public discussions on racial identity (O Globo, 2014, 2019b). Politically, the policy also became increasingly contentious, with support for the Law of Quotas decreasing among those on the right end of the ideological spectrum (see Figure E.7).

³The precise allocation of seats in universities would be as follows: For each major in a federal university, half of the admission seats would be reserved for students having completed high-school entirely in a public school. The rest of seats would be open to competition. Within public high-school admission seats, half of them would be reserved for students from public schools with gross monthly family income per capita lower than 1.5 minimum salaries. Part of these admission seats, both below or above 1.5 minimum salaries, would further be reserved for non-White individuals, proportionally to the state's population share. Universities could also implement these quotas progressively, guaranteeing that at least 12.5% of admission seats were reserved for targeted groups in 2013, 25% in 2014, 37.5% in 2015, and reaching 50% in 2016. See Figure A.2 for an example of the mandated quotas in universities located in the state of Pernambuco and Santa Catarina.

3 Data and Descriptive Trends

This section provides an overview of the datasets used in the analysis. I start by describing the education datasets that I employ to construct university enrolment rates across municipalities and the measure of exposure to non-White quotas. Then, I outline the datasets I use to construct political outcomes, along with a brief overview of the machine learning model I train to classify candidates into racial groups based on their official photographs. Finally, I present descriptive statistics on the key variables to contextualize the findings.

3.1 Education

To construct municipality enrolment rates across racial groups and the measure of exposure to the policy, I combine the following datasets. First, I employ the Census of Higher Education (CES) microdata to construct municipality-level enrolment rates to universities. This dataset contains restricted-access individual-level data on the universe of students enrolled in all Brazilian tertiary education institutions from 2009 to 2019. I restrict my sample to incoming students enrolling through a regular selection process in any tertiary education institution.⁴ For each student, the CES provides information on the program and institution where they are enrolled, the type of vacancy they fill, and demographic information. Importantly, it reports the municipality of birth of the student, which I use to construct university enrolment rates across municipalities.

This dataset has two important limitations. First, information on students' municipality of birth is missing for 30% of the sample. Second, prior to the Law of Quotas, only 40% of students reported their racial group, a number that rose to 80% after the policy was implemented. Using as outcome variables local enrolment rates with such measurement errors can generate consistent but imprecise regression estimates. However, deriving pre-policy enrolment patterns — a key component of the exposure measure— using this data would introduce significant bias in my estimates. To address this, I derive pre-policy enrolment patterns using the 2010 Census microdata.⁵

Additionally, I combine data on the implementation of reserved admission seats across federal universities from 2010 to 2015 from Mello (2022) and from 2016 to 2018 from the centralized admission system, SISU. These datasets provide information on the share of reserved seats in each university program for the targeted categories. By combining this information with data on program vacancies from CES, I calculate the total number of reserved seats offered across programs, which I aggregate to obtain the total number of reserved seats at the university level. Finally, I use microdata from the national centralized admission exam (ENEM) and the Census of Basic Education to construct municipality-level variables, including the average ENEM scores obtained by students in the municipality, high-school

⁴I exclude those that enter through transfers or other special programs, as the latter are not subject to quotas. I also exclude from my sample those individuals enrolled in an online program.

⁵These data limitations can be addressed by matching two restricted-access datasets, the CES and ENEM Microdata, following Mello (2022). However, this process requires access to a safe data room in Brazil, which I plan to pursue in the future.

enrolment rates, and the share of public high schools.

Figure 2 shows the racial composition of incoming student cohorts at federal universities in Brazil from 2010 to 2018, based on CES data and students’ self-reported racial identity. Prior to the Law of Quotas, approximately 40% of incoming students were non-White, with White students making up 60%. Although the policy targeted not only non-White students but also public high-school and low-income students regardless of race, we observe a shift in the racial composition of incoming cohorts from the first year of implementation. The share of White students declined and stabilized around 2016, when federal universities reached the mandated quota levels. By that time, the racial composition had inverted, with non-White students comprising 60% of incoming cohorts and White students around 40%.

3.2 Politics

Measuring race-specific voting behaviour in this context is particularly challenging, as voting outcomes by racial group cannot be observed due to the anonymity of the vote, and candidates’ racial identity was not disclosed prior to the reform. To address these limitations, I combine datasets from the Superior Electoral Court (TSE) and Census 2010 Microdata.

First, I retrieve vote counts from all elections held between 2006 and 2020, available at both the municipality and ballot box levels (“seção eleitoral”). I aggregate the ballot box results into over 90,000 polling stations across Brazil, and create a panel of these stations using the mapping developed by Hidalgo (2021). Each polling station serves, on average, one thousand eligible voters. My analysis focuses on legislative elections, since they use an open-list proportional system that allows voters to directly select one candidate from a party list. This provides a way to differentiate between party policy positions and candidates’ racial identity. For each office, I use data from two elections before the introduction of the Law of Quotas, and two elections after. Finally, as the Brazilian party system is highly fractionalized, with more than 30 parties contesting in elections, I classify parties into the left-right ideological spectrum using the index constructed by Zucco and Power (2021).⁶

Second, I use information on voters demographics, including age, gender, and educational attainment of the electorate assigned to each polling station. However, the TSE does not provide data on voters’ racial or income profiles. To address this, I estimate the socioeconomic composition of polling stations’ catchment areas, defined as the set of census tracts —typically containing around 300 households— that are served by each polling station.⁷

Finally, I incorporate data on candidates’ demographics and related election variables, including age,

⁶For further details on this classification, refer to Appendix E.2

⁷Although the TSE does not specify a clear rule for assigning voters to polling stations, it indicates that voters are permanently assigned to the closest polling station within the judicial or administrative boundaries of their residence (*Código Eleitoral - Lei nº 4.737, de 15 de julho de 1965*). Hence, I assume that voters in each census tract are served by the closest polling station.

gender, educational attainment, occupation, and campaign spending. While the TSE provides information on candidates’ self-reported racial group, this data is only available starting in 2014, after the Law of Quotas was implemented. To address the lack of pre-policy information on candidates’ race, I classify candidates using their official photographs and an automated machine-learning model for image classification. These candidate photos are primarily sourced from the TSE, with missing images supplemented through web-scraping from the official government website “Divulgação de Candidaturas e Contas Eleitorais”.⁸ In total, I collected images of nearly two million candidates running for elections between 2006 and 2022. Appendix E.1 provides a detailed description of the methodology used to classify candidates into Brazil’s main racial groups.

Next, I examine patterns of racial voting –defined as the extent to which a voter’s racial identity predicts their support for candidates of a particular race or ideology– prior to the implementation of racial quotas in federal universities. Figure 3 illustrates the relationship between the share of the non-White population and support for left-wing and non-White candidates across federal and local legislative elections. Panels (a) and (b) show that, before the policy, municipalities with a larger non-White population did not systematically exhibit greater support for left-wing parties at either electoral level. However, when it comes to racial alignment between voters and candidates, the patterns differ. Panel (c) reveals that federal elections did not display strong racial voting tendencies, as support for non-White candidates was not significantly higher in municipalities with larger non-White populations. In contrast, panel (d) shows that in local elections, municipalities with a larger share of non-White voters exhibited noticeably greater support for non-White candidates, indicating a clearer racial voting pattern at the local level. As shown in Figure A.3, non-White candidates were present in both federal and local races, ruling out the possibility that the observed differences simply reflect variation in the candidate pool.

3.3 Complementary Sources

I complement this dataset with additional variables from the 2010 Census. These include total population, population density, gender and age distributions, racial composition, literacy rate, and average monthly nominal income. I construct geographic characteristics of municipalities, including their distance to the closest regional capital, and whether there is a federal university in the municipality.

4 Definition of Local Exposure to Racial Quotas

To assess whether the expansion of non-White quotas in federal universities shifted political behaviour across racial groups, I construct a measure of exposure to this policy at the municipality level. This measure leverages pre-policy enrolment patterns across municipalities in the spirit of a shift-share IV, based on the idea that students’ decisions about enrolling in universities are influenced by the

⁸<https://divulgacandcontas.tse.jus.br/divulga/>

educational trajectories in their social environment, as they enclose information about institutions, shape their preferences, and reduce attendance costs (Altmejd et al., 2021; Barrios-Fernández, 2022). Accordingly, the measure is designed to capture how reserved seats at each university would have been distributed across municipalities if they had followed pre-policy enrolment patterns. Formally:

$$q_{mt} = \sum_u \frac{s_{mu}}{s_u} \cdot \frac{Q_{ut}}{pop_m^{18-24}} \quad (1)$$

where q_{mt} represents exposure to non-White quotas in municipality m at time t . The advertised number of racial quotas at a university u in academic year t is denoted by Q_{ut} . In my main specification, I define racial quotas as admission seats reserved to non-White students that attended a public high school, regardless of their income group. s_{mu} is the number of students enrolled in federal university u in 2010 from municipality m . s_u is the total number of students that enrol in university u in 2010. Hence, $\frac{s_{mu}}{s_u}$ captures baseline enrolment rates, defined as the share of students enrolled in u that come from municipality m in 2010.⁹ ¹⁰ Finally, pop_m^{18-24} is the number of individuals in the typical enrolment age (18–24 years old) in 2010. In the terminology of the shift-share literature, I define the shares as $\frac{s_{mu}}{s_u} \frac{1}{pop_m^{18-24}}$ and the shift as Q_{ut} .

In essence, the measure provides the number of seats apportioned to a municipality relative to its population at risk, assuming that seats were distributed across municipalities precisely based on pre-reform enrolment patterns. The measure reflects changes in the availability of university admission seats for different racial groups in each municipality. Non-White students experience a positive supply shock, while White students face a reduction in the seats available to them. In Appendix D, I formalize how Equation 1 can be decomposed into own-race exposure to non-White quotas, weighted by the racial composition of municipalities.

4.0.1 Characteristics of Local Exposure

Table B.1 summarizes the main characteristics of the exposure measures for the period 2010-2018, along with descriptive statistics of their main components. On average, universities reserved 800 seats for non-White students each year, which were distributed across 84% of municipalities in Brazil as predicted by pre-reform enrolment patterns — the remainder municipalities are not treated, as no student enrolled in a federal university at baseline. On average, a single municipality accounts for 2% of total enrolment in a university, with few universities having one municipality contributing more than 80% of their student body. Altogether, this results in a municipality being exposed to an average of 3 reserved seats per 1,000 university-age students, which ranges from 0 to 119 reserved seats. Moreover, my measure of treatment is not concentrated in specific geographic clusters, as displayed in Figure

⁹Municipality enrolment rates across universities are measured in the initial period because contemporaneous local enrolment rates are likely endogenous to the expansion of reserved admission seats.

¹⁰In a setting with acute racial disparities like Brazil, one could argue that members of different racial groups exhibit different enrolment patterns across universities, requiring the use of race-specific shares. In Appendix C.1.1 I document that while the non-White group had, on average, lower enrolment rates than Whites at baseline, they attended similar institutions.

B.1. A closer look to other states evidences that exposure varies significantly throughout the country, and is not only clustered around municipalities with federal university headquarters.

To provide a more concrete description of the cross-sectional variation I am using, I computed the Rotemberg weights, which, intuitively, indicate the universities whose variation drives the overall shift-share estimator (Goldsmith-Pinkham et al., 2020). Because of that, these weights identify which shares to focus on when assessing the plausibility of the identification assumption that I will discuss in the next section. Table C.4 reports summary statistics of these weights (Panel A), together with the five universities with the highest Rotemberg weights (Panel B).¹¹ From the 94 federal institutions that conform the instrument, 25 have non-zero Rotemberg weights. Still, these institutions cover all major regions of Brazil. For those universities contributing to the variation of the exposure measure, the distribution of weights is very concentrated, with no single university explaining a large part of the variation in the data that the estimator uses, except for the Federal University of Bahia. This university has the largest Rotemberg weight, followed by the Federal University of Pará, which together only account for 28% of the overall variation in the overall shift-share estimate. This is reassuring, as enrolment in federal universities is largely local, indicating that my research design is not relying on variation from just a few universities.

To further describe the cross-sectional variation of the exposure measure and the potential threats to its validity, I report the correlation between 2010 municipality characteristics and the share of the top-5 Rotemberg weight universities and total exposure in Table C.5. The first thing to notice is that the R^2 is significantly small for the individual university shares (columns (1) to (5)), indicating that baseline municipality characteristics cannot explain a large part of the cross-sectional variation in those shares. In the last column, the dependent variable is the average exposure measure over the period. As shown, it is correlated with almost all the covariates and explain almost 70% of the variation of the measure. While this indicates that municipalities were not randomly exposed to racial quotas, my identification strategy is designed to flexibly control for these factors, which could also influence changes in local enrolment rates and election outcomes.

5 The Effect of Racial Quotas on Local Enrolment

In this section, I outline the empirical strategy employed to assess the impact of the expansion of non-White quotas in federal universities on local enrolment across different racial groups and address the main identification challenges. I show that the measure of local exposure to the reform accurately captures how reserved seats were allocated across municipalities and how this translated into

¹¹To compute the Rotemberg weights I used the Stata command developed by Goldsmith-Pinkham et al. (2020), *bartik_weight*, using my main regression specification (Equation 2) and a dataset aggregated in 3-year periods to overcome computational power constraints. The dependent variable is enrolment of non-White students, defined as the number of non-White student from each municipality that enrol in a federal university relative to the university-age population in the municipality.

differential enrolment patterns among students.

5.1 Empirical Model

In order to estimate the causal effect of the expansion of racial quotas on educational outcomes, I estimate the following model:

$$y_{mt} = \beta \cdot q_{mt} + X'_m \cdot \delta_t + \alpha_m + \alpha_{st} + \varepsilon_{mt} \quad (2)$$

where mt refers to a generic municipality m at year t . y_{mt} refers to the enrolment rate, defined as the number of students that enrol in university in year t relative to the university-age population in the municipality m . q_{mt} is the local exposure to quotas, as defined in Equation 1. I include X'_m , a set of baseline municipal characteristics interacted with a year fixed effect, δ_t , which allow municipalities with different characteristics at baseline to follow different trends.¹² Finally, α_m absorbs time-invariant characteristics of municipalities, while α_{st} captures any state-specific shock affecting municipalities within the state. Standard errors are clustered at the municipality level.

The coefficient of interest is β , which captures the effect of changes in exposure to non-White quotas on changes in enrolment. If enrolment patterns across federal universities prior to the policy were perfectly predictive of how quotas were distributed across municipalities, one would expect the effect on non-White enrolment, β^{nw} , to be close to 1. There is the possibility, however, that quotas merely reclassified non-White students to “quota holders”, in which case β^{nw} would be zero. The local effects of the policy on White enrolment, β^w , is unclear. Given the redistributive nature of affirmative action, one may expect β^w to be negative. However, since the supply of admission seats is not local, this may not necessarily be the case, as the cost of the policy might be borne by the White group regardless of the gains enjoyed by non-Whites in the same municipality. Total enrolment effects in the municipality will thus be approximately the sum of β^{nw} and β^w .¹³

The identification of β requires changes in local exposure to non-White quotas to be orthogonal to innovations in the error term, after controlling for X_m . In a shift-share research design, this exogeneity condition can be stated in terms of the exogeneity of baseline shares (pre-enrolment patterns) or the exogeneity of shocks (expansion of reserved admission seats) (Borusyak et al., 2022). Given that my exposure design emphasizes differences in treatment intensity stemming from the cross-sectional variation of the measure, I follow the identification framework of Goldsmith-Pinkham et al. (2020). This condition is satisfied if each of the exposure shares, $\frac{s_{mu}}{s_u} \cdot \frac{1}{pop_m^{18-24}}$, are exogenous to changes in

¹²See Table B.2 for a list of control variables included in the regression, together with the main summary statistics.

¹³The municipality enrolment rate in any federal university can be interpreted as the probability that an individual from municipality m enrolls in a university u in year t . This probability can be expressed as the race-specific probability of enrolling, weighted by the probability of belonging to a given racial group, their population share. If these enrolment probabilities are accurately captured by the exposure measure, q_{mt}^r , then total enrolment can be expressed as $Pr(u|m, t) = \beta^{nw} q_{mt}^{nw} \times \frac{pop_{nw,m}^{18-24}}{pop_m^{18-24}} + \beta^w q_{mt}^w \times \frac{pop_{w,m}^{18-24}}{pop_m^{18-24}} + \alpha_m + \alpha_{st} + \varepsilon_{mt} = (\beta^{nw} + \beta^w) q_{mt} + X'_m \gamma_t + \alpha_m + \alpha_{st} + \varepsilon_{mt}$, abstracting from cross-race effects of exposure and control variables for simplicity.

the outcome of interest, y_{mt} (Goldsmith-Pinkham et al., 2020). In other terms, municipalities with low vs high pre-reform enrolment rates in each university would have followed similar trends, had the university not reserved seats for non-White students from 2013.¹⁴

The main threat to this identifying assumption is the existence of other contemporaneous educational policies that may affect enrolment rates through the exposure shares. During the study period, there were three major educational policies that might confound the effect of racial affirmative action in federal universities. These are the expansion of non-racial quotas under the Law of Quotas from 2012, the expansion of admission seats in some universities, and the staggered implementation of a centralized admission system (SISU) that started in 2010. To control for the variation in local enrolment rates attributable to these confounding policies, I include them in the regression using a similar shift-share design. In Appendix C, I provide a detailed description of these policies, explain how I construct the corresponding exposure measures, and show that the enrolment effects remain robust after their inclusion.

A related threat to the validity of my design is whether any baseline municipality characteristics that correlates with the exposure measure in Table C.5 could also explain changes in the trajectory of enrolment. For instance, the share of students enrolled in public high schools is negatively correlated with exposure to racial quotas, which highlights that private school students were typically more successful in accessing federal universities. Since enrolment in public high schools was a prerequisite to benefit from seats reserved by the Law of Quotas, transitions from private to public high schools, as documented by Mello (2023), could affect students' subsequent chances of gaining access to a federal university and confound the effect of racial quotas. To account for this, I include the share of public high school students (as well as the other variables specified above) at baseline, interacted with a year fixed effect.

5.2 Results

Table 1 reports the effect of the expansion of non-White quotas on local enrolment in federal universities. The dependent variable is the number of students that gained admission to a federal university, relative to the population at university age (18 to 24 years old) in the municipality. The main explanatory variable is Exposure to non-White quotas. I provide estimation results of the most saturated specification, where I include municipality and state-year fixed effects, local exposure to contemporaneous educational policies and baseline municipality controls interacted with a year fixed effect.

In column (1), I start by investigating whether the measure of local exposure to the reform captures how quotas were distributed across municipalities. The dependent variable is the enrolment rate

¹⁴Note that municipality pre-policy enrolment patterns may be endogenous to local enrolment levels. However, in a difference-in-differences framework, the identifying assumption depends on baseline shares being exogenous to changes in the outcome variable rather than its levels. It is nonetheless informative to assess whether these shares correlate with observable municipality characteristics that could also predict changes in the outcome, helping to control for omitted variable bias.

through non-White quotas. The coefficient is close to one, showing that reserved seats for non-Whites in federal universities were closely reapportioned and took up across municipalities according to pre-reform enrolment patterns. In terms of magnitude, for each reserved seat per 1,000 university-age students assigned to a municipality (as predicted by the exposure measure), almost 1 student per 1,000 university-age student enrolled in a federal university through a non-White reserved seat. There is the possibility, though, that these quotas did not increase the enrolment of non-White students, but that they merely reclassified them into “quota holders”. In column (2), I show that the expansion of racial quotas lead to an almost one-to-one increase in the enrolment of non-White students, as intended by the policy.

Next, I assess whether the local increase of enrolment of non-White students in federal universities displaced non-eligible students from those institutions. In particular, I estimate the effect of local exposure to non-White quotas on the enrolment rate of White students across municipalities. Results are reported in column (3). The coefficient is small and negative, but imprecisely estimated. This indicates that past enrolment patterns fail to predict the incidence of the policy on White students. One possibility is that, since students could apply to universities nationwide, the local increase in non-White enrolment did not necessarily reduce enrolment opportunities for White students within the same municipality. In other words, White students might have borne the costs of the policy in aggregate, regardless of the local enrolment gains experienced by non-Whites, as predicted by pre-policy enrolment patterns. This frames the interpretation of the policy’s political effects as being driven by the direct policy gains of non-Whites, as well as broader shifts in perceptions of displacement or empowerment across racial groups triggered by exposure to the policy.

Another dimension to consider is whether the observed gains in enrolment for non-White students stem from federal universities attracting these students into the tertiary education system or reallocating them from lower-tier state and private universities. In Table B.3, I explore the broader impact of the policy within the Brazilian university system. The dependent variable is the enrolment rate in state and private universities, by racial group. All coefficients are statistically insignificant. However, they suggest that the expansion of racial quotas in federal universities may have lead to a fall in enrolment of non-White students in state and private institutions, a decrease that was more than offset by an increase in enrolment in federal universities. This would imply that non-White students did not only reallocate from state and private universities to federal institutions, but that the policy also attracted new non-White students into the tertiary education system. As for White enrolment, while the effect is not statistically significant, the positive coefficient observed in private universities suggests that White students may have been displaced into these institutions. This aligns with Otero et al. (2021), which argues that non-eligible students from affirmative action may have better outside options, such as attending higher-quality private universities.

In Appendix C.1, I present a set of robustness checks to support the validity of my identification assumptions. These include showing the absence of pre-trends, confirming that baseline enrolment

patterns across racial groups did not differentially predict changes in university enrolment prior to the reform. Additionally, I show that White and non-White students followed similar enrolment patterns at baseline and that using race-specific exposure shares yields qualitatively similar results to those obtained with race-neutral shares. I also provide evidence that the expansion of racial quotas across federal institutions was not influenced by university characteristics or political motivations. Moreover, I demonstrate that the treatment effect of each individual share on local enrolment is homogeneous (Figure C.1). Finally, I show that the results are not driven by changes in students' racial self-reporting behavior.

6 The Effect of Racial Quotas on Racial Voting

If the introduction of racial affirmative action heightened the relevance of race in policymaking decisions, one might expect voters in exposed municipalities to see greater value in supporting candidates who promote policies that benefit their own racial group. I start by analysing the effects of the policy in federal legislative elections, the government level responsible for implementing the reform and where race was explicitly included in policymaking. I then explore whether the political effects of the policy extended to other electoral levels, where such conflict was a priori less salient.

6.1 Municipality-Level Evidence

The analysis begins at the municipality level, acknowledging a key limitation: unlike enrolment data, voter choices are not available by race due to the anonymity of the vote, thereby restricting the analysis to studying aggregate voting outcomes. I begin by investigating whether municipalities more exposed to the policy, and where enrolment gains for the non-White population were larger, changed their voting behaviour. In particular, I estimate the following model:

$$v_{mt} = \gamma \cdot q_{mt} + X'_m \cdot \delta_t + \alpha_m + \alpha_{st} + \varepsilon_{mt} \quad (3)$$

where v_{mt} represents the vote share for candidates of either a specific racial group or ideology in municipality m at year t . The main explanatory variable is local exposure to non-White quotas, q_{mt} , instead of local enrolment of racial groups. This is to account for the possibility that voter realignment along racial identities may not be solely driven by the direct enrolment effects of the policy but also by perceived threats of displacement or empowerment across racial groups that the policy might trigger. The rest of variables are defined as in Equation 2. Crucially, the inclusion of state-year fixed effects accounts for endogenous changes in the supply of federal candidates, as these run at the state level. This helps ensure that the results in federal elections capture changes in voter preferences for candidates due to the policy, rather than simply reflecting shifts in the availability of candidates running for office.

The coefficient of interest is γ and reflects changes in voting outcomes resulting from changes in

local exposure to non-White quotas. As the results in the previous section showed that this measure primarily captures the policy gains experienced by non-White students, one might anticipate only the non-White electorate in those municipalities to react. Nonetheless, existing literature suggests that the socioeconomic advancement of disadvantaged groups can provoke a backlash among the White population if they perceive these gains as a threat to their traditional socioeconomic dominance (Trebbi et al., 2008; Kuziemko and Washington, 2018; Bernini et al., 2023). The estimated value of γ , thus, reflects the net impact of the expansion of racial quotas on changes in voting behaviour, which will depend on the relative responses of both groups to the policy, and their relative population sizes.¹⁵ The identification of γ is essentially the same as the one described in Section 5, with supporting evidence of lack of pre-trends in Appendix C.2.

I begin by assessing the effect of exposure to non-White quotas on support for candidates across ideological groups. Given that left-wing parties were perceived as more vocal in supporting racial quotas in public universities and advocating for non-White communities compared to other parties (see Appendix E.2), they may have become more appealing to the non-White electorate benefiting from the policy. At the same time, as discussed above, this effect could have been counterbalanced by a conservative shift among the White electorate in response to the policy. In Table 2 I report the estimates from the most saturated specification, which includes municipality and state-year fixed-effects, municipality characteristics interacted with a year fixed effect, and municipality exposure to contemporaneous educational policies that could confound the effects of the expansion of non-White quotas. I show that the expansion of racial quotas did not result in an ideological shift in vote shares across parties in federal elections (columns (1) to (3)). This finding is consistent with either a null effect of the policy or an ideological polarization along racial lines that, on aggregate, did not significantly alter overall vote shares.

Next, I examine whether exposure to the policy influenced support for federal candidates based on their racial identity (Table 3). In column (1), I report the aggregate effects, while columns (2) through (4) disaggregate the vote shares by candidate ideology, based on their party affiliation. Panel A shows that the policy did not result in an increase in support for non-White candidates overall, and the results remain statistically insignificant across different ideological groups.¹⁶ However, Figure 4 reveals that this null effect can be explained by an increase in support for Black candidates specifically, which came at the expense of candidates from other racial groups.

In Panel B, I focus specifically on Black candidates. Given that the reform led to a roughly one-to-one increase in non-White enrolment, the results imply that for every additional non-White student admitted through quotas per 100 eligible students, there were approximately 4 additional votes per 100 for Black candidates (0.01×4.00). This represents a 50% increase in their electoral success

¹⁵See Appendix D for a simple model that guides the expected political effects.

¹⁶In Table C.11 I use as dependent variable the weighted vote share for candidates, where the weights correspond to the probability that a candidate exhibits socially prescribed traits of Black Brazilians, as estimated using Vertex AI. The results obtained with this continuous measure are consistent with my main specification.

relative to their baseline support (8%). Moreover, the results indicate that half of this effect is driven by increased support for Black candidates running in left-wing parties (column (2)), while the other half is attributed to greater support for right-wing Black candidates, although the coefficient for the latter is not precisely estimated.

My main hypothesis on the greater electoral success of Black candidates is that they placed more emphasis on race-related issues in their electoral platforms, making them more attractive to non-White communities that benefited from affirmative action, including Mixed-race, Black, and Indigenous peoples. To support this, I conducted a text analysis on over 40,000 electoral programs of candidates running for president, state governors, and mayors from 2010 to 2020, which illustrates how prominently race-related issues were featured in candidates' political agendas. Figure B.2 shows that Black candidates in left-wing parties have historically included more racial issues in their electoral platforms than candidates from other racial groups. Interestingly, after the reform, Black candidates running in center- and right-leaning parties also began to incorporate more race-related topics into their platforms compared to other candidates, albeit to a lesser extent. This greater emphasis on race-related issues likely acted as a signal to non-White voters about candidates' policy stances and their commitment to addressing racial inequalities. If these patterns in executive elections also applied to legislative elections, it would suggest that non-White communities were increasingly drawn to candidates they believed were better positioned to implement policies favouring their racial group.

Another possibility is that the observed shift in support for Black candidates may not stem solely from changes in voter preferences, but rather from changes in the political supply of candidates, which could have constrained voter choices prior to the reform. In federal elections, the estimates account for shifts in candidate supply by including state-specific time trends, as discussed above. Relatedly, left-wing parties also had a higher proportion of Black candidates in their lists, as shown in Figure B.3. However, since I do not observe an aggregate shift toward the left, I can rule out the possibility that the effect on Black candidates was merely a mechanical consequence of their greater representation on left-wing lists.

Elections for city councillor exhibit a similar pattern (Table 4). There was a general increase in support for non-White candidates, primarily driven by Black candidates gaining support in municipalities more exposed to the policy. Given that Black candidates already had a stronger presence in city councils compared to the Federal Congress prior to the reform (21% vs. 8%), the resulting gains in representation are smaller at the local level ($0.01 \times 2.3 / 0.21 = 11\%$). Nonetheless, this increase is still significant enough to enhance the likelihood of non-White candidates securing seats in city councils, as shown in Table B.5. Notably, the increase in support is concentrated among Black candidates running in left-wing parties (column (2)), aligning with a broader shift toward the left (Table 2, columns (4) to (6)). However, in local council elections, where candidates run at the municipality level, estimates might still reflect both voter preferences and shifts in the racial composition of the candidate pool. Indeed, I show that, in municipalities more affected by the policy, the share of Black candidates

running for city councillors increased, especially within left-wing parties (Table B.6). This suggests that while voter demand likely drove support for Black candidates in federal elections, local council election results may also reflect changes in candidate supply. Nonetheless, this evidence indicates that the expansion of racial quotas in federal universities improved the electorate prospects of Black candidates, even in contexts where race was not directly embedded in policymaking.

Overall, these results show that municipalities more exposed to racial quotas, and hence where the non-White population experienced larger gains from the policy, shifted their support toward minority candidates who placed greater emphasis on race-related issues in their campaign programs. If these effects were driven by racial polarization, this would imply that the response from the non-White electorate was stronger than that of White voters. In the next section, I extend the analysis to the polling station level to further assess whether these findings are indeed the result of voter polarization along racial lines.

6.2 Inferring Racial Voting Behaviour using Polling-Station Data

To shed light on the differential response to racial affirmative action across racial groups, I exploit variation across highly granular geographical units: polling stations. On average, there are more than 250 stations per municipality, each serving around one thousand voters. This level of analysis allows to identify how exposure to the reform led to different responses among racial groups within a municipality, as inferred from the racial composition of polling stations. In particular, I estimate the following model:

$$v_{lmt} = \sum_{\tau=1}^5 \theta_{\tau} \cdot q_{mt} \cdot shNW_{l,\tau} + S'_{lmt} \cdot \delta + \alpha_l + \alpha_{st} + \varepsilon_{lmt} \quad (4)$$

where lmt denotes a generic polling station l located in municipality m at year t . v_{lmt} represent the vote share for non-White candidates, and q_{mt} refers to exposure to racial quotas at the municipality level. The racial composition of polling stations is captured by $shNW_{l,\tau}$. This variable equals 1 if the share of non-White population in the polling station's catchment area is in the τ -th quintile and 0 otherwise. S'_{lmt} is a set of socioeconomic characteristics of the electorate in polling station l at baseline, including age, gender, education, race, and income profiles, which I interact with a year fixed effect. Polling station fixed effects, α_l , and state-year fixed effects, α_{st} , are included, which control for time-invariant polling stations characteristics and state-specific trends, respectively. The error term, ε_{lmt} is clustered at the municipality level.

The coefficients of interest are θ_{τ} , which capture the differential effect of the expansion of racial quotas on voting patterns depending on the polling station racial composition. Specifically, coefficients θ_1 and θ_2 approximately capture the effect of exposure to non-White quotas on the voting behaviour of predominantly White polling stations, while θ_4 and θ_5 capture the response predominantly of the non-White electorate.

Figure 5 shows that the expansion of non-White quotas lead to the ideological polarization of voters along racial lines in elections for the Federal Congress. Panel (a) presents the estimates of Equation 4, where the dependent variable is the vote share for candidates affiliated with left-wing parties. Panels (b) and (c) display the corresponding estimates for the policy’s effect on voter support for candidates from center-leaning and right-wing parties, respectively. I find that while the non-White electorate increased support for candidates running in left-wing parties, the White electorate responded by increasing their support for more conservative parties.¹⁷ This result is consistent with existing literature documenting the political impacts of targeted programs, particularly the use of such programs by voters to infer parties’ policy preferences (Manacorda et al., 2011; Zucco, 2013; Caprettini et al., 2021). Moreover, the shift of the White electorate towards more conservative positions aligns with the literature documenting a White backlash from the socioeconomic advancement of racial minorities in the U.S. (Trebbi et al., 2008; Kuziemko and Washington, 2018; Bernini et al., 2023).

Having established that the expansion of affirmative action in federal universities became politically salient for both racial groups in exposed municipalities, I next examine whether it also influenced preferences for candidates’ racial identity. Figure 6 presents the estimates of Equation 4, focusing specifically on support for Black candidates in federal elections, as these candidates received the most electoral gains in municipalities with greater exposure to the policy. In Panel (a) I include all Black candidates, while in Panels (b) to (d) I split them by ideology. The results reveal an overall increase in support for Black candidates across racial groups, with the effect being primarily driven by polling stations where the majority of the electorate is White. When examining the results by ideology, I find that the policy mobilized the non-White electorate to support Black candidates affiliated with left-wing parties, whereas White voters shifted their support toward conservative Black candidates, particularly those running in right-wing parties.

As discussed in the municipality analysis, the increased support for Black candidates among the non-White electorate can be explained by these candidates’ stronger emphasis on race-related issues in their electoral platforms, signalling a commitment to advocate for the interests of non-White voters. However, the simultaneous shift in support toward left-wing parties among the non-White electorate raises the possibility that this effect is driven by the greater presence of Black candidates on these parties’ lists, rather than an explicit preference for Black candidates themselves. In Figure B.6, I show that exposure to the policy resulted in a greater increase in votes for left-wing Black candidates compared to left-wing candidates overall, indicating that the non-White electorate disproportionately supported Black candidates within left-wing parties. Combined with the shift toward the left, which came largely at the expense of center parties, this indicates that non-White voters crossed party lines to support candidates from their racial group.

The behaviour of the White electorate to the expansion of non-White quotas suggest a complex

¹⁷Note that I find no evidence that this increase in support for the left was driven by greater support for the PT (Figure B.5), ruling out a direct reward from the non-White electorate to the incumbent party for implementing the policy.

response to the policy. Coefficients from polling stations in the first and second quintiles of non-White population share indicate a significant increase in support for right-wing Black candidates among White voters (Figure 6d). Specifically, the coefficients imply that for each non-White student benefiting from the policy out of a 100 eligible students, 6 votes out of 100 were cast by White voters in favour of these candidates. As shown in Figure 5, this shift was not driven by an increase in support for right-wing parties as a whole, but by a surge in support for center-leaning conservative candidates, particularly at the expense of the left and, most notably, the PT (Figure B.5). This suggests two stories: first, White voters became more conservative in municipalities where non-Whites benefited more from the policy; and second, already conservative White voters increasingly supported Black conservative candidates, as further supported by Figure B.6.

While the first point is consistent with the literature documenting a White backlash from the socioeconomic advancement of racial minorities (Trebbi et al., 2008; Kuziemko and Washington, 2018; Bernini et al., 2023), the increased support for conservative Black candidates adds a new dimension to this backlash. In this paper, I provide a potential explanation for this: some conservative Black candidates actively opposed racial affirmative action and rejected the racial justice agenda promoted by the left, making them appealing to White conservative voters. Black candidates such as Hélio Lopes, a close ally of Jair Bolsonaro, illustrates this point. He was the most-voted federal deputy in the state of Rio de Janeiro in 2018 and has been vocal in his opposition to racial quotas and other policies aimed at remedying the historical disadvantages experienced by Afrodescendants (Folha de S. Paulo, 2023). Another prominent example is Sérgio Camargo, appointed by Bolsonaro in 2019 as the president of the Palmares Cultural Foundation, a state-owned organization that aims to promote Afro-Brazilian culture. Camargo, who ran for federal deputy in São Paulo in 2022, has openly denied the existence of racism in Brazil, and accused Black voters of victimism and being misinformed by the left (O Globo, 2019a).

The elections for city councillors reveal a similar pattern. As shown in Figure 8, non-White voters increasingly shifted their support toward non-White candidates affiliated with left-wing parties, while White voters supported more conservative Black candidates, primarily within center-leaning parties. It is important to acknowledge the potential for a mechanical effect, as the electorate appears to have polarized along ideological lines, with non-White voters favouring the left and the White electorate aligning with center-leaning parties (Figure 7). Moreover, these observed effects do not exclude the possibility of supply-driven factors influencing voting behaviour. Nevertheless, the results suggest that the increase in the number of non-White candidates running for local council within the left was predominantly met with support from the non-White electorate. This indicates that even if preferences for same-race candidates were not explicitly activated by the policy, they nonetheless existed and were exercised once candidates from their racial group became available.

In Appendix C.2, I present a series of robustness tests. First, I show that the effects of the policy remain consistent when using the continuous measure of race derived from Vertex AI. Second, I include

municipality-year fixed effects to control for time-varying unobservable factors at the municipality level that could correlate with the exposure measure. The results remain consistent with the main findings. Third, I demonstrate that the racial composition of polling stations is not confounded by other electorate characteristics, such as income or educational attainment, further confirming that the results are driven by differences in voting patterns across racial groups. Finally, I provide evidence of the absence of pre-trends, supporting the validity of my empirical design.

7 The Trade-offs of Racial Voting

So far, I documented that the expansion of racial affirmative action across federal universities led to increased support for Black candidates in federal and local elections and improved their descriptive representation in city councils. While this progress in descriptive representation has the potential to enhance the representation of the non-White electorate in policymaking, the racial polarization documented in previous sections raises important concerns about potential trade-offs. In this section, I explore two of such trade-offs: first, whether voters prioritized racial identity of candidates over their competence, and second, how the racialization of politics may have influenced public goods provision, as voters may have become less inclined to support policies they perceive as benefiting other racial groups.

7.1 Quality of Policymakers

The use of heuristics to make political choices has been documented in low-information environments and as a way to simplify complex political decisions (Casey, 2015; Aguilar et al., 2015). In Brazil’s legislative elections, the large number of candidates — ranging from an average of 75 in local elections to over 200 in federal elections — makes the voting process particularly challenging. If racial affirmative action led voters to prioritize candidates’ racial identity over other relevant attributes relevant for policymaking, this could result in the selection of less-qualified individuals (Banerjee and Pande, 2007).

To explore this concern, I examine the effect of exposure to racial affirmative action on two measures of candidate competence: tertiary education attainment and prior experience in office. Candidate’s education level is the most common proxy for candidate competence in the literature of political selection (Dal Bó and Finan, 2018), but could also reflect their socioeconomic background. Likewise, differences in candidates’ political experience can also reflect factors influencing their chances of gaining office, such as differences in party endorsements, campaign financing, or voters’ attitudes toward different racial groups. Consequently, if non-White candidates consistently displayed lower educational levels or political experience, a rise in racial voting could mechanically lead to a decline in the average selection of competent candidates, as proxied by these measures.

Table 5 presents the findings at the municipality level. In Panel A, columns (1) and (2), I assess the impact of local exposure to racial quotas on the vote share of candidates with tertiary education in both federal and local elections. Although the results are not statistically significant, the coefficients exhibit different patterns across election types: a positive coefficient appears in federal elections, but a negative one in local elections. This could reflect different barriers to entry, with federal elections attracting more qualified candidates. In columns (3) and (4), I analyse support for candidates with prior political experience—defined as those who previously held federal, state, or local office. Here, while the coefficients lack statistical significance, both federal and local elections show negative effects.

In Panel B, I focus on the support for Black candidates, the group that electorally benefited most from the implementation of racial affirmative action. All coefficients significantly increase in magnitude but remain not statistically significant. The first column shows that the increase in support for graduate candidates in federal election is mostly driven by an increase in the support for Black candidates. The negative coefficient in local elections also increases in magnitude but remains negative. Regarding political experience, while the coefficients are not statistically significant, they are both positive. This suggests that while overall there might have been a decline in the selection of candidates with prior political experience, affirmative action appears to have encouraged voters to support Black candidates with high educational attainment and prior experience in office, lowering concerns regarding a potential decline in candidates' competence.

Despite these findings, the overall selection of competent policymakers at the municipal level remains statistically insignificant and tends to be negative. Given the historical dominance of White candidates in Brazilian politics, this outcome was somewhat expected. Nevertheless, while trade-offs regarding political experience may exist, the long-term benefits of enhancing non-White representation in office, thus promoting racial equality in political representation, are likely to outweigh these initial drawbacks.

7.2 Public Goods Provision

The second concern regards the provision of public goods. Seminal papers in political economy argue that fragmented societies tend to exhibit lower levels of public goods, as racial groups may have different policy preferences or exhibit preferences for redistribution within their own group (Alesina et al., 1999; Fong and Luttmer, 2009). In other words, as political divisions deepen along racial lines, voters may become less willing to support funding for public goods if they perceive that these resources will disproportionately benefit other racial groups. To investigate this potential consequence of racial voting, I follow Alesina et al. (1999)'s empirical exercise and analyse the effects of the implementation of racial affirmative action on local government spending per capita, as well as its composition. For the analysis, I use data from the Brazilian Public Sector Accounting and Tax Information System (Siconfi) covering the period from 2006 to 2020.

Table 6 reports the results. In column (1), I report the effect of exposure to racial quotas on total spending per capita, expressed in logarithmic terms. The findings indicate a negative, albeit sta-

tistically insignificant, effect on overall spending per capita. This suggests that the introduction of racial quotas may not have led to a substantial decrease in local government expenditures. Furthermore, in columns (2) through (8), I disaggregate spending into major categories. The results show no statistically significant effects on each category of spending within municipalities. One concern arises from the negative coefficient observed in the share of spending allocated to education. A key requirement for eligibility in the Law of Quotas was that applicants had attended high-school in a public institution. This could have led to a reluctance among White voters to invest in these schools if they perceive that the benefits are primarily accruing to non-White students. Alternatively, there may be a strategic incentive for voters to increase funding for public schools, motivated by the desire to improve their own children’s chances of gaining admission to university. Investigating this is work in progress and is not included in the current version of the paper.

8 Conclusion

Affirmative action policies are often proposed as solutions to address longstanding inequalities among disadvantaged groups in a wide range of settings. Evidence shows that, indeed, targeted policies, particularly those focused on race, are more effective at promoting diversity than income- or merit-based approaches. This paper aims to contribute to an informed debate on the broader effects of affirmative action policies by highlighting a crucial unintended trade-off: while these initiatives can promote the socioeconomic advancement of marginalized groups through the reallocation of scarce resources, they can also inadvertently deepen political divisions along identity lines.

This paper provides a first attempt at exploring this unintended consequence of affirmative action by exploring the political effects of the implementation of Brazil’s Law of Quotas, a major national educational reform that mandated the reservation of admission seats in federal universities for low-income and non-White students. Employing a municipality exposure measure in the spirit of a shift-share IV, I document that the expansion of seats across universities significantly increased the enrolment of non-White students in federal universities, as intended. Moreover, the results suggest that the cost of this policy might have been borne by all Whites students in Brazil, irrespective of the gains enjoyed by the non-Whites in the same municipality.

Furthermore, I find that the policy led to the ideological polarization of the electorate along racial lines in the federal legislative office, where the policy conflict over race was more pronounced, but that also extended to lower government levels. Moreover, both targeted and non-targeted racial groups increased their support for Black candidates, but with opposing ideological stances. Specifically, the non-White electorate increased its support for Black candidates running in progressive parties, which have traditionally emphasized race-related issues in their platforms. In contrast, I document that the White electorate, reacting to the policy gains of non-Whites, shifted their support toward Black conservative candidates. This finding reveals a novel mechanism: potential losers of group-based

policies may perceive minority candidates in conservative parties as more credible opponents of such policies, offering a way to oppose them without being perceived as exclusionary or anti-minority.

Understanding these consequences of affirmative action is of economic interest due to their implications for policymaking. A key concern highlighted in the literature is whether the racialization of politics might undermine the competence of elected candidates. While the results are imprecisely estimated, they suggest that voters' preferences for candidates' racial identity may have come at the expense of candidate quality, particularly in terms of political experience. Another concern is whether the resulting polarization of the electorate could diminish the level or quality of public goods provision. My findings provide no evidence to support this concern, suggesting that affirmative action can achieve its intended goals without significantly compromising the overall quality of policymaking.

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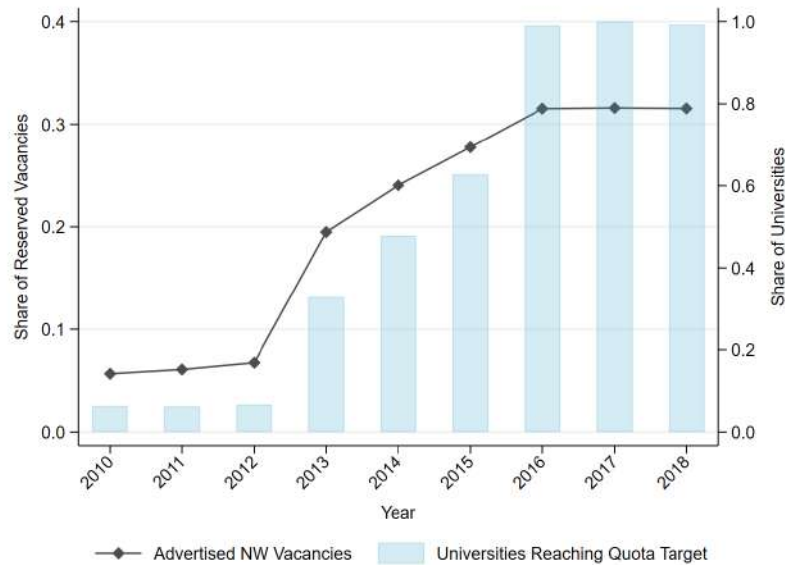
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10 Main Tables and Figures

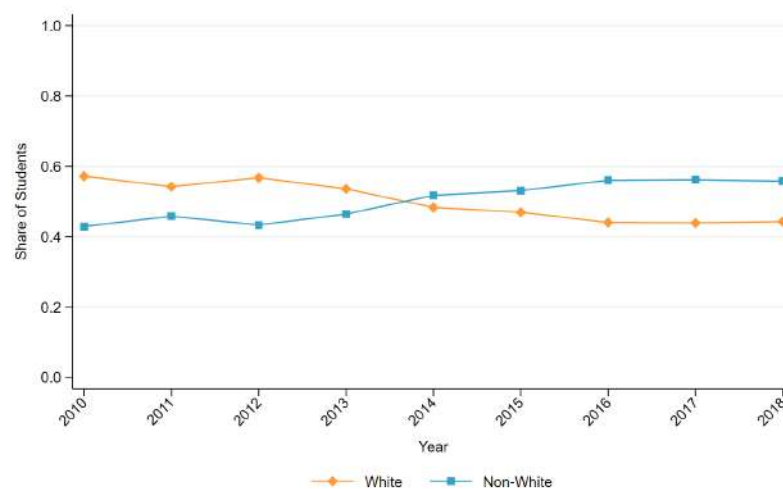
10.1 Descriptives

Figure 1: Implementation of Non-White Quotas, 2010-2018



Note: This graph plots the implementation of non-White quotas in federal universities from 2010 to 2018. The left axis displays the advertised share of seats reserved for non-White students. The right axis displays the share of universities that achieved the targeted level of non-White quotas mandated by the Law. Universities are weighted according to their size.

Figure 2: Enrolment in Federal Universities by Racial Group, 2010-2018

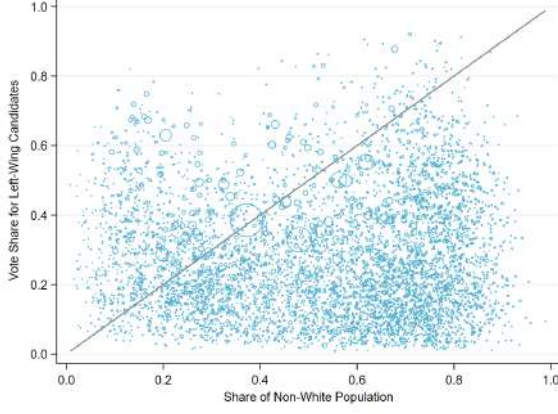


Note: This figure shows the share of incoming students by racial group in federal universities from 2010 to 2018, conditional on self-reporting their racial group at enrolment. The orange line represents White students, and the blue line represents non-White students.

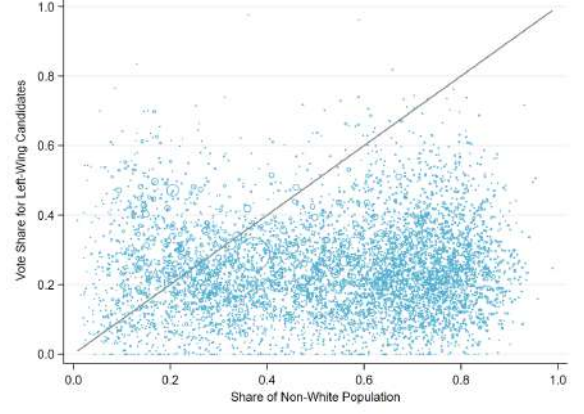
Figure 3: Racial Voting before the Law of Quotas

I. Vote for Left-Wing Candidates

(a) Federal Deputies

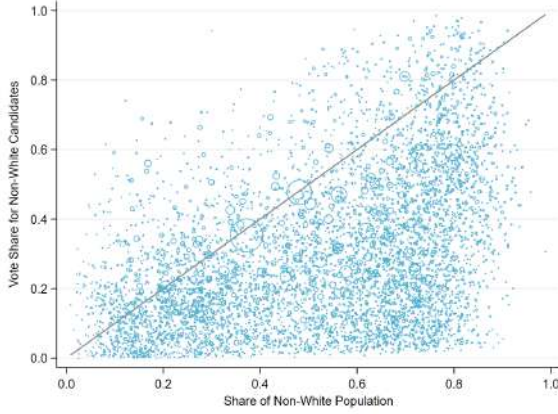


(b) City Councillors

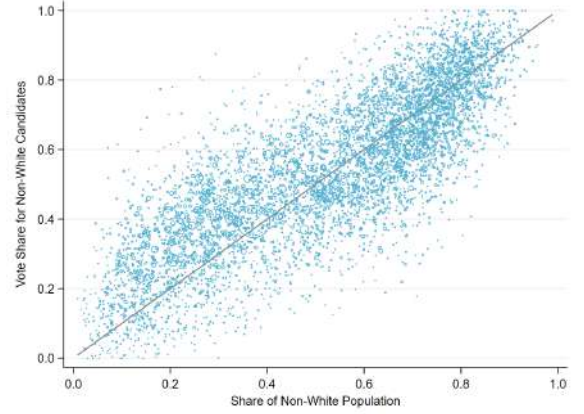


II. Vote for Same-Race Candidates

(c) Federal Deputies



(d) City Councillors



Note: This figure illustrates patterns of racial voting before the implementation of the Law of Quotas. Panels (a) and (b) show the relationship between the vote share for left-wing candidates and the share of the non-White population at the municipality level for the 2010 federal and 2012 local legislative elections, respectively. Panels (c) and (d) depict the relationship between the vote share for non-White candidates and the non-White population share for the same elections. The gray line represents a 45° line for reference. Each circle corresponds to a municipality, with the size of the circles proportional to the municipality's population.

10.2 Education Results

Table 1: The Effect of Racial Quotas on Enrolment in Federal Universities

DV: Enrolment Rate	Non-White Quota	Non-White	White
	(1)	(2)	(3)
Exposure to Non-White Quotas	0.874*** (0.260)	0.798** (0.362)	-0.384 (0.407)
Avg DV at baseline	.001	.007	.009
Municipalities	5,460	5,460	5,460
Observations	49,140	49,140	49,140

Note: This table reports the estimates of Equation (2). The dependent variable is the municipality enrolment rate, defined as the number of students of each category who enrol in a federal university relative to the university-age population in the municipality. All regressions include municipality and state-year fixed effects, as well as the baseline municipality characteristics described in Section 5 interacted with a year fixed-effect. Regressions are weighted by municipality population between 18 to 24 years old at baseline. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

10.3 Political Results

10.3.1 Municipality Level

Table 2: The Effect of Racial Quotas on Municipality Voting along the Ideological Spectrum

DV: Vote Share	<i>Federal Deputies</i>			<i>City Councillors</i>		
	Left	Center	Right	Left	Center	Right
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure to Non-White Quotas	-0.461 (1.653)	1.790 (1.990)	0.363 (1.636)	2.797*** (0.928)	-0.407 (1.074)	-2.760*** (1.058)
Avg DV at baseline	0.324	0.299	0.329	0.256	0.277	0.328
Municipalities	5,460	5,460	5,460	5,460	5,459	5,459
Observations	21,840	21,840	21,840	21,840	21,810	21,810

Note: This table reports the estimates of Equation (3). The dependent variable is the vote share for candidates running in left-, center-, and right-wing parties, as classified by Zucco and Power (2021). I exclude the vote share for unclassified parties. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

Table 3: The Effect of Racial Quotas on Municipality Support for Non-White Federal Candidates

	Total	By Ideology		
		Left	Center	Right
	(1)	(2)	(3)	(4)
<i>Panel A. Vote Share for Non-White Candidates</i>				
Exposure to Non-White Quotas	0.555 (2.188)	1.622 (1.243)	-0.918 (1.070)	1.575 (1.706)
Avg DV at baseline	0.330	0.126	0.073	0.110
<i>Panel B. Vote Share for Black Candidates</i>				
Exposure to Non-White Quotas	4.001** (1.694)	1.942*** (0.657)	0.356 (0.419)	1.460 (1.552)
Avg DV at baseline	0.075	0.038	0.011	0.017
Municipalities	5,460	5,460	5,460	5,460
Observations	21,840	21,840	21,840	21,840

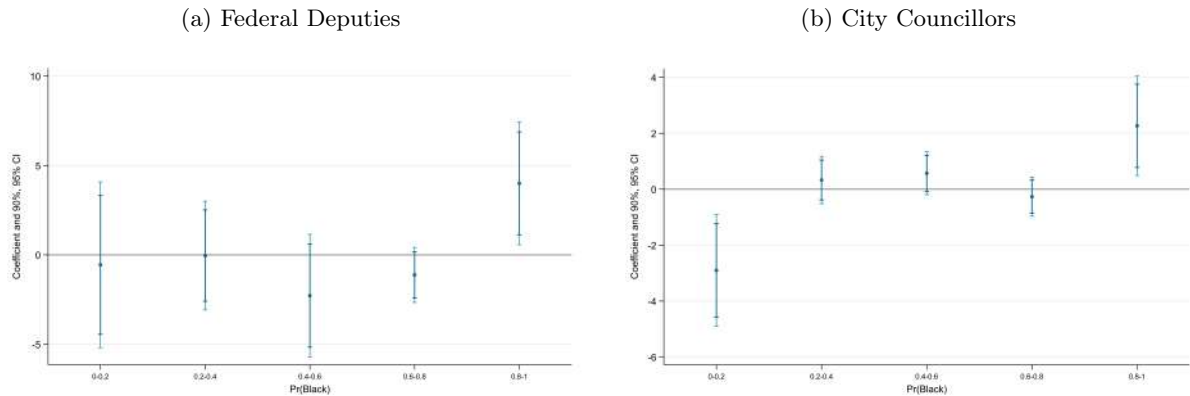
Note: This table reports the estimates of Equation (3). In Panel A, the dependent variable is the vote share for non-White candidates in federal legislative elections, while in Panel B the dependent variable is the vote share for Black candidates. Column (1) includes all candidates, and columns (2) to (4) break down candidates by those running in left-, center-, and right-wing parties. I exclude candidates in unclassified parties. All regressions include municipality and state-year fixed effects, as well as the baseline municipality characteristics interacted with a year fixed-effect. Regressions are weighted by municipality population at baseline. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

Table 4: The Effect of Racial Quotas on Municipality Support for Non-White Local Candidates

		By Ideology		
	Total	Left	Center	Right
	(1)	(2)	(3)	(4)
<i>Panel A. Vote Share for Non-White Candidates</i>				
Exposure to Non-White Quotas	2.891***	2.168***	0.332	-0.456
	(1.014)	(0.721)	(0.709)	(0.732)
Avg DV at baseline	0.534	0.142	0.132	0.173
<i>Panel B. Vote Share for Black Candidates</i>				
Exposure to Non-White Quotas	2.309***	1.394**	0.602*	-0.251
	(0.885)	(0.576)	(0.366)	(0.427)
Avg DV at baseline	0.213	0.060	0.048	0.067
Municipalities	5,459	5,459	5,459	5,459
Observations	21,680	21,680	21,680	21,680

Note: This table reports the estimates of Equation (3). In Panel A, the dependent variable is the vote share for non-White candidates in local legislative elections, while in Panel B the dependent variable is the vote share for Black candidates. Column (1) includes all candidates, and columns (2) to (4) break down candidates by ideology. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

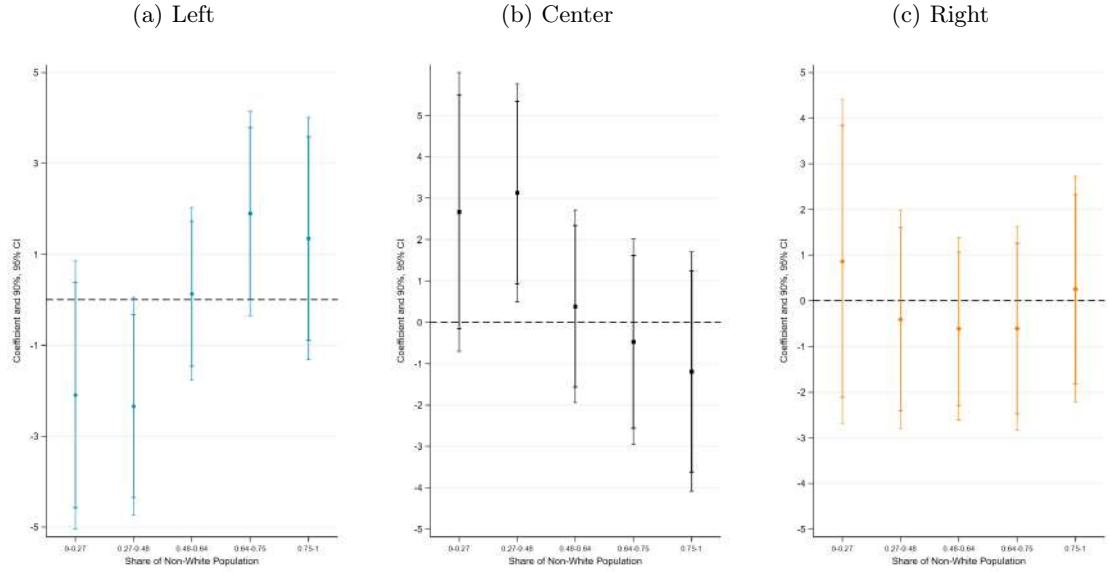
Figure 4: The Effect of Racial Quotas on Municipality Support for Candidates, by Pr(Black)



Note: These figures report the estimates of Equation (3). The dependent variable is the vote share for candidates with different predicted probabilities of conforming with the socially-prescribed Black phenotypical traits (probability quintile, along the horizontal axis). See also notes for Table 3. Confidence intervals are provided at the 90% and 95% levels.

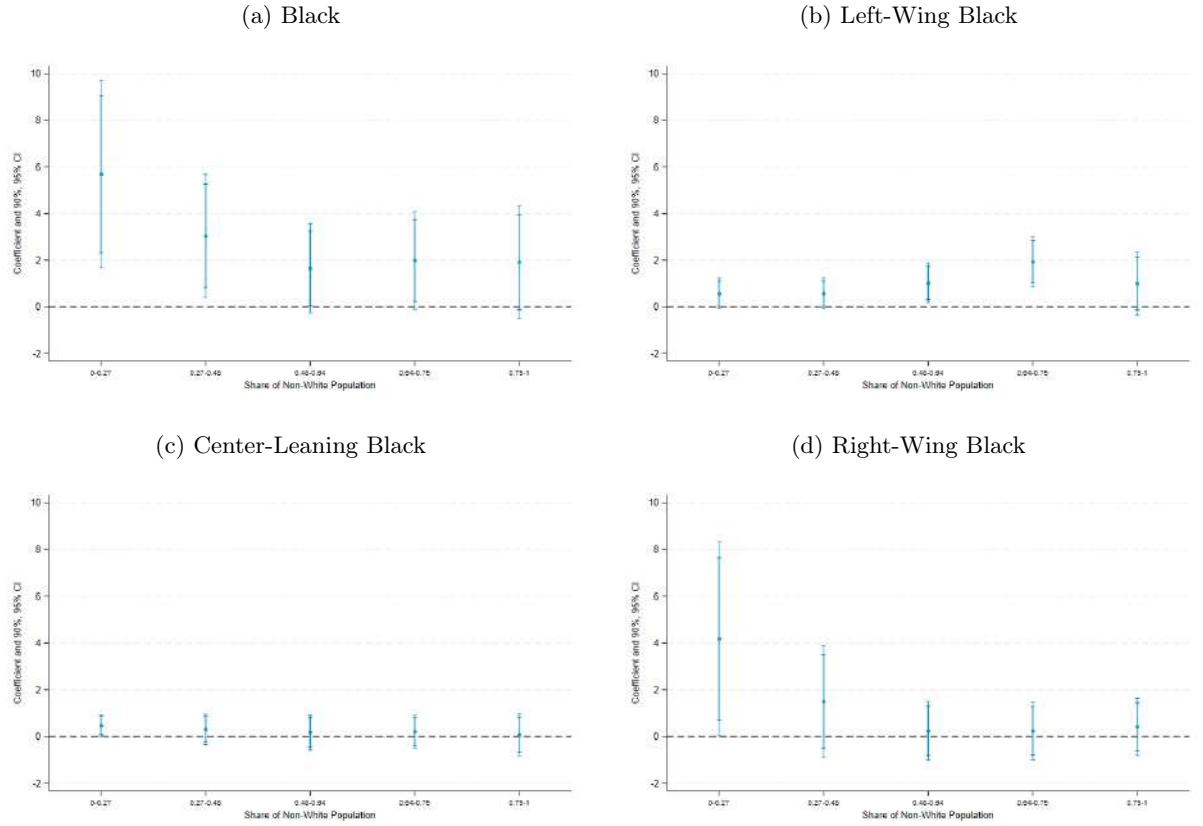
10.3.2 Polling-Station Level

Figure 5: The Effect of Racial Quotas on Polling-Station Support for Federal Candidates by Ideology



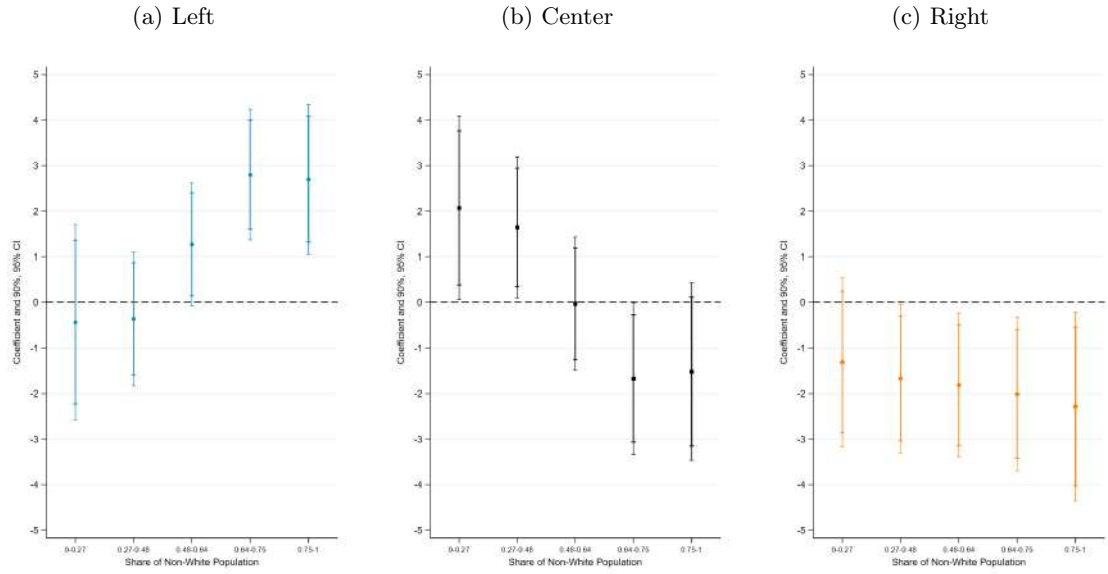
Note: This figure reports the estimates of Equation (4). The dependent variable is the vote share for candidates running for federal deputies in left-, center-, and right-wing parties, as classified by Zucco and Power (2021). Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis). All regressions include polling-station and state-year fixed effects, as well as baseline polling-station and municipality characteristics interacted with year fixed effects. Regressions are weighted by baseline polling-station population. Confidence intervals are provided at the 90% and 95% levels.

Figure 6: The Effect of Racial Quotas on Polling-Station Support for Black Federal Candidates by Ideology



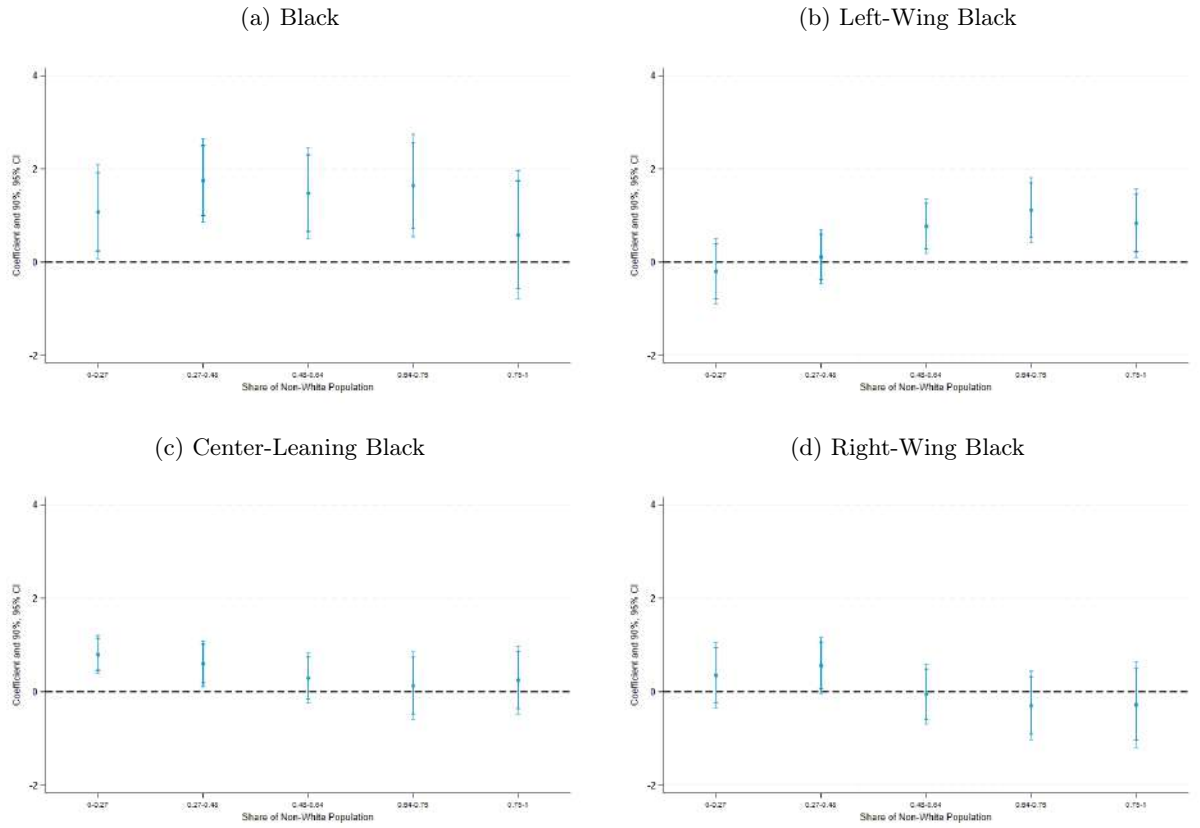
Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in federal legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis). Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure 7: The Effect of Racial Quotas on Polling-Station Support for Local Candidates by Ideology



Note: This figure reports the estimates of Equation (4). The dependent variable is the vote share for candidates running for city councillors in left-, center-, and right-wing parties, as classified by Zucco and Power (2021). Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis). See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure 8: The Effect of Racial Quotas on Polling-Station Support for Black Local Candidates by Ideology



Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in local legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis). Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

10.3.3 Trade-offs

Table 5: The Effect of Racial Quotas on Candidates' Competence

	<i>Graduate Candidates</i>		<i>Political Experience</i>	
	Federal	Local	Federal	Local
	(1)	(2)	(3)	(4)
<i>Panel A. All Candidates</i>				
Exposure to Non-White Quotas	0.497 (2.032)	-1.234 (0.823)	-1.887 (1.613)	-1.137 (0.961)
Avg DV at baseline	0.746	0.348	0.784	0.411
<i>Panel B. Black Candidates</i>				
Exposure to Non-White Quotas	3.109* (1.665)	-0.215 (0.456)	2.079 (1.692)	0.773 (0.542)
Avg DV at baseline	0.043	0.050	0.050	0.068
Municipalities	5,460	5,459	5,460	5,460
Observations	21,840	21,810	21,840	21,810

Note: This table reports the estimates of Equation (3). In the columns under “Graduate Candidates”, the dependent variable is the vote share for candidates with tertiary education. In the columns under “Political Experience”, the dependent variable is the vote share for candidates with political experience, defined as candidates who have held office at some point before the election. Panel A includes all candidates, while Panel B restricts the sample to Black candidates. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

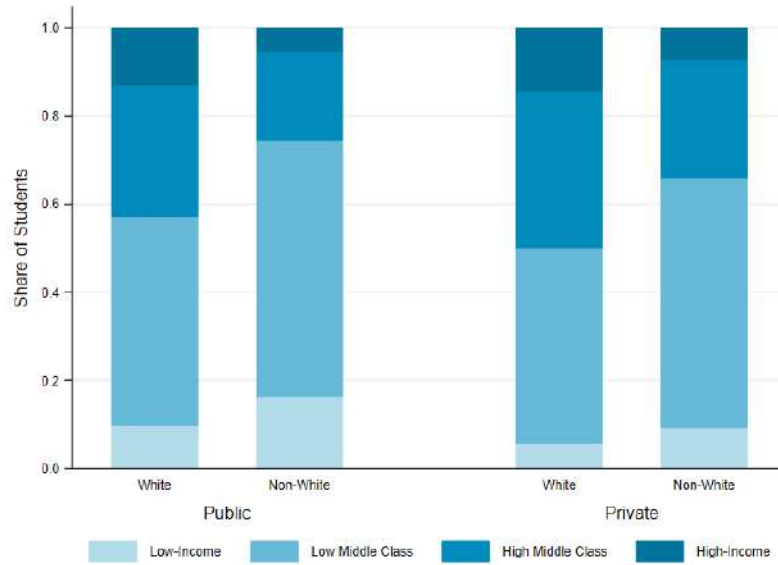
Table 6: The Effect of Racial Quotas on Local Spending: Total and Composition

	<i>Spending Categories (shares)</i>							
	Total	Social Assistance	Health	Education	Urbanism	Sanitation	Transport	Other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exposure to Non-White Quotas	-0.919 (1.629)	0.013 (0.054)	0.022 (0.246)	-0.120 (0.143)	0.037 (0.248)	0.161 (0.157)	0.063 (0.102)	-0.160 (0.285)
Avg DV at baseline	8.712	.010	.079	.091	.037	.009	.008	.767
Municipalities	5,459	5,458	5,458	5,458	5,457	5,445	5,447	5,459
Observations	42,041	41,948	41,965	41,970	41,598	33,407	37,207	42,041

Note: This table reports the estimates of Equation (3). In column (1), the dependent variable is log-spending per capita by municipal governments. In columns (2) to (9), spending is disaggregated into categories, with the dependent variable expressed as a share of total spending. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

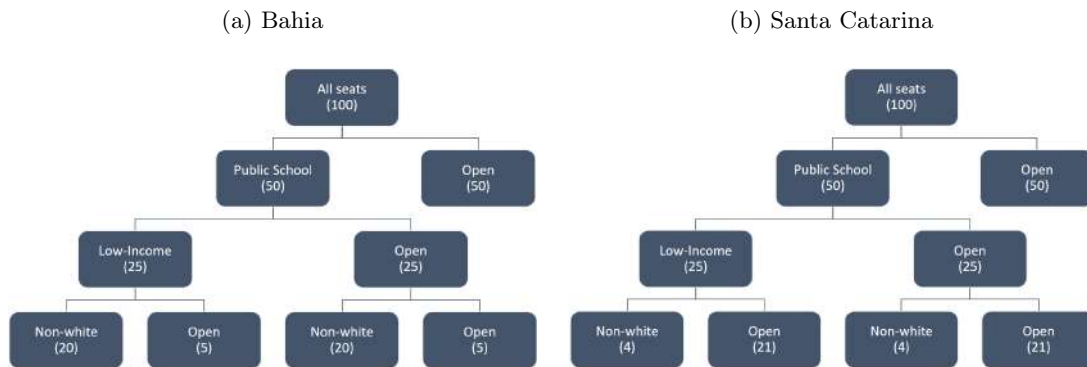
A Supplementary Figures and Tables: Context

Figure A.1: Income and Racial Composition of Brazilian Universities, 2010



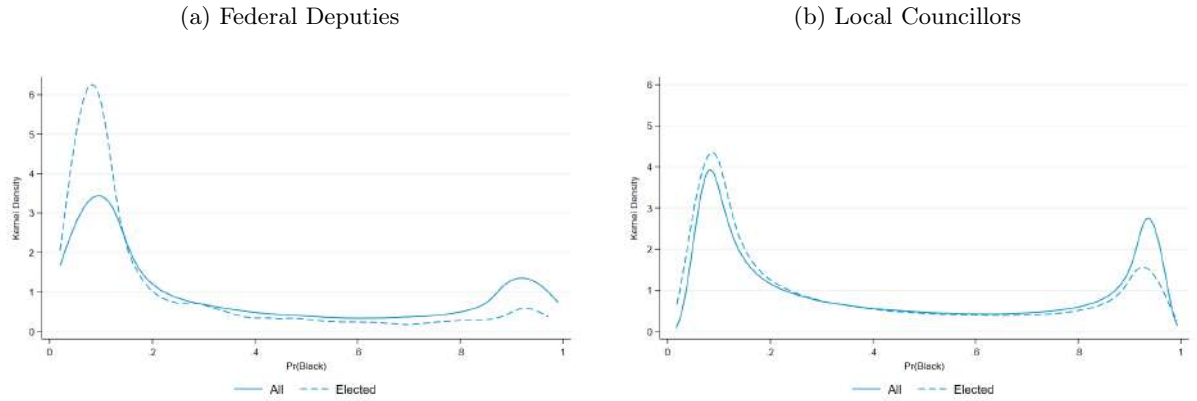
Note: This figure displays the income and racial composition of students enrolled in public (panel (a)) and private universities (panel (b)) as of year 2010. I follow the definition of income groups from Clément et al. (2020). I define low-income students as those coming from families with household income of less than 0.5 minimum salaries per capita.

Figure A.2: Mandated Implementation of the Law of Quotas



Note: These diagrams represent the targeted allocation of admission seats, as mandated by the Law of Quotas. Panel (a) depicts the distribution of seats for federal universities located in the state of Bahia, where 77% of the population is non-White. Panel (b) shows the allocation of quotas in the state of Santa Catarina, where 16% of the population is non-White as of 2010.

Figure A.3: Racial Composition of the Candidate Pool, 2006-2020



Note: These figures present the Kernel density estimates for the probability that candidates exhibit the physical traits associated with Black Brazilians. The solid lines represent the distribution of the entire candidate pool, while the dashed lines depict the distribution of elected officials only. Panel (a) shows the distribution for federal deputies, and Panel (b) for city councillors. The Kernel function used is Epanechnikov.

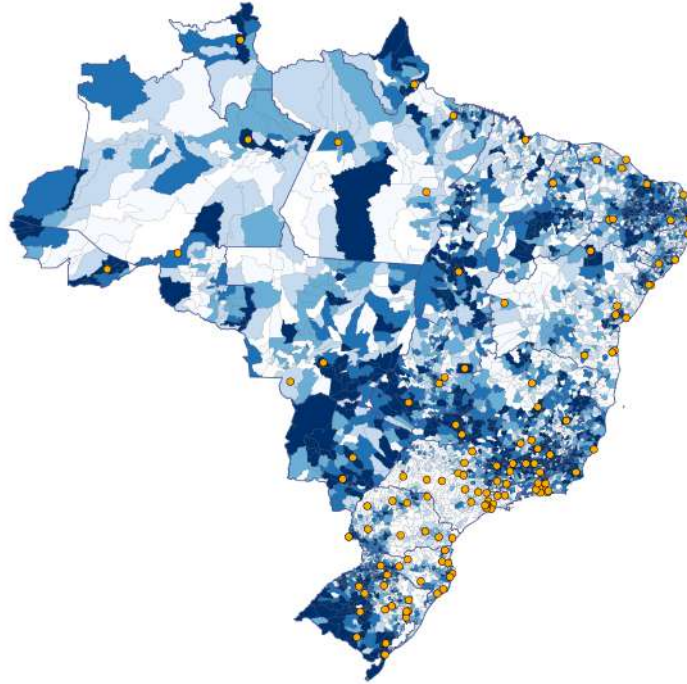
B Supplementary Figures and Tables: Analysis

Table B.1: Summary Statistics of Exposure to Non-White Quotas

	Mean	SD	Min	Median	Max
Admission seats reserved to non-Whites:					
Q_{ut}	831	938	0	550	4,758
Enrolment patterns (non-zero):					
s_{mu}/s_u	0.024	0.104	0.000	0.002	0.944
Population at risk in m:					
pop_m^{18-24}	148,142	327,322	83	16,332	1,328,138
Exposure to non-White quotas in m:					
q_{mt}	0.003	0.005	0.000	0.001	0.119

Note: Observations: 4,709,682. Observations are weighted by municipality population in 2010. Summary statistics for the measures of exposure to non-White quotas and the number of reserved admission seats are calculated for the period 2010-2018. The rest of variables are as of 2010. As the matrix of enrolment patterns display a large number of zeros, I present summary statistics conditional on them being non-zero.

Figure B.1: Average Exposure to Racial Quotas, 2013-2018



Note: This map displays the average exposure to racial quotas implemented in federal universities across municipalities for the period 2013-2018. Darker colours represent higher exposure. The orange dots indicate the municipality where the headquarter of a federal university is located. Grey lines represent municipality limits, and dark blue lines represent state limits.

Table B.2: Summary Statistics of Control Variables, 2010

	Mean	SD	Min	Max
Demographics:				
Population	1,237,604	2,793,564	805	11,253,503
Population Density	5.361	2.261	0.123	9.314
Share of Population 18-24	0.125	0.011	0.063	0.273
Share of Women	0.510	0.016	0.189	0.542
Share Non-Whites	0.512	0.208	0.008	0.989
Education:				
Literacy Rate	0.890	0.085	0.544	0.983
Share of Population with Tertiary Education	0.071	0.048	0.001	0.240
Share of Public HS students	0.883	0.092	0.269	1.000
High-School Enrolment	4.912	0.987	0.000	15.494
Average ENEM Grade (std.)	-0.151	0.251	-1.728	2.272
Geographic:				
Municipalities with a University Campus	0.538	0.499	0.000	1.000
Ln(Distance to Big City)	2.848	2.113	0.000	6.806
Economy:				
Household Median Income	1,283.718	507.586	134.000	3,000.000
Share of Middle-Class Households	0.617	0.165	0.103	0.947

Note: Observations: 5,565. Observations are weighted by municipal population.

Table B.3: The Effect of Racial Quotas on Enrolment in State and Private Universities

DV: Enrolment Rate	<i>State</i>		<i>Private</i>	
	Non-White	White	Non-White	White
	(1)	(2)	(3)	(4)
Exposure to Non-White Quotas	-0.101 (0.076)	-0.114 (0.071)	-0.166 (0.229)	0.012 (0.295)
Avg DV at baseline	.004	.007	.005	.015
Municipalities	5,460	5,460	5,460	5,460
Observations	49,140	49,140	49,140	49,140

Note: This table reports the estimates of Equation (2). The dependent variable is the university enrolment rate, defined as the number of students of each group who enrol in state (columns (1) to (2)) or private universities (columns (3) to (4)) relative to the university-age population size in the municipality. See also notes for Table 1. Significance levels: *** 1%, ** 5%, * 10%.

Table B.4: The Effect of Racial Quotas on Enrolment in Federal Universities, by Non-White Group

DV: Enrolment Rate	Mixed-Race	Black	Indigenous
	(1)	(2)	(3)
Exposure to Non-White Quotas	0.773*** (0.285)	-0.009 (0.083)	0.034 (0.036)
Avg DV at baseline	.004	.003	0.0008
Municipalities	5,460	5,460	5,460
Observations	49,140	49,140	49,140

Note: This table reports the estimates of Equation (2). The dependent variable is the non-White enrolment rate by non-White racial subgroup, defined as the number of students of each group in a municipality who enrol in a federal university relative to the university-age population size in the municipality.

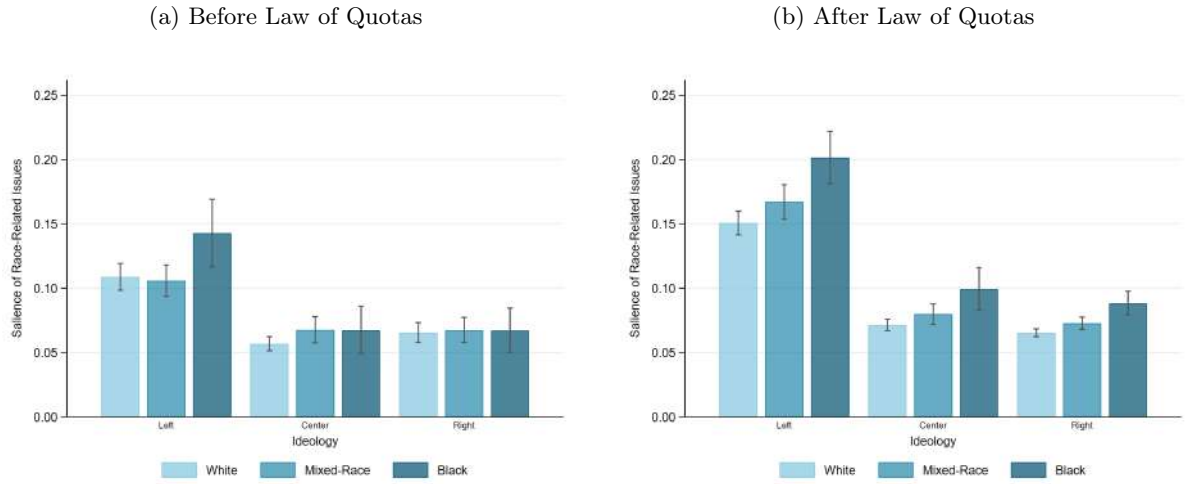
See also notes for Table 1. Significance levels: *** 1%, ** 5%, * 10%.

Table B.5: The Effect of Racial Quotas on the Share of Non-White Local Elected Candidates

	Total	By Ideology		
		Left	Center	Right
	(1)	(2)	(3)	(4)
<i>Panel A. Share of Non-White Elected Candidates</i>				
Exposure to Non-White Quotas	3.957*** (1.428)	2.724** (1.147)	0.856 (0.967)	-0.045 (1.080)
Avg DV at baseline	0.484	0.135	0.121	0.163
<i>Panel B. Share of Black Elected Candidates</i>				
Exposure to Non-White Quotas	2.995** (1.346)	2.382** (1.073)	0.664 (0.568)	-0.339 (0.652)
Avg DV at baseline	0.171	0.049	0.040	0.057
Municipalities	5,459	5,459	5,459	5,459
Observations	21,680	21,680	21,680	21,680

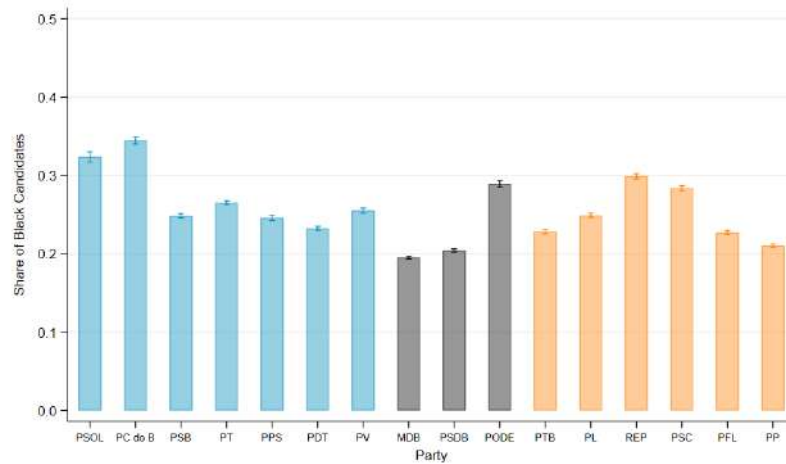
Note: This table reports the estimates of Equation (3). In Panels A and B, the dependent variable is share of non-White and Black elected candidates in local legislative elections, respectively. Column (1) includes all candidates, and columns (2) to (4) break down candidates by ideology. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

Figure B.2: Salience of Race-Related Topics in Campaign Programs



Note: These figures present the average frequency with which candidates across racial groups, categorized by ideology, mention race-related issues in their electoral programs before and after the implementation of the Law of Quotas in 2013 (panels (a) and (b), respectively). Frequency is defined as the percentage of race-related terms relative to the total number of terms, excluding common stopwords. See Appendix E.3 for the list of race-related terms used. The analysis includes candidates running for president, state governors, and mayors. Observations: 46,248.

Figure B.3: Racial Composition of Main Parties, 2006-2022



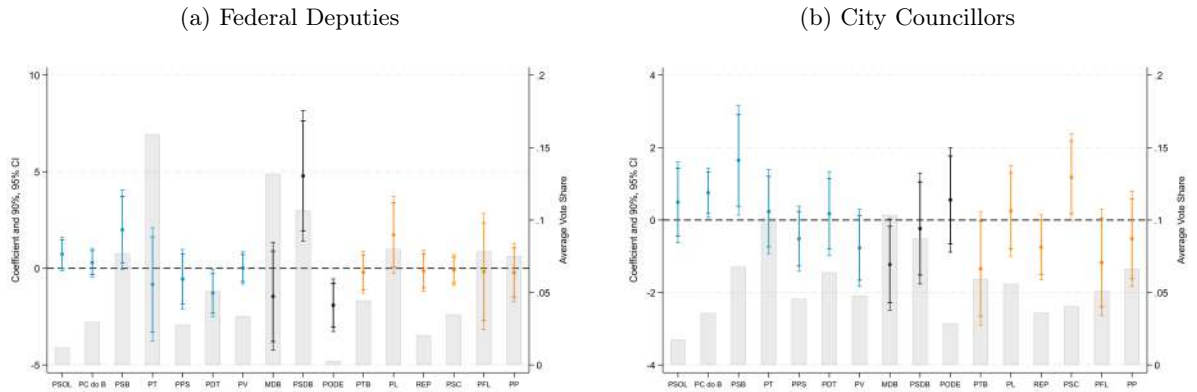
Note: This figure displays the share of Black candidates within the candidate pool of parties for elections conducted between 2006 and 2022. The bars are colour-coded to represent parties classified as Left (blue), Center (black), and Right (orange).

Table B.6: The Effect of Racial Quotas on the Supply of Candidates in Local Elections

	Total	By Ideology		
		Left	Center	Right
	(1)	(2)	(3)	(4)
<i>Panel A. Share of Non-White Candidates</i>				
Exposure to Non-White Quotas	1.085	0.992	-0.386	-0.426
	(0.887)	(0.750)	(0.584)	(0.613)
Avg DV at baseline	.611	.156	.144	.179
<i>Panel B. Share of Black Candidates</i>				
Exposure to Non-White Quotas	1.457**	0.914**	0.006	-0.227
	(0.675)	(0.463)	(0.319)	(0.372)
Avg DV at baseline	.279	.074	.062	.079
Municipalities	5,459	5,459	5,459	5,459
Observations	21,700	21,700	21,700	21,700

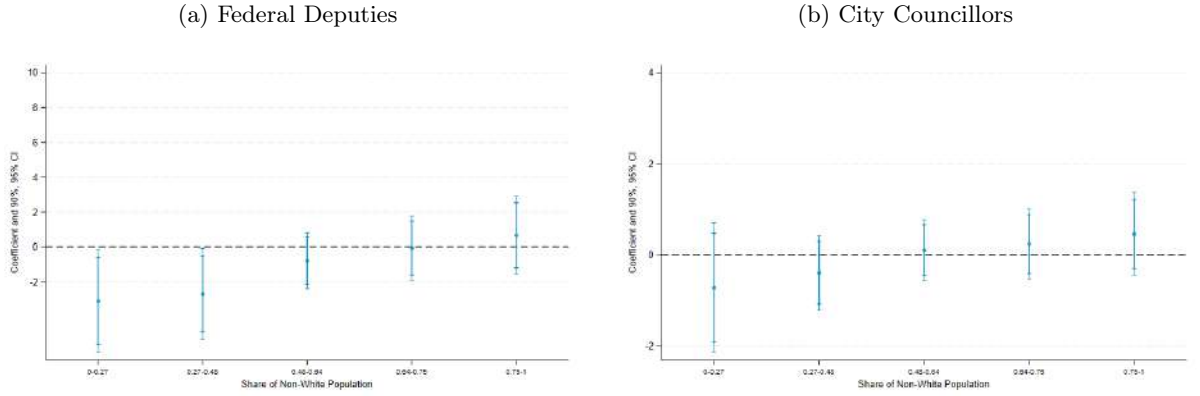
Note: This table reports the estimates of Equation (3). The dependent variable is the proportion of non-White candidates running for city councillors, relative to the total number of candidates. Column (1) includes all candidates, and columns (2) to (4) break down candidates by those running in left-, center-, and right-wing parties. I exclude candidates in unclassified parties. See also notes for Table 3. Significance levels: *** 1%, ** 5%, * 10%.

Figure B.4: The Effect of Racial Quotas on Municipality Support for Major Parties



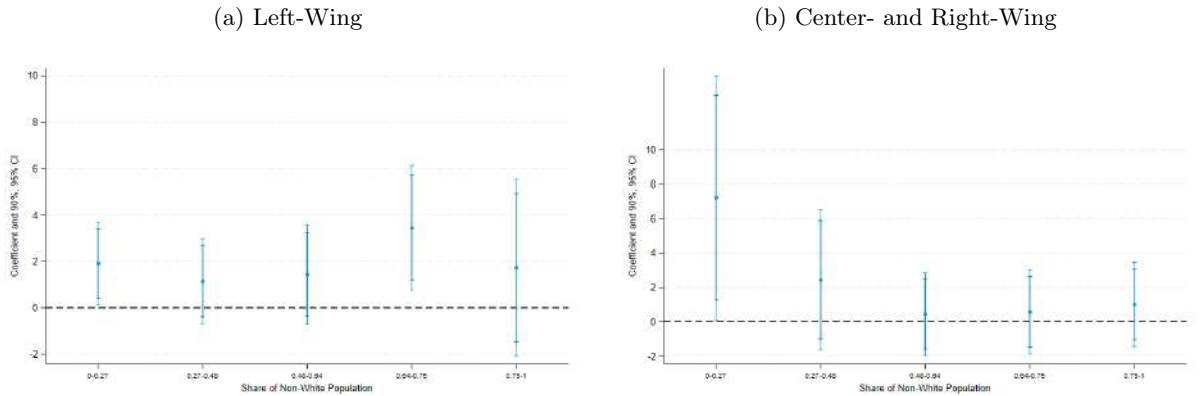
Note: These figures report the estimates of Equation (3). The dependent variable is the vote share for the main parties in Brazil. Parties are ordered left-to-right using Zucco and Power (2021)'s ideological score. The colour of the coefficients represents parties classified as Left (blue), Center (black), and Right (orange). The bars represent the average party vote share at baseline. See also notes for Table 3. Confidence intervals are provided at the 90% and 95% levels.

Figure B.5: The Effect of Racial Quotas on Polling-Station Support for the Workers' Party



Note: These figures report the estimates of Equation (4). The dependent variable is the share of votes for the Workers' Party in federal (panel (a)) and local (panel (b)) elections. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure B.6: The Effect of Racial Quotas on Polling-Station Support for Black Federal Candidates, conditional on Ideology



Note: These figures report the estimates of Equation (4). The dependent variable is the share of votes for Black candidates within each ideological group, expressed as a fraction of total votes for that group. Panel (a) reports the estimates for left-wing candidates, while panel (b) presents the estimates for center- and right-wing candidates. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

C Robustness of Results

C.1 Education

In this section, I discuss potential threats to the exogeneity of my exposure measure and present a series of tests to demonstrate the robustness of my main results.

C.1.1 Race-Specific Shares

Here I evaluate whether municipality enrolment patterns in federal universities differ across racial groups prior to the policy. I begin by comparing municipality-university enrolment patterns of White and non-White students in a regression setting. Table C.1 reports the correlation between race-specific enrolment rates, progressively introducing municipality and university fixed effects (columns (2) and (3), respectively). As shown, White and non-White students from the same municipality tend to enrol in similar universities.

To further substantiate that they followed similar enrolment patterns, I extend the dataset to include observations at the municipality-university-race level to assess whether they attend universities with different characteristics. The analysis is restricted to municipalities with at least one student enrolled in a federal university. For each municipality-university-race pair, I calculate the fraction of students of a given race enrolling in that university. I estimate the following regression:

$$share_{mu}^r = \beta_1 D_{mur}^{NW} + \beta_2 (D_{mur}^{NW} \times X_u) + \alpha_{mu} + \varepsilon_{mur} \quad (5)$$

where $share_{mu}^r = \frac{s_{mu}^r}{s_u^r}$ is the share of students of racial group r enrolled in university u in 2010 that are originally from municipality m . D_{mur}^{NW} is a dummy variable that equals 1 if the racial group of the student is non-White. Hence, β_1 captures the average difference in enrolment shares across both racial groups within a municipality. I also include the interaction between D_{mur}^{NW} and university characteristics, X_u , to evaluate whether non-Whites students attended federal institutions of different characteristics. These include the number of advertised admission seats, quality of the institution, a dummy that equals 1 if the university is the biggest in the state, a dummy that equals 1 if the university is among the top-five universities in Brazil in terms of enrolment, average marks in the ENEM exam, average program demographics (including share of public high-school students, share of non-Whites, and share of low-income students), and distance from the municipality to the university. I control for municipality-university fixed effects to absorb any variation common for students in both racial groups. Results are reported in Table C.2. In column (1), I only include the dummy for the non-White group, which indicates that, on average, the race-specific enrolment shares are equal across racial groups, despite the fact that non-Whites students had a lower university enrolment rate. In column (2) I include all university-specific characteristics in the regression. The results suggest that, while White students tend to attend federal universities more frequently, both racial groups enrolled in similar universities at baseline.

This reassures that the measure of exposure to racial quotas using total enrolment patterns, without separating them by race, is correctly specified to capture the gains and losses across racial groups. Moreover, in Table C.3, I show that the enrolment effects are similar to those found with my main specification.

Table C.1: Comparison of Enrolment Patterns at Baseline

DV: White Enrolment Patterns	(1)	(2)	(3)
Non-White Enrolment Patterns	1.017*** (0.000)	1.017*** (0.011)	1.016*** (0.010)
Municipality FE	-	✓	✓
University FE	-	-	✓
Observations	523,110	523,110	523,110

Note: The dependent variable is the share of White students from a municipality enrolled in each university. The independent variable is the share of non-White students from the same municipality enrolled in each university. Regressions are weighted by their municipality population between 18 to 24 years old at baseline. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

Table C.2: Race-specific Enrolment in Federal Universities

DV: Share of Students	(1)	(2)
Non-White	-0.001 (0.000)	-0.001 (0.011)
× Ln(# Admission Seats)		-0.000 (0.001)
× Quality (CPC Index)		0.001 (0.001)
× Biggest Fed. University in State		0.000 (0.001)
× Biggest 5 Fed. Universities in Brazil		0.000 (0.001)
× Avg ENEM Score		-0.002 (0.002)
× Share Public High-School Students		-0.002 (0.002)
× Share Non-white Students		-0.000 (0.002)
× Share Low-Income Students		-0.003 (0.006)
× Ln(Distance to University)		0.000 (0.001)
Constant	0.026*** (0.000)	0.023*** (0.000)
R ²	0.998	0.998
Observations	28,342	22,680

Note: The unit of analysis is a municipality-university-race triplet. The dependent variable is the share of students from racial group in a municipality that enrolls in a given university in 2010. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

Table C.3: The Effect of Racial Quotas on Enrolment in Federal Universities, Race-specific Exposure

DV: Enrolment Rate	Non-White (1)	White (2)
Own-Race Exposure to Non-White Quotas	1.498** (0.6353)	-0.244 (0.5697)
Avg DV at baseline	.013	.020
Share of Population	.544	.456
Municipalities	5,460	5,460
Observations	49,140	49,140

Note: The dependent variable is enrolment rate in federal universities, defined as the number of students of each racial group in a municipality that enrolls in any federal university, relative to the own-race population at risk. All regressions are weighted by race-specific population between 18 to 24 years old in 2010. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

C.1.2 Rotemberg Weights

Table C.4: Rotemberg Weights

Panel A. Summary Statistics

	Sum	Mean	SD	Min	Max
α_u	1	0.040	0.042	0.002	0.195

Panel B. Top Five Rotemberg Weight Universities

University	α_u	g_u	β_u	Share
Federal University of Bahia	0.195	10,025	3.538	0.041
Federal University of Pará	0.092	5,927	3.526	0.038
Federal University of Sergipe	0.092	7,935	3.466	0.039
Federal University of Maranhao	0.078	9,860	3.548	0.036
Federal University of Alagoas	0.064	5,968	3.544	0.040

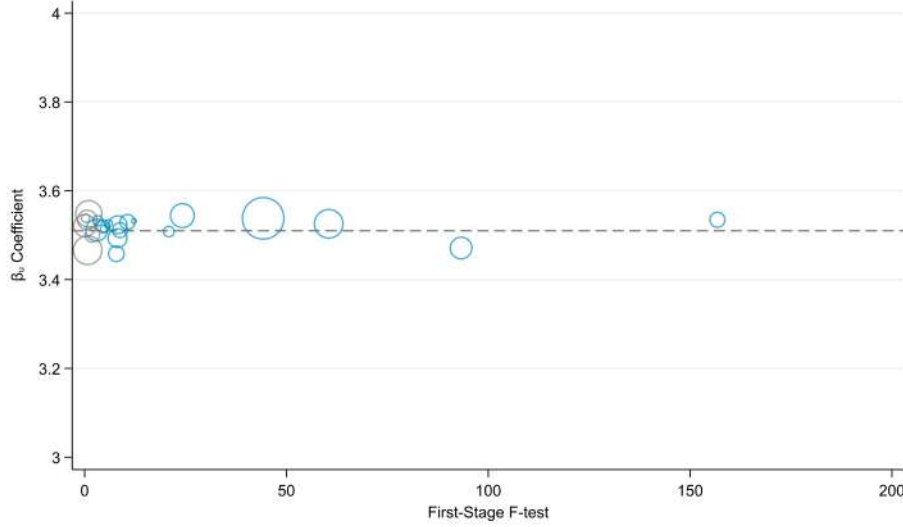
Note: Panel A reports summary statistics of non-zero Rotemberg weights (26% of federal universities). Panel B reports the top 5 Rotemberg weight industries. α_u is the Rotemberg weight, g_u is the total number of vacancies reserved between 2013 and 2015 for each university, β_u is the coefficient of the TSLS regression using each university share independently, and *Share* is the average university share, which I multiply by 1,000,000 for legibility.

Table C.5: Baseline Correlation Between University Shares and Municipality Characteristics

University Shares	UFBA	UFS	UFPA	UFMA	UFAL	Exposure
	(1)	(2)	(3)	(4)	(5)	(6)
Ln(Population)	-0.002 (0.010)	-0.093 (0.070)	-0.010 (0.011)	-0.006 (0.008)	-0.033* (0.018)	-0.001*** (0.000)
Ln(Population Density)	0.009 (0.008)	0.047 (0.031)	0.004 (0.008)	-0.002 (0.007)	0.016** (0.007)	-0.000** (0.000)
Share of Population 18-24	0.162 (0.914)	0.345 (0.699)	0.364 (1.022)	8.725 (6.880)	-2.191* (1.217)	0.046*** (0.012)
Share of Women	3.124 (2.276)	1.680* (0.902)	-2.041*** (0.602)	2.406 (1.982)	3.051 (2.331)	0.076*** (0.009)
Share of non-Whites	0.446 (0.326)	0.251 (0.221)	0.078 (0.130)	-0.154 (0.124)	-0.040 (0.041)	0.005*** (0.001)
Literacy Rate	1.222 (0.919)	0.106 (0.319)	1.305** (0.566)	1.338 (1.138)	-1.038*** (0.284)	0.018*** (0.003)
Share of Population with Tertiary Education	2.640 (2.160)	-0.328 (0.678)	-1.544* (0.880)	1.494 (1.676)	-2.599* (1.475)	0.038*** (0.007)
Share of Public HS students	0.468 (0.419)	-1.437 (1.099)	-0.252 (0.207)	0.115 (0.254)	-1.217 (0.943)	-0.010*** (0.003)
High-School Enrolment	-0.177 (0.131)	-0.297** (0.136)	0.127 (0.116)	0.179 (0.166)	-0.170 (0.177)	-0.002** (0.001)
Average ENEM Grade (std.)	-0.236 (0.179)	-0.382 (0.269)	0.139** (0.060)	-0.109 (0.077)	-0.218 (0.170)	-0.000 (0.001)
Ln(Median HH Income)	-0.565 (0.439)	0.193 (0.171)	0.105 (0.071)	-0.551 (0.483)	0.214** (0.084)	-0.006*** (0.001)
Share of Middle Class HHs	1.004 (0.797)	-0.147 (0.216)	-0.923*** (0.309)	0.400 (0.442)	-0.113 (0.100)	0.006*** (0.002)
University in the Municipality	0.003 (0.011)	0.043 (0.029)	0.041** (0.018)	-0.010 (0.025)	0.048** (0.023)	0.003*** (0.000)
Ln(Distance to Big City)	-0.010 (0.009)	-0.015 (0.014)	-0.008 (0.018)	-0.013 (0.013)	-0.031* (0.019)	-0.000*** (0.000)
Avg DV at baseline	0.041	0.039	0.038	0.036	0.040	0.003
R ²	0.198	0.078	0.099	0.110	0.093	0.676
Observations	5,461	5,461	5,461	5,461	5,461	5,461

Note: Each column reports the coefficients of a separate regression of baseline municipality characteristics on university shares for the top-5 Rotemberg-weights universities. These universities are: Federal University of Bahia (UFBA), Federal University of Sergipe (UFS), Federal University of Pará (UFPA), Federal University of Maranhão (UFMA), and the Federal University of Alagoas (UFAL). For legibility, the university shares are multiplied by 1,000,000. The dependent variable in the final column is the average exposure measure over the period 2010-2018. Observations are weighted by municipality population in 2010.

Figure C.1: Heterogeneity of β_u



Note: This figure reports the relationship between each instrument β_u , the first-stage F-test and the Rotemberg weights. β_u is the coefficient from a regression of each individual municipality-university share on local enrolment. The size of the points are scaled by the Rotemberg weights, and their colour correspond to whether they are statistically significant (blue if yes, gray if no). The figure excludes instruments with F-test higher than 200 for illustration purposes.

C.1.3 Confounding Educational Policies

I evaluate whether other contemporaneous educational policies implemented in federal universities might confound my main results. I identify three of such policies: the introduction of race-neutral quotas under the Law of Quotas, the expansion of admission seats, and the introduction of the centralized admissions system.

First, the Law of Quotas mandated not only the reservation of quotas for non-White individuals but also included race-neutral quotas targeting public high school and low-income students, regardless of their race.¹⁸ Second, the effects of the policy on enrolment documented in Table 1 may be explained by the expansion of the federal system, also affecting enrolment through pre-reform enrolment patterns. Indeed, the federal tertiary education system expanded during the period of study.¹⁹ If the expansion of admission seats is correlated with the introduction of racial quotas, this could mitigate the potential displacement of White students and inflate the enrolment gains of non-White students from racial quotas. The last policy I consider is the implementation of the centralized admission system in public institutions of tertiary education, the *Sistema de Seleção Unificada (SISU)* from 2010. The main

¹⁸Non-White students could also apply for these quotas, but their racial group was not considered part of their admission process.

¹⁹In 2013, eight new institutions began their operation, and the universities that were already in activity expanded the number of admission seats between 2012 and 2014. Provided that I leverage historical enrolment patterns, my exposure measure does not capture the implementation of racial quotas in new universities.

purpose of the system was to reduce the costs of the admission process and, as the Law of Quotas, it was introduced gradually and by 2016 more than 90% of students enrolled through the system. It has been documented that this new system induced sophisticated students from high socioeconomic backgrounds to strategize over their tertiary educational choices, at the detriment of students from low socioeconomic backgrounds (Mello, 2022). However, its interaction with the expansion of racial quotas was complementary for low-SES. An obvious concern is whether the expansion of SISU might be explaining the change in enrolment patterns that I accrue to the expansion of racial quotas.

In Table C.6 I evaluate the robustness of the results once I control for these policies using an exposure-type expression:

$$q_{mt} = \sum_u \frac{s_{mu}}{s_u} \cdot \frac{educpol_{ut}}{pop_m^{18-24}} \quad (6)$$

where $educpol_{ut}$ refers to the number of admission seats reserved to students other than non-White, the total number of admission seats offered in each academic year in universities, or the number of admission seats that the university advertised through the centralized admissions' system. I report estimates from Equation 2. The analysis shows that the effect of the reform on non-White enrolment remains robust even after accounting for these additional policies. Furthermore, the positive correlation between local exposure to the expansion of vacancies and White enrolment suggests that this policy may have mitigated potential displacement effects for White students. After accounting for this policy, the effect on White enrolment becomes statistically insignificant, indicating that the gains in non-White enrolment driven by the policy did not necessarily reduce enrolment opportunities for Whites within the same municipalities.

Table C.6: The Effect of Racial Quotas on Enrolment in Federal Universities, controlling for
Confounding Educational Policies

DV: Enrolment Rate	Non-White	White
	(1)	(2)
Exposure to Non-White Quotas	0.798** (0.375)	-0.384 (0.429)
Exposure to Race-Neutral Quotas	-0.511** (0.240)	-0.185 (0.414)
Exposure to Seat Expansion	0.163 (0.173)	0.480** (0.230)
Exposure to SISU Expansion	-0.054 (0.193)	-0.195 (0.281)
Avg DV at baseline	0.007	0.009
Municipalities	5,460	5,460
Observations	49,140	49,140

Note: This table reports the estimates of Equation (2), where I control for confounding education policies. The dependent variable is university enrolment rate by type of student, defined as the number of students who enrol in any a federal university relative to the university-age population size in the municipality. See also notes for Table 1. Significance levels: *** 1%, ** 5%, * 10%.

C.1.4 Disclosure of Race in the CES

Here, I explore whether the observed increase in enrolment of non-White students might be driven by changes in the disclosure of race. As discussed in the data section, prior to the implementation of the Law of Quotas, around 60% of students failed to report their racial group. This significantly dropped to 20% over the course of the study period. This could be problematic if this shift in reporting behaviour was correlated with local exposure to racial quotas and confounded the real effect of the policy. While I plan to address this issue by matching the restricted-access microdata from the CES and the ENEM in future work, this analysis has not yet been conducted. In this draft, I characterize the missing data and provide evidence that changes in reporting behaviour do not account for my main results.

I begin by documenting the likelihood of disclosing the racial identity across racial groups. To that end, I combine enrolment data from the CES and the demographic Census of 2010. I restrict the analysis to municipalities with at least one student enrol in a federal university. In Table C.7, I regress the share of students enrol according to the Census 2010 against the share of students enrol

in 2010 by racial group (columns (1) and (2)). In column (3), I examine the correlation between the share of students without information on their race and the share of non-White students that enrol in federal universities according to the demographic census. The results indicate that the propensity to disclose race was not systematically related to students' racial identities. Moreover, in Table C.8 I find that most of the variation in race underreporting is attributable to universities. This suggests that universities may have increasingly required students to disclose their racial group at the time of enrolment after the introduction of the Law of Quotas.

Finally, in Table C.9 I explore whether the positive effect on non-White enrolment could be attributed to changes in racial reporting behaviour in exposed municipalities. To assess this, I examine whether my exposure measure predicts changes in the enrolment of students whose racial information was not reported across different types of universities. The results show that all coefficients are statistically insignificant and, if anything, tend to be positive. This indicates that the increase in non-White enrolment in federal universities is unlikely to be driven by selection bias related to racial reporting, but rather reflects actual shifts in enrolment patterns.

Table C.7: Propensity to Underreport Race across Municipalities

DV: Share of Students (CES)	Non-White (1)	White (2)	Not Reported (3)
Share non-White Students (Census 2010)	0.602*** (0.071)		0.141 (0.121)
Share White Students (Census 2010)		0.539*** (0.051)	
Constant	-0.041* (0.022)	-0.067*** (0.023)	0.480*** (0.044)
Avg DV at baseline	.22	.24	.54
R ²	0.309	0.280	0.012
Observations	3,482	3,482	3,482

Note: This table displays the coefficient from a regression of the baseline share of NW students enrol in a federal university from each municipality on the share of students that do not report race when enrolling.

Table C.8: Source of Variation in Race Reporting

	University x Year FE (1)	Municipality x Year FE (2)	Both (3)
Adjusted R ²	0.643	0.109	0.598
Observations	156,974	146,255	146,255

Note: This table reports the adjusted R² from a regression analyzing the relationship between the specified fixed effects and the share of students with no racial group reported. The unit of analysis is municipality-university-year.

Table C.9: Selection in Race Reporting

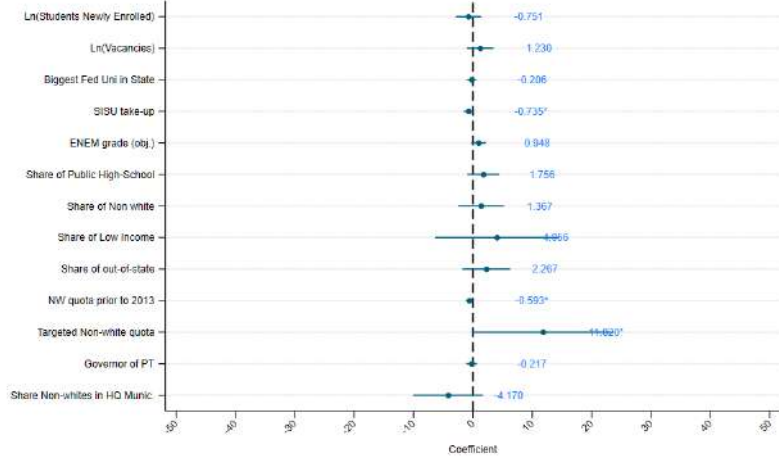
DV: Enrolment Rate of Students with No Race Reported	Federal (1)	State (2)	Private (3)
Exposure to Non-White Quotas	0.403 (0.300)	0.140 (0.167)	0.459 (0.562)
Avg DV at baseline	.0232	.0135	.071
Municipalities	5,460	5,460	5,460
Observations	49,140	49,140	49,140

Note: This table reports the estimates of Equation (2). The dependent variable is the university enrolment rate of students with no reported racial group, defined as the number of students of who enrol in federal, state, or private universities and that do not report race relative to the university-age population size in the municipality. See also notes for Table 1. Significance levels: *** 1%, ** 5%, * 10%.

C.1.5 Strategic Implementation of Non-White Quotas

Another concern relates to the strategic implementation of racial quotas across federal universities. Since the policy allowed for a gradual introduction of reserved seats, universities with more favorable racial attitudes might have chosen to implement the quotas more rapidly. Although my analysis relies on the exogeneity of pre-reform enrolment patterns, proving the exogeneity of the shift further strengthens the validity of my exposure measure. I show that federal universities did not strategically time their implementation of quotas. Figure C.2 reveals that universities which voluntarily adopted affirmative action prior to the Law of Quotas reached the mandated level of racial quotas more quickly. However, the larger the targeted quota size, the slower the implementation process. Additionally, universities with a higher use of the centralized admission system (SISU) implemented quotas earlier. Importantly, I find no evidence that state politics, including having a governor from the Workers' Party (PT), the party responsible for the reform, influenced the speed of implementation. Therefore, I conclude that factors such as university size, baseline student composition, or political motivations did not affect the expansion of racial quotas across federal institutions.

Figure C.2: Determinants of Implementation of Racial Quotas



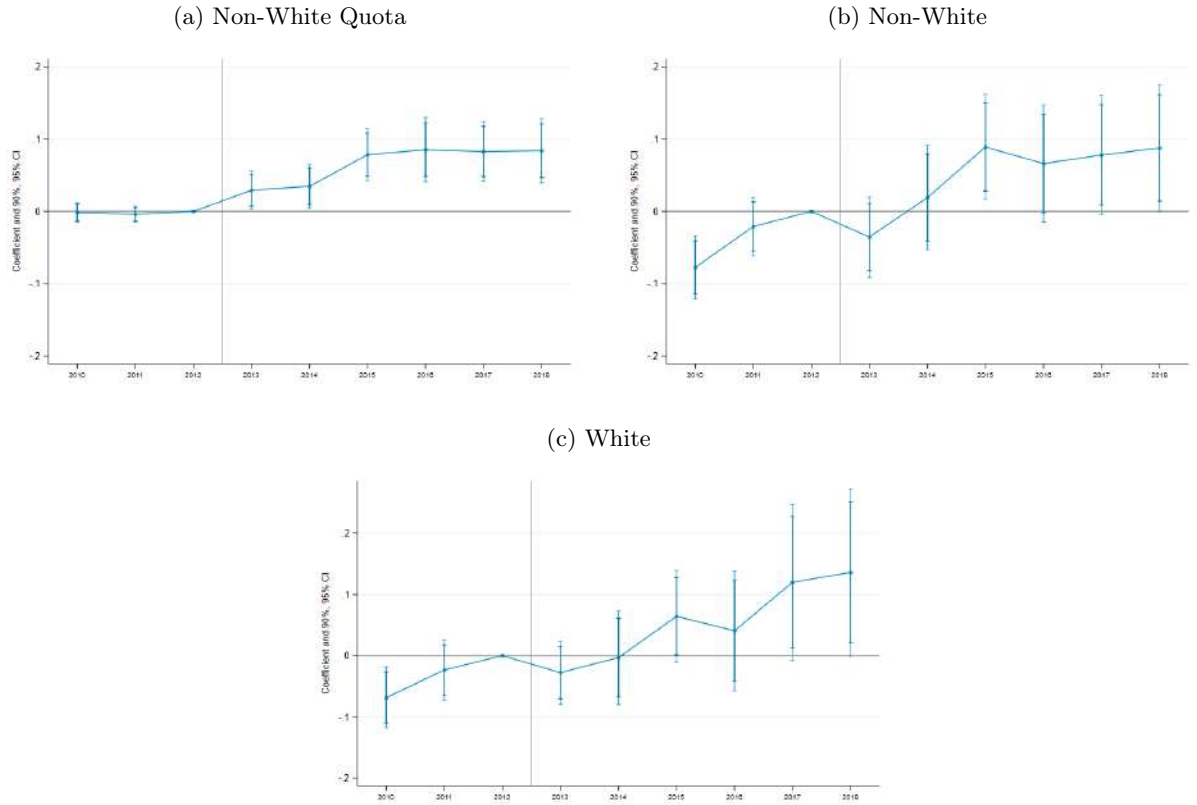
Note: This figure displays the coefficients of the regression of years to reach the targeted level of racial quotas on university characteristics in 2010. The regression is at the university level. Confidence intervals at the 95% level are displayed. Observations are weighted by university size. Significance levels: *** 1%, ** 5%, * 10%.

C.1.6 Pre-trends

The main identification assumption in my analysis is that pre-policy enrolment patterns are orthogonal to innovations in the error term. In line with my identification strategy, I assume that the exogeneity of my shift-share design depends on the exogeneity of the baseline shares. Specifically, this requires that the initial shares of municipality-university enrolment do not predict changes in enrolment prior to the policy's implementation, conditional on municipality-level controls. In this paper, I focus on total municipality enrolment across federal universities at baseline, defined as $\sum_u \frac{s_{mu}}{s_u} \frac{1}{pop_m^{18-24}}$. To test this assumption, I regress these baseline shares on the main outcomes of interest for the period 2010 to 2018. I use the same specification as in Equation 2, substituting the municipality exposure measure with total pre-policy enrolment patterns.

Figure C.3 presents the results of the pre-trend analysis. In panel (a), I test for pre-trends in enrolment through non-White quotas; in panel (b), the dependent variable is non-White student enrolment; and in panel (c), I show enrolment of White students. The figures indicate that pre-policy enrolment patterns only predict enrolment through non-White quotas after the policy is implemented. For non-White and White enrolment, the 2010 coefficients suggest that baseline shares predict enrolment levels in that year, which is expected since I am using contemporaneous shares. While the increase in non-White enrolment aligns with the goals of the racial quotas policy, the upward trend in White enrolment suggests that these shares may also be capturing the effects of other concurrent educational policies, as discussed in Appendix C.1.3.

Figure C.3: Pre-trends - Enrolment in Federal Universities



Note: These figures report the estimates of Equation 2, where I substitute exposure to non-White quotas with the baseline shares interacted with a year fixed effect. The dependent variables are: (a) enrolment rate through non-White quotas, (b) enrolment rate of non-White students, and (c) enrolment rate of White students. Each annual coefficient is displayed. See also notes for Table 1. Confidence intervals at the 90% and 95% significance levels are reported.

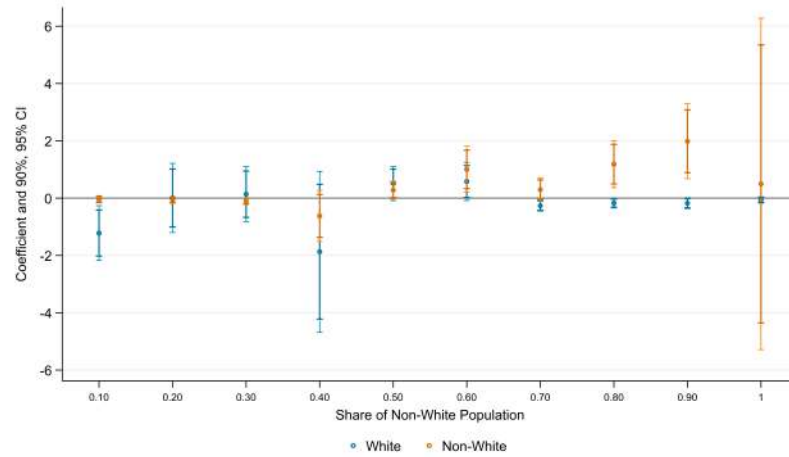
C.1.7 Disaggregation of the Exposure Measure

Table C.10: Likelihood of Enrolment in Federal Universities

DV: Enrolment Rate	Non-White Quotas (1)	Non-White (2)	White (3)
Own-Race Exposure to Non-White Quotas	0.667*** (0.236)	0.399 (0.314)	-0.215 (0.281)
Avg DV at baseline	0.002	0.013	0.020
Share of Population	-	.544	.456
Municipalities	5,460	5,460	5,460
Observations	49,140	49,140	49,140

Note: This table reports the estimates of Equation (2). The dependent variable is race-specific enrolment rate in federal universities, defined as the number of students of racial group r that enrolls in any federal university relative to the population size of r in the municipality. See also notes for Table 1. Significance levels: *** 1%, ** 5%, * 10%.

Figure C.4: Likelihood of Enrolment in Federal Universities, by Racial Composition of Municipality



Note: This figure exhibits the effect of own-race exposure to non-White quotas across municipalities with different shares of non-White population. Blue coefficients represent the effect of non-White quotas on White enrolment, while orange coefficients represent the effect of the policy on the enrolment of non-Whites. See also notes for Table 1. Confidence intervals at the 90% and 95% confidence levels are reported.

C.2 Politics

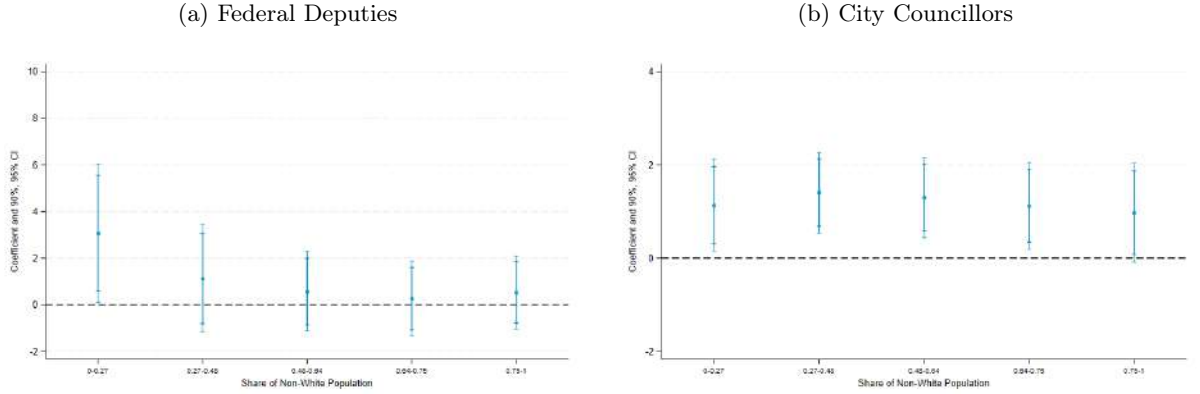
C.2.1 Political Effects using Continuous Measure of Race

Table C.11: The Effect of Racial Quotas on Municipality Support for Candidates Using the Continuous Measure of Race

	Total	By Ideology		
		Left	Center	Right
	(1)	(2)	(3)	(4)
<i>Panel A. Federal Deputies</i>				
Exposure to Non-White Quotas	1.246	1.052	0.024	0.932
	(1.366)	(0.751)	(0.664)	(1.313)
Avg DV at baseline	0.244	0.094	0.054	0.080
<i>Panel B. City Councillors</i>				
Exposure to Non-White Quotas	2.036***	1.792***	0.134	-0.319
	(0.771)	(0.592)	(0.467)	(0.522)
Avg DV at baseline	0.383	0.104	0.095	0.124
Municipalities	5,460	5,460	5,460	5,460
Observations	21,840	21,840	21,840	21,840

Note: This table reports the estimates of Equation (3). In Panel A, the dependent variable is the weighted average of vote shares for candidates in federal legislative elections, where the weights are the predicted probability of the candidate being Black, as estimated by Vertex AI. Panel B focuses on candidates for city council. Column (1) includes all candidates, and columns (2) to (4) break down candidates by those running in left-, center-, and right-wing parties. I exclude candidates in unclassified parties. All regressions include municipality and state-year fixed effects, as well as the baseline municipality characteristics interacted with a year fixed-effect. Regressions are weighted by municipality population at baseline. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

Figure C.5: The Effect of Racial Quotas on Polling-Station Support for Candidates Using the Continuous Measure of Race



Note: These figures report the estimates of Equation (4). In panel A, the dependent variable is the weighted average of vote shares for candidates in federal legislative elections, where the weights are the predicted probability of the candidate being Black, as estimated by Vertex AI. Panel B focuses on candidates for city council. See also notes for Figure 6. Confidence intervals are provided at the 90% and 95% levels.

C.2.2 Controlling for Time-Varying Unobserved Municipality Factors

In this section, I test the robustness of the polling station results by incorporating municipality-year fixed effects. Their inclusion addresses concerns that exposure to non-White quotas may correlate with time-varying unobservable factors at the municipality level. These factors could include endogenous changes in the supply of candidates driven by the policy, as discussed in Section 6, or shifts in the political discourse of candidates within municipalities over time that could influence elections. To account for these potential confounders, I estimate the following Equation:

$$v_{lmt} = \sum_{\tau=1}^5 \theta_{\tau} \cdot q_{mt} \cdot shNW_{l,\tau} + S'_{lmt} \cdot \delta_t + \alpha_l + \alpha_{mt} + \varepsilon_{lmt} \quad (7)$$

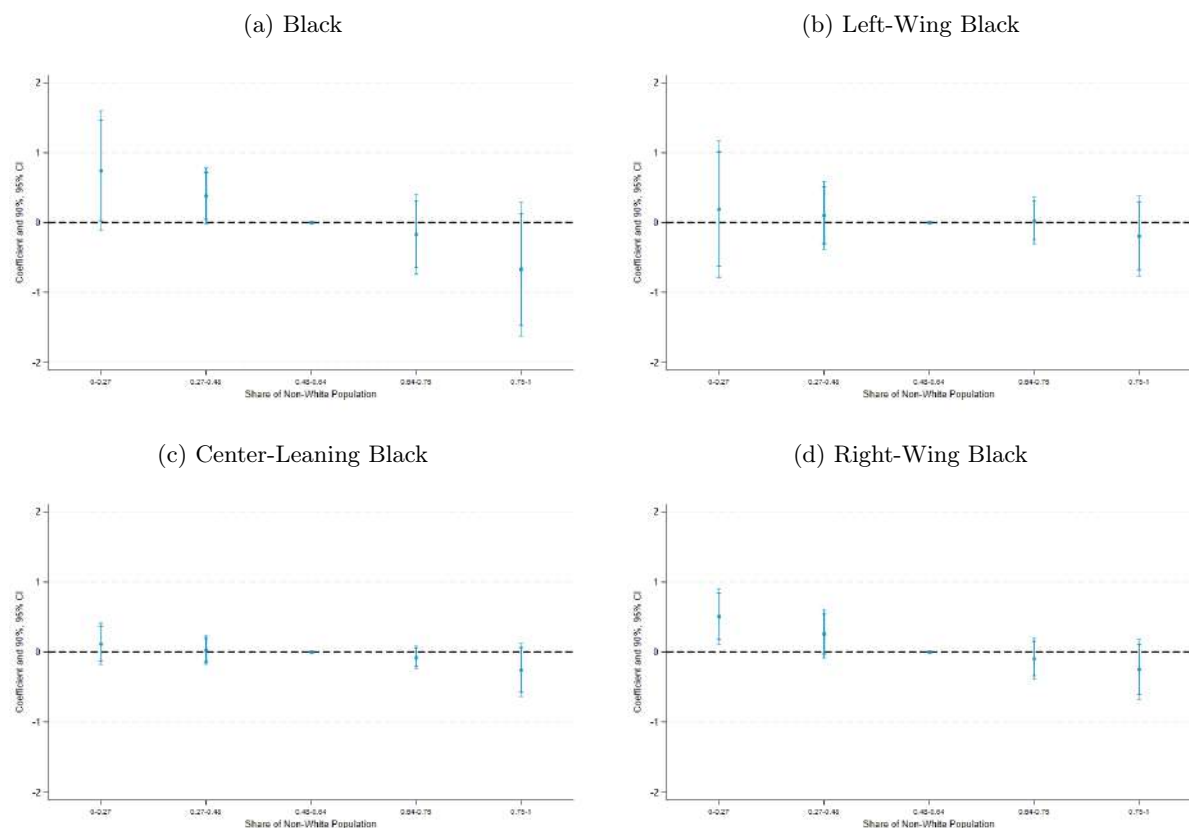
where the notation follows from Equation 4, and α_{mt} represents municipality-year fixed effects. The inclusion of these fixed effects absorbs municipality-level variation, making not possible the identification of all θ_{τ} 's due to perfect collinearity with the municipality-year effects. Thus, I set as base category the third quintile, which are polling stations composed by similar shares of White and non-White voters. The interpretation of θ_{τ} will reflect the effect of the policy across poling stations relative to the effect in racially balanced units.

Figure C.6 present the results for federal elections, focusing on the impact of the policy on support for Black candidates, both overall and by ideology. Consistent with the main results without municipality-year fixed effects, there is an increase in support for right-wing Black candidates among White voters in comparison to racially balanced municipalities. For non-White candidates from left-wing parties,

there is no clear pattern in the results, and I cannot rule out that the overall effect may be close to zero, or that there is no significant difference compared to racially balanced polling stations. Figure C.7 confirms a clear ideological divide along racial lines: non-White voters increased their support for left-wing parties, even if they did not necessarily show a significant shift toward Black candidates. This suggests that non-White voters gravitated toward progressive parties, which had increasingly advocated for their rights post-reform. Meanwhile, White voters shifted their support toward center- and right-wing parties, reinforcing the backlash documented in Section 6.2.

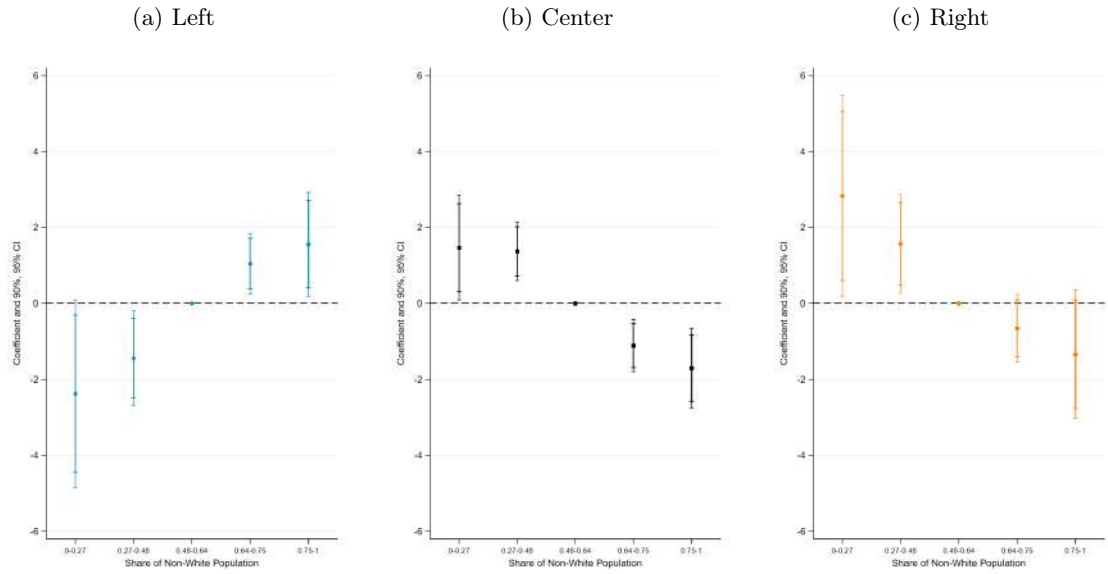
Figures C.8 and C.9 present results for council elections, which are consistent with the main findings of the paper. In particular, polling stations with a higher share of non-White voters showed increased support for Black candidates within left-wing parties (panel (b)) with respect to racially-balanced polling stations. However, there is no clear evidence of a shift in the voting behavior of the White electorate in terms of support for Black candidates in center- and right-wing parties. Despite this, the results clearly indicate ideological polarization along racial lines, mirroring the patterns observed in the federal elections.

Figure C.6: The Effect of Racial Quotas on Polling-Station Support for Black Federal Candidates,
Controlling for Time-Varying Municipality Factors



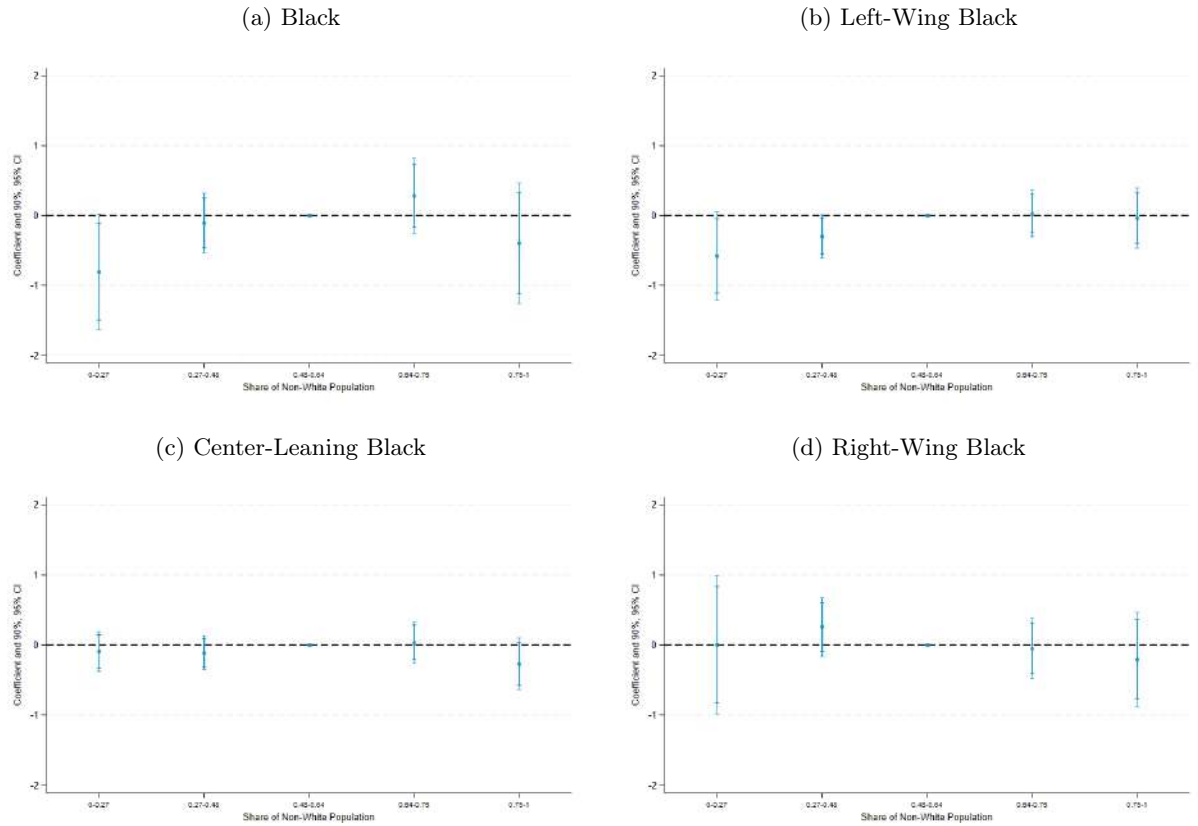
Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in federal legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.7: The Effect of Racial Quotas on Polling-Station Support for Federal Candidates by Ideology, Controlling for Time-Varying Municipality Factors



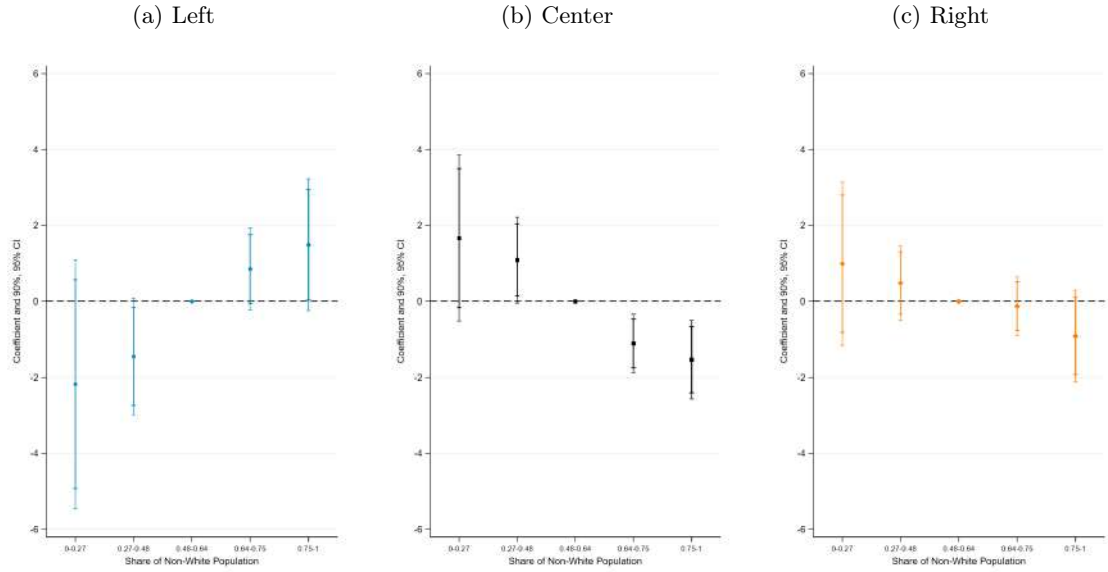
Note: This figure reports the estimates of Equation (4). The dependent variable is the vote share for candidates running for federal deputies in left-, center-, and right-wing parties, as classified by Zucco and Power (2021). Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.8: The Effect of Racial Quotas on Polling-Station Support for Black Local Candidates, Controlling for Time-Varying Municipality Factors



Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in local legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.9: The Effect of Racial Quotas on Polling-Station Support for Local Candidates by Ideology, Controlling for Time-Varying Municipality Factors

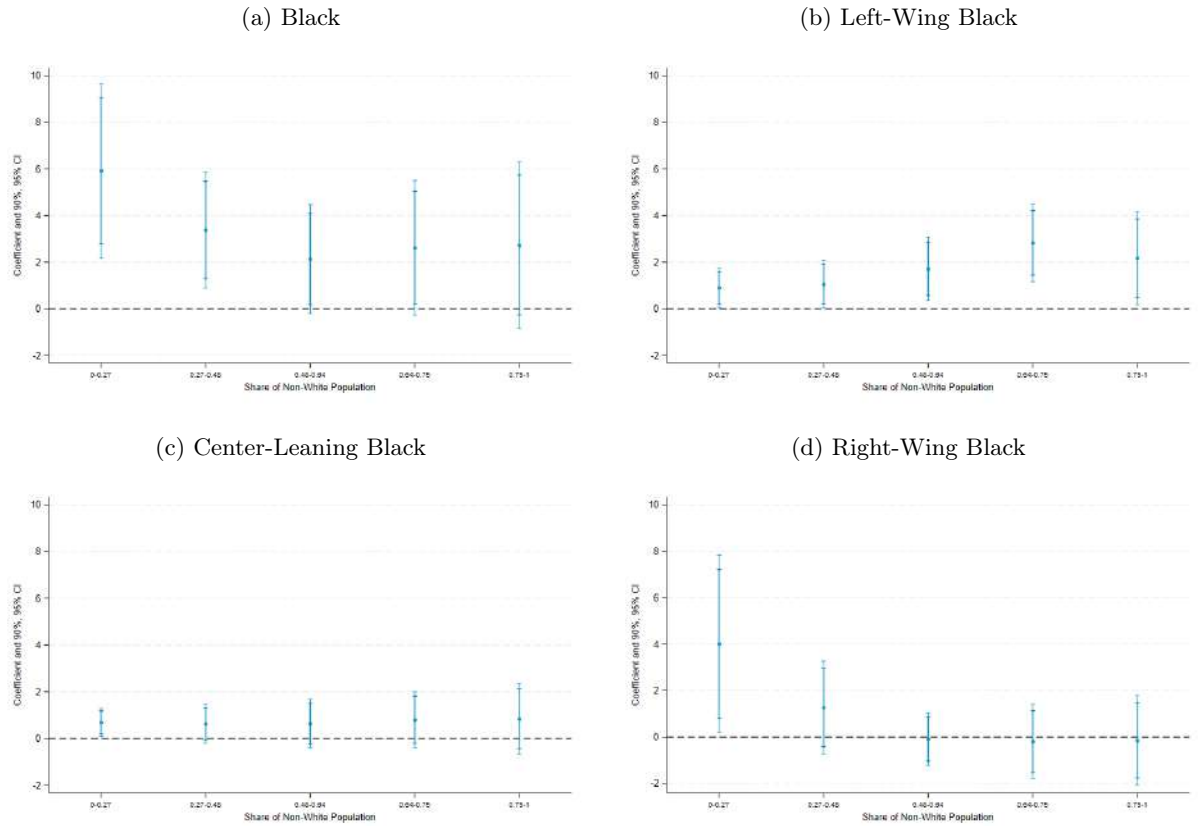


Note: This figure reports the estimates of Equation (4). The dependent variable is the vote share for candidates running for city councilor in left-, center-, and right-wing parties, as classified by Zucco and Power (2021). Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

C.2.3 Controlling for Other Demographics

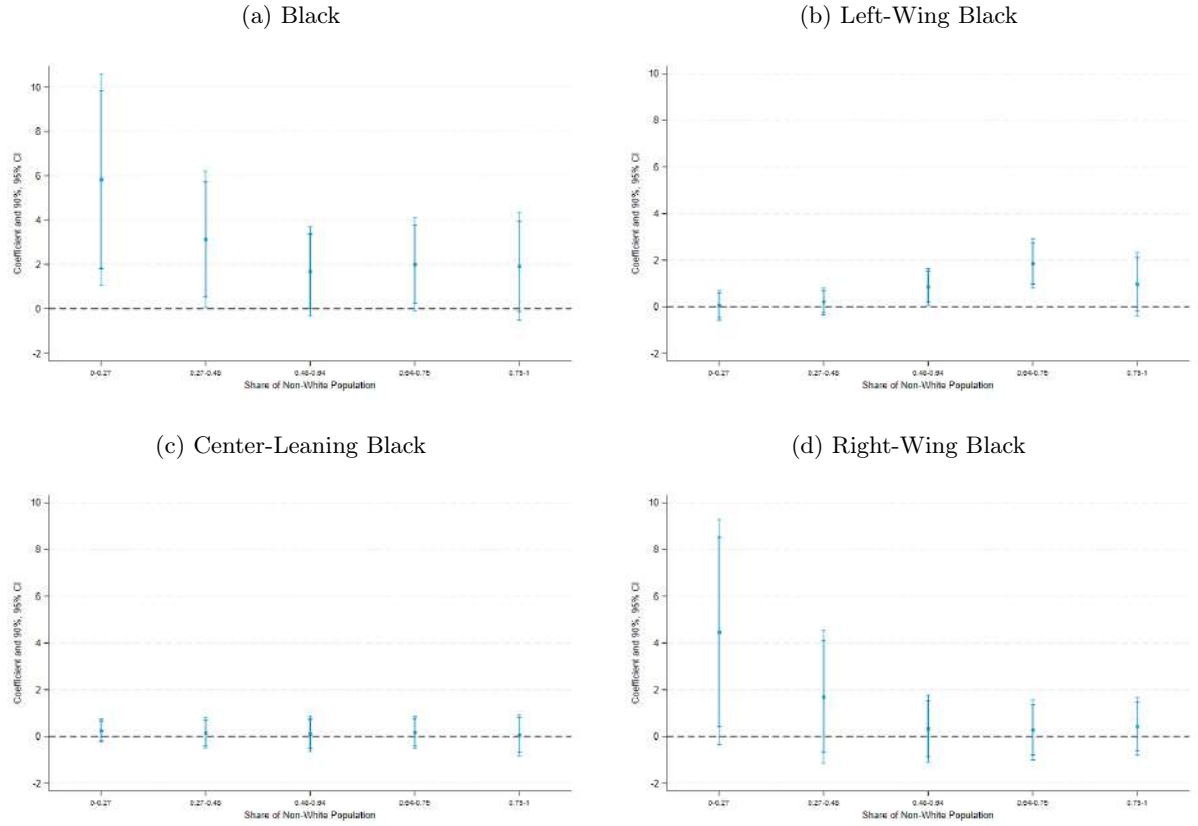
Here I test whether the heterogeneity effect of the measure across polling stations with different racial compositions might be picking up other characteristics of the electorate, such as differences in income or educational attainment across racial groups that could also affect their voting behavior. In the figures below, I provide estimates of Equation 4 for support for Black candidates across ideologies, and include as a control the interaction of exposure to non-White quotas with the share of low-income households in the polling station (Figures C.10 and C.12) and the share of the electorate with tertiary education (Figures C.11 and C.13), separately. The coefficients confirm that the inclusion of these other interactions do not significantly affect my main results.

Figure C.10: The Effect of Racial Quotas on Polling-Station Support for Black Federal Candidates,
Controlling for Exposure-Income Interaction



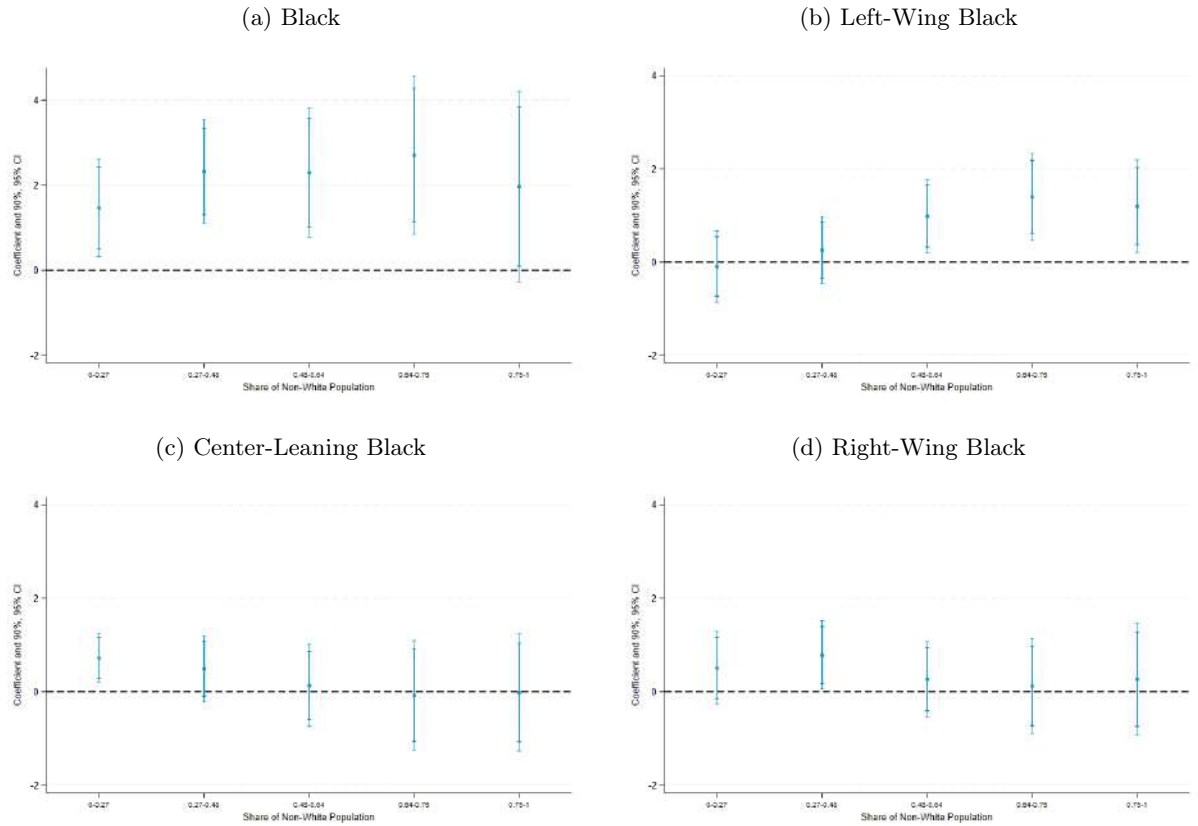
Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in federal legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. I include the interaction of the exposure measure with the share of the low-income households assigned to the polling station. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.11: The Effect of Racial Quotas on Polling-Station Support for Black Federal Candidates,
Controlling for Exposure-Education Interaction



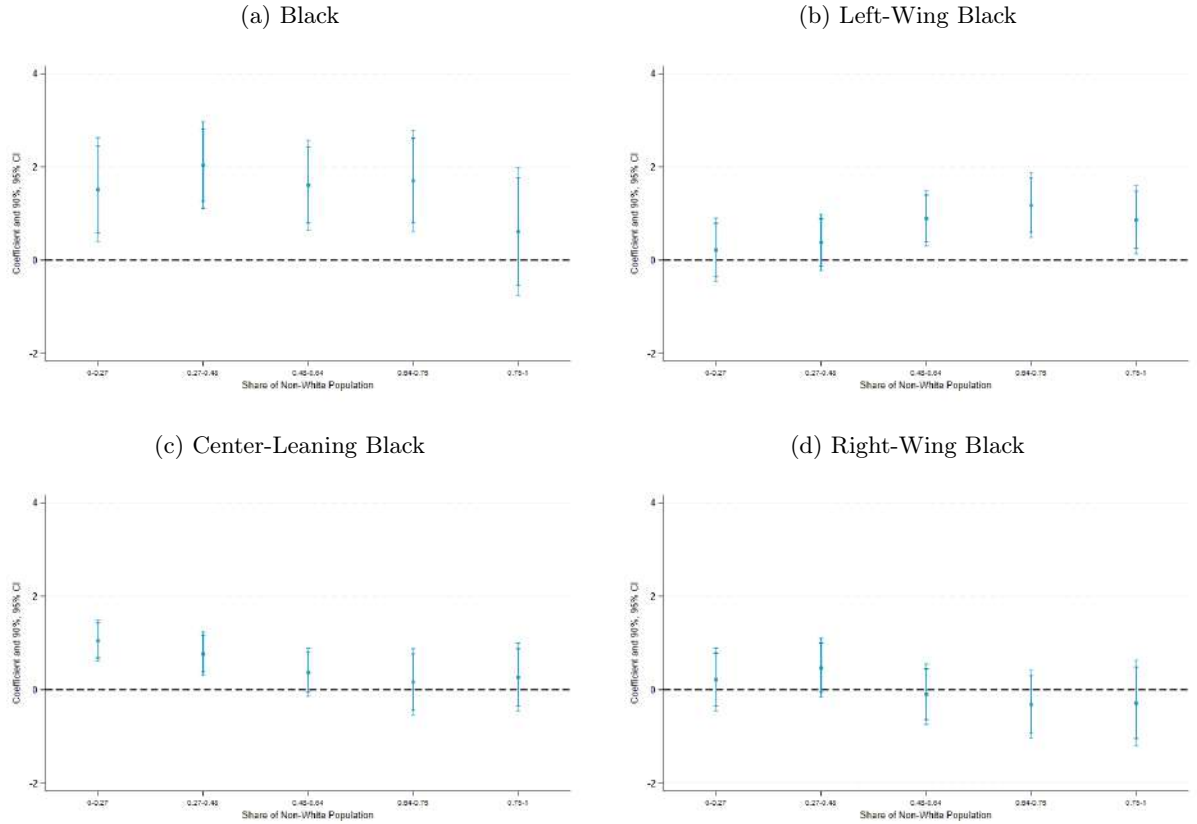
Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in federal legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. I include the interaction of the exposure measure with the share of the electorate with graduate education. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.12: The Effect of Racial Quotas on Polling-Station Support for Black Local Candidates,
Controlling for Exposure-Income Interaction



Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in local legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. I include the interaction of the exposure measure with the share of the low-income households assigned to the polling station. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

Figure C.13: The Effect of Racial Quotas on Polling-Station Support for Black Local Candidates, Controlling for Exposure-Education Interaction



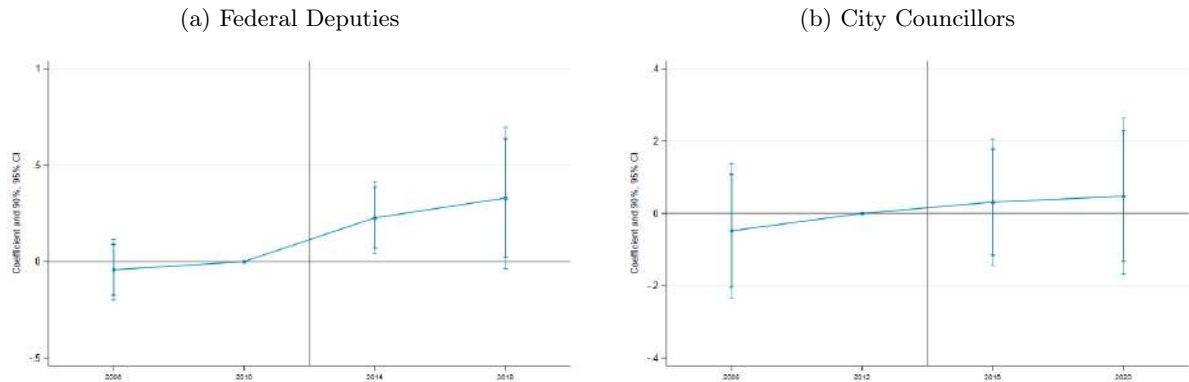
Note: These figures present the estimates from Equation (4). The dependent variable is the vote share for Black candidates in local legislative elections. Each coefficient displayed reflects the effect of non-White exposure to racial quotas across quintiles of non-White population share (shown on the horizontal axis), with the third quintile as base category. Panel (a) includes all Black candidates, while panels (b) to (d) focus on the effect on support for Black candidates across ideologies. I include the interaction of the exposure measure with the share of the electorate with graduate education. See also notes for Figure 5. Confidence intervals are provided at the 90% and 95% levels.

C.2.4 Pre-trends

Here I present the results of the pre-trend analysis for municipality and polling station results, following a similar approach as in the education pre-trend analysis. I focus on the effect on support for Black candidates in both federal and local elections. Figure C.14 presents the municipality-level results, with panel (a) showing federal deputy elections and panel (b) showing city council elections. In both cases, there is no evidence that baseline local enrolment shares in federal universities influenced support for Black candidates prior to the reform. After the reform, however, there is a noticeable upward trend in support for Black candidates in federal elections, while the coefficients for city council elections remain statistically insignificant.

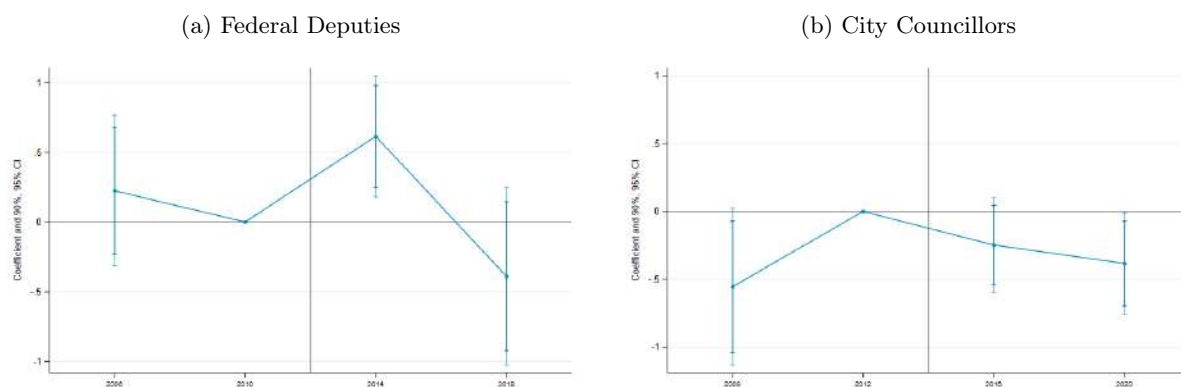
In Figure C.15, I examine pre-trends at the polling station level, focusing on the differential effect of the policy on voting behavior across racial groups by interacting baseline enrolment shares with the share of non-White population in the polling stations over time. Panel (a) suggests no significant pre-trends in federal deputy elections regarding the differential effect across racial groups. After the policy's implementation, the coefficient turns positive but then becomes negative in the 2018 elections, potentially indicating that while non-White voters initially increased their support for Black candidates, this was followed by a backlash from White voters, leading to a negative coefficient in subsequent elections. Panel (b) shows that the results for city council elections should be interpreted with caution, as the coefficient for 2008 is negative and statistically significant, suggesting possible pre-trends before the policy was implemented.

Figure C.14: Municipality Pre-trends - Vote for Black Candidates



Note: These figures reports the estimates of Equation 3, where I substitute exposure to non-White quotas with the baseline shares interacted with a year fixed effect. The dependent variable is the vote share for Black candidates in federal (panel (a)) and local legislative elections (panel (b)) at the municipality level. Each annual coefficient is displayed. See also notes for Table 3. Confidence intervals at the 90% and 95% significance levels are reported.

Figure C.15: Polling-Station Pre-trends - Vote for Black Candidates



Note: These figures reports the estimates of Equation 4, where I substitute exposure to non-White quotas with the baseline shares interacted with a year fixed effect and the share of non-White population in the polling station. The dependent variable is the vote share for Black candidates in federal (panel (a)) and local legislative elections (panel (b)) at the polling station level. Each annual coefficient is displayed, with only the interaction term reported. See also notes for Figure 5. Confidence intervals at the 90% and 95% significance levels are reported.

D Model

Here, I develop a simple model to guide the analysis on how racial affirmative action implemented in federal universities lead to changes in enrolment and affected voting patterns across racial groups.

D.1 Education

As defined in Section 4, municipalities are affected to racial quotas via their pre-reform enrolment patterns across institutions. Let's begin with a general measure of exposure, which uses race-specific enrolment patterns at baseline. For exposition, I drop the time component t :

$$q_m^r = \sum_u \frac{s_{mu}^r}{s_u^r} \cdot \frac{Q_u}{pop_{r,m}^{18-24}} \quad (8)$$

where $\frac{s_{mu}^r}{s_u^r}$ represent race-specific enrolment patterns, which capture the likelihood that a student from race r enrolled in u comes from municipality m . Each component of the measure is defined as in Section 4. If pre-reform enrolment patterns were predictive of how quotas were actually distributed across municipalities, one would expect this measure to affect the likelihood that a student of each racial group enrolls in university. Formally, the probability that a student of racial group r in municipality m enrolls in a federal university due to racial quotas is approximated by:

$$Pr(enrol|m, r) = \beta_0 + \beta_1^r q_m^r + \beta_2^r q_m^{-r} \quad (9)$$

where β_1^r reflects the effect of racial quotas on enrolment stemming from information of their own racial network, and β_2^r the additional effect of racial quotas on enrolment stemming from information transmission from the other race network, captured by q_m^{-r} . Notice that if both racial groups have similar enrolment patterns, $\frac{s_{mu}^w}{s_u^w} \approx \frac{s_{mu}^{nw}}{s_u^{nw}} \approx \frac{s_{mu}^r}{s_u^r}$, then $\beta_2^r \approx 0$, as there is no additional information shared by the other racial group that could affect enrolment of r . Total enrolment in federal universities due to racial quotas is thus given by

$$Pr(enrol|m) = Pr(enrol|m, nw)Pr(nw|m) + Pr(enrol|m, w)Pr(w|m) \quad (10)$$

Plugging 9 in the expression above, and noticing that $P(r|m)$ can be approximated by the population share of racial group r in the municipality, one gets the following expression for total enrolment

$$Pr(enrol|m) = \beta_0 + [\beta_1^{nw} q_m^{nw} + \beta_2^{nw} q_m^w] \frac{pop_m^{nw}}{pop_m} + [\beta_1^w q_m^w + \beta_2^w q_m^{nw}] \frac{pop_m^w}{pop_m} \quad (11)$$

Assuming race-neutral enrolment patterns (and hence $\beta_2^r \approx 0$), the expression for total enrolment becomes

$$Pr(enrol|m) = \beta_0 + [\beta_1^{nw} + \beta_1^w] q_m \quad (12)$$

where $q_m = \sum_u \frac{s_{mu}}{s_u} \times \frac{Q_u}{pop_m^{18-24}}$. Therefore, this expression indicates that local changes in enrolment accrued to racial quotas will depend on the relative gains and losses across racial groups.

Provided that racial quotas are intended to increase enrolment of non-White students, one would expect $\beta_1^{nw} > 0$ and $\beta_1^w \leq 0$. However, it is important to note that racial quotas might not necessarily lead to an increase in the enrolment of non-White students. For instance, in a municipality where a large proportion of non-White students were already enrolling at baseline, the introduction of racial quotas might merely reclassify these students as “quota holders” without actually altering the racial composition of the student body enrolling in the university. This remains an empirical question.

In the empirical analysis, I find support for the assumptions I make in this derivation: 1) race-neutral enrolment patterns (Appendix C.1.1), and 2) non-White quotas were closely reapportioned across municipalities following pre-policy enrolment patterns (Column (1) of Table 1). Moreover, my main results show that β_1^{nw} is close to 1, and β_1^w is 0. This will allow simplifying the analysis of how racial quotas affect voting patterns.

D.2 Political Effects at the Municipality Level

The effect of the reform on voting patterns will follow a similar framework. The ideal exercise would measure the effect of racial quotas on the likelihood that people vote for candidates of their same racial group. This probability is defined as:

$$Pr(v_r|m, r) = \gamma_0^r + \gamma_1^r q_m^r + \gamma_2^r q_m^{-r} \quad (13)$$

where γ_0^r captures baseline preferences for candidates of own’s racial group (e.g. racial loyalty in voting), and γ_1^r represents how actual policy effects (in terms of enrolment) change voters’ preferences towards same-race candidates. Instead, γ_2^r captures the effect of the other-race exposure to racial quotas on r ’s voting behaviour. This can reflect changes in voting behaviour driven by competitive social preferences. In other words, voters might not only care about their direct gains or losses from the policy, but also about the impacts on other racial groups. For example, γ_2^w might reflect the resentment of White voters towards non-White benefiting from the policy, even if Whites in the municipality were not directly affected. Similarly, γ_2^{nw} might reflect the satisfaction of non-White groups witnessing the decrease in White enrolment due to the policy. I expect $\gamma_1^r \geq 0$ and $\gamma_2^r \geq 0$.

Because individual voting data is not available in this context, I only am able to estimate total voting outcomes at the municipality level. Hence, the probability of a voter in a municipality m to support a non-White candidate is given by

$$Pr(v_{nw}|m) = Pr(v_{nw}|m, nw)Pr(nw|m) + Pr(v_{nw}|m, w)Pr(w|m) \quad (14)$$

Plugging 13 in the expression above, and noticing that $Pr(v_{nw}|m, w) = 1 - Pr(v_w|m, w)$:

$$Pr(v_{nw}|m) = [\gamma_0^{nw} + \gamma_1^{nw} q_m^{nw} + \gamma_2^{nw} q_m^w] \frac{pop_m^{nw}}{pop_m} + [1 - \gamma_0^w - \gamma_1^w q_m^w - \gamma_2^w q_m^{nw}] \frac{pop_m^w}{pop_m} \quad (15)$$

Rearranging this in terms of exogenous variables, we get the following expression:

$$Pr(v_{nw}|m) = \xi + [\gamma_1^{nw} - \gamma_1^w] q_m + \gamma_2^{nw} q_m \frac{pop_m^{nw}}{pop_m^w} - \gamma_2^w q_m \frac{pop_m^w}{pop_m^{nw}} \quad (16)$$

where $\xi = \gamma_0^{nw} \frac{pop_m^{nw}}{pop_m} + \frac{pop_m^w}{pop_m} - \gamma_0^w \frac{pop_m^w}{pop_m}$. Since my regression specification includes municipality fixed effects, ξ will be absorbed. The expression above, however, does not allow identifying the four structural parameters of the model separately without additional assumptions. Drawing from the educational results in Section 5, the evidence indicates that while non-White students benefited from the policy in exposed municipalities, this was not necessarily at the expense of White students in the same municipality. Based on this, it is reasonable to assume that White voters will not react to the policy due to direct losses. Similarly, I do not expect non-White voters to react to the policy's effects on Whites. Translating this into the model's parameters, I assume $\gamma_1^w = 0$ and $\gamma_2^{nw} = 0$, which simplifies Equation 16 to:

$$Pr(v_{nw}|m) = \xi + \gamma_1^{nw} q_m - \gamma_2^w q_m \frac{pop_m^w}{pop_m^{nw}} \quad (17)$$

This provides a more tractable Equation for identifying the political responses of both White and non-White groups to the policy's implementation. Which of the two effects dominate at the municipality level will depend on the strength of each racial group's reactions to non-White gains, together with the racial composition of the municipality.

D.2.1 Model Estimates

Table D.1 presents the estimates from Equation 17. Additionally, I include a set of baseline municipality characteristics interacted with a year fixed effect, municipality and state-year fixed effects. The results show that the coefficient γ_1^{nw} is positive but statistically significant only in local elections, suggesting that the policy led to increased support for non-White candidates among non-White voters, in line with the main results of this paper. The coefficient γ_2^w , which captures the response of White voters, is positive but not statistically significant. This aligns with the evidence found in Section 6.2 that some White voters increased support for Black candidates affiliated with right-wing parties as part of a backlash.

Table D.1: Model Estimates at the Municipality Level

DV: Vote Share	Federal Deputies (1)	Local Councillors (2)	Elected Local Councillors (3)
Exposure to Non-White Quotas	0.931 (1.204)	1.974** (0.773)	2.591** (1.055)
Exposure to Non-White Quotas $\times \frac{pop_m^w}{pop_m^{nw}}$	0.306 (0.217)	0.057 (0.072)	0.121 (0.111)
Avg DV at baseline	.244	.383	.348
Municipalities	5,460	5,459	5,459
Observations	21,840	21,810	21,664

Note: This table reports the estimates of Equation (16). The dependent variable in columns (1) and (2) is the weighted average of vote shares for candidates in federal and local legislative elections, respectively, where the weights are the predicted probability of the candidate being Black, as estimated by Vertex AI. In column (3) the dependent variable is the average predicted probability that elected councillors are Black, relative to the number of councillors elected. The interaction term is the exposure to non-White quotas interacted with the ratio of White to non-White population in the municipality, $\frac{pop_m^w}{pop_m^{nw}}$. All regressions include municipality and state-year fixed effects, as well as the baseline municipality characteristics interacted with a year fixed-effect. Regressions are weighted by municipality population at baseline. Robust standard errors clustered by municipality are reported in parenthesis. Significance levels: *** 1%, ** 5%, * 10%.

E Data Appendix

E.1 Racial Classification of Candidates using Vertex AI

In this section, I describe the methodology employed to categorize candidates into the main Brazilian racial groups.²⁰ In particular, I train an Automated Machine Learning (AutoML) image classification model in Google Vertex AI. AutoML allows researchers to customize models to their specific needs using a data-driven, objective, and automated approach that identifies the most suitable model for the chosen application (Hutter et al., 2019). In order to acknowledge that the socially-prescribed physical features that characterize each racial group might be context-specific, I train the model with a subset of Brazilian candidates' official pictures and the racial categories they identify with. The model then identifies the phenotypical traits that candidates commonly associate with self-identifying as Black.

In the remainder of the section, I describe the training sample, how I classify candidates into racial groups using the output of Vertex AI, and evaluate its performance.

E.1.1 Training Sample

I follow the guidelines provided by Vertex AI on how to build the training dataset for achieving a good performance. The training dataset should comprise a set of images along with their corresponding classifications into the objective categories, in this case, racial groups. Selecting the training sample involves determining the type of images to include, specifying the racial categories for the model to classify, and an annotation dataset, which maps the selected images into the corresponding racial categories.

First, regarding the selection of images, Vertex AI recommends training the model using images that are visually similar to the ones used when making predictions.²¹ Given that I will use official candidate photos to classify them into racial groups, the training sample will consist of a random subset of such images. Second, regarding racial categories, I classify candidates into two categories: White and Black. While the Brazilian Census classifies individuals into five categories (White, mixed-race, Black, Indigenous, and Asian), in the model I classify candidates along the White-Black continuum. This is for two reasons. First, the model is unable to correctly predict the racial group of candidates that self-report mixed-race. They are often classified as White or Black with similar probabilities (See Table E.1), suggesting that they exhibit phenotypes common to both the White and Black racial groups. Including such a fluid racial category in the training sample would reduce the precision of my model. Moreover, training the model using only the Black and White categories enables me to

²⁰In discussing race, it is essential to recognize that while this concept lacks a biological basis, it carries significant social meaning that influences people's experiences and interactions. A major concern is that, although necessary for the analysis, racial categorizations can inadvertently perpetuate the inequalities that research aims to address. Yet, in contexts like Latin America, national narratives of racial harmony can often obscure the realities of structural racism and entrenched racial hierarchies (Cleland, 2022).

²¹<https://cloud.google.com/vertex-ai/docs/image-data/classification/prepare-data>

characterize mixed-race candidates as those with intermediate probabilities. The second reason speaks to one technical requirement of Vertex AI: in order to achieve accurate predictions, the dataset must contain a sufficiently large and balanced set of images across racial categories (at least 1,000 pictures per category). The number of Indigenous and Asian candidates is substantially small, and I would need to include most of them in the training sample to comply with this recommendation. Hence, I exclude these two racial categories.

The last component of the training sample is the annotation dataset, which associates each candidate image in the sample with its racial categorization. This step is particularly challenging due to the subjectivity of racial categories in Brazil, and by the absence of a systematic measure of candidates' race prior to the implementation of racial quotas in federal universities. To construct the annotation dataset, I rely on the racial group that candidates report to the TSE when registering for the 2014 elections. I choose 2014 because it is the first year that the TSE required the disclosure of race by candidates. This classification is, however, potentially limited in different margins. First, I classify candidates based on the racial group that they report after the Law of Quotas was implemented. If candidates perceive that aligning with one group offers higher electoral rewards after the policy, they may opportunistically alter their stated racial identity to capitalize on racial politics. I argue that this reporting issue is alleviated by using the immediate election after the introduction of the Law of Quotas and the random selection of candidates in the training dataset. The 2014 elections were the first one after the implementation of racial quotas, so politicians might learn the significance of race in the political setting only after observing the election results of 2014. Hence, strategic reporting behaviour is likely to be more pronounced in the subsequent elections. Moreover, strategic reporting behaviour is documented to be conducted by 25% of the candidates running in both the 2014 federal and state and the 2016 municipality elections (Janusz, 2021). By selecting a random sample of candidates in 2014 I expect to have a truthful reported race, on average.

The second concern related to using candidates' self-reported race is whether this is the relevant measure to classify candidates. In a context where ancestry is not systematically used to determine racial categories, voters' perceptions of candidates' race may be a more relevant measure.²² I argue that self-reported race does not significantly differ from others' perceived racial group. To do so, I match the candidates in my training sample to the classification made by a representative sample of Brazilians. The latter is constructed using data from a survey carried out by Bueno and Dunning (2017), where respondents classified candidates using different racial measures based on a black-and-white official picture. These measures of race included the five census categories, whether they perceive them as Afro-descendant, whether they perceive them as White (*branco*) or Black (*preto*), and their position in a colour scale.²³ Overall, I am able to match 138 candidates. As reported in Table E.2, over 70% of candidates that self-report Black in TSE are perceived to have African ancestry, and a

²²This opposes settings like the US, where racial classification was historically defined by the "one-drop rule of hypodescent", which designated as African American anyone with traceable African descent (Daniel and Lee, 2014).

²³For more details on the design of the survey, see Bueno and Dunning (2017).

similar proportion is also classified as Black. There is, however, considerable disagreement among survey respondents regarding the census racial categorization of these candidates. While the majority of self-declared Black candidates are also perceived as Black according to the Census, 30% of them are classified as being Mixed-race. Nonetheless, self-declared Blacks are also perceived as exhibiting a darker-than-average skin colour.

Overall, the training dataset is composed of 3,000 candidates. To make sure that this is a representative sample of candidates, I compare the characteristics of candidates in the training sample with the rest of politicians running in the 2014 elections. The candidates included in the training sample are strongly balanced in most characteristics for all election periods (Table E.3). Regarding the geographical distribution of the candidates in the training sample, candidates from South-East regions are overrepresented. This is primarily driven by the disproportionate presence of candidates running in the states of Rio de Janeiro and São Paulo in my training sample (Figure E.1). Finally, I randomly split the sample into Training, Validation, and Test groups, in the proportions required by the platform.

E.1.2 Training Details

I train a multi-label image classification model using the training dataset. The version details are summarized in the Table E.2.

E.1.3 Classification of Candidates into Racial Groups

Using the model I trained, I predict the race of candidates running for federal, state, and local elections between 2006 and 2022. The output of the model is a pair of probabilities, or similarity, for each of the targeted racial groups. That is, for each candidate, it provides the probability that they conform with prescribed White and Black physical traits, respectively. Figure E.3 displays the Kernel densities of such probabilities. They are almost symmetrical, and exhibit the same two peaks. Most of the candidates running in the Brazilian elections are classified by the model as being White with a probability higher than 80% (or with a probability of being Black lower than 20%). A substantially smaller mass of candidates are predicted to conform with the physical traits characteristic of Black Brazilians, and the rest of candidates have an intermediate probability of belonging to each of the racial groups.

To simplify the analysis, I use these patterns in the data to classify candidates into White, Black, and Mixed-race groups using the probability of conforming with Black phenotypical traits. Specifically, I classify candidates as White when such probability is lower than 20%, as Black if the probability is higher than 80%, and Mixed-race if the candidates are assigned an intermediate probability. In machine learning, this is essentially using a 0.8 confidence threshold to assign predicted labels to each candidate.

In the next section, I evaluate the model performance and discuss how this classification compares

with self-reported racial groups of candidates.

E.1.4 Evaluation of the Model

What does the Model Capture?

AutoML models are highly advantageous because they significantly simplify the complex process of training machine learning models while achieving optimal performances. However, a notable drawback is that they often operate as a black box, meaning users cannot see the internal workings of the model. To address this, Vertex AI offers *Feature-based explanations*, which visually highlight which pixels of an image contribute most to the model's prediction.

In Figure E.4 I selected 6 pictures of candidates from the testing dataset that the model correctly predicted. For each picture, Vertex AI overlays a pink-green gradient, where greener areas indicate the parts of the picture that contributes most to correctly predicting the race of the candidate. This feature helps the researcher to evaluate whether the model picks up the correct parts of the picture to predict the race of candidates. As shown in the figure, the model uses the pixels of the central part of the face of candidates, specifically the shape of the nose and mouth, to predict candidates' race. This reassures that the model is using relevant facial features of candidates to predict the race, aligning with the criteria used by monitoring panels in federal universities to verify that students with racial quotas complied with the phenotypical traits commonly attributed to the non-White population.²⁴

Main Evaluation Metrics

In this section, I provide a discussion on the performance of the model using the main evaluation metrics of machine learning models, precision and recall. *Precision* measures the proportion of true positive predictions out of all the positive predictions made by the model. In this specific setting, precision tells what share of the candidates that the model predicts to be White also self-report as White. If the model predicts that 100 candidates are White, and 90 of them indeed self-report as White, then the precision would be 90%. *Recall*, the complement of the Type II error rate, captures the fraction of the times that the model correctly predicts the specified label. In this specific setting, it tells us what share of all candidates who self-report as White are correctly predicted to be White by the model.

Table E.4 reports precision and recall measures for all labels, as well as separately for White and Black racial groups. Precision and recall metrics are calculated relative to a 0.8 confidence threshold, meaning that the model assigns a label when its confidence for that label is above 80%. *Average precision* measures how the model performs across all confidence thresholds. The model I trained performs significantly well across labels, with precision measures above 90% and recall around 70%.

²⁴For an example of the criteria used in such panels, see the Decisão Nº 212/2017 in the Federal University of Rio Grade do Sul.

Considering that a model with no false positives would have a precision of 100%, this demonstrates the high performance of the model I trained. The model performs especially well with Black candidates, achieving the highest metrics.

Discrepancy with Self-reported Race in TSE

I proceed to evaluate how the racial classification predicted by the model differs from the race that candidates report in TSE at the time of registering their candidacy. To do so, I construct a confusion matrix for all candidates running from 2014, the first year when the race of candidates was disclosed. This analysis helps identify any discrepancies between the model’s predictions and the candidates’ self-reported race, allowing for a comprehensive understanding of the model’s accuracy and potential biases. Results are reported in Table E.5.

The confusion matrix provides insight into the recall measure, highlighting the percentage of times the predicted racial classification conforms with the candidate’s self-reported group and when it did not, indicating which racial group was predicted instead. Moreover, I report the average probability of conforming with White and Black traits per self-reported groups as well as the Kernel densities of $\Pr(\text{Black})$ for each self-reported group in Figure E.5. Self-reported White candidates are 60% of the times correctly assigned to their racial group. There is, however, 32% of the times that they are classified Mixed-race. Notwithstanding, the average probability of conforming with White traits is 72%, which highlights that these candidates are close to the 80% cut-off that I use to binarize racial categories. As discussed earlier, the model performs especially well for candidates that self-report Black, with the most common misclassification being Mixed-race. Again, as with White candidates, these are presumably those close to the 80% cut-off. The classification of mixed-race candidates aligns with the fluidity of this racial group in Brazil: 40% are classified as mixed-race, while the remaining 60% are classified as White and Black in similar proportions, with a slight inclination toward Black classification.

One limitation of the classification derived from the AutoML predictions is that it excludes two Census racial categories: Indigenous and Asian. Consequently, individuals belonging to these categories are assigned to either White, Mixed, or Black groups. I evaluate how the model classifies those individuals, and I find that Indigenous candidates are predominantly (89%) classified as either Mixed-race or Black. This outcome is reassuring, as Indigenous peoples are considered by the Census as part of the non-White racial group, together with Mixed-race and Black individuals. Finally, Asian candidates are more likely to be classified as White, as exhibited by the confusion matrix and the probability of conforming with Black traits.

Racial categories are not discrete identities but rather exist along a continuum, especially in contexts marked by substantial racial miscegenation, like Brazil. Therefore, I anticipated that the model would not be a perfect classifier due to the inherent complexity and fluidity of racial identities. However,

despite these challenges, the model performs admirably well in distinguishing between different racial groups.

Consistency of Predictions within Candidates

To further validate the predictions from Vertex AI, I test whether the probabilities assigned to candidates are consistent over time. For this purpose, I restrict the sample to candidates who have run in more than one election round. These candidates account for 30% of the total candidates ever running in elections, out of which 94% ran for local office. I then compute the difference between the probability of conforming to Black traits in each election round compared to the probability in the first year they ran for office. The main summary statistics are reported in Table E.6 and the histogram of within-candidate difference in probabilities in Figure E.6. Errors are centred around zero, with over 50% being below 0.2. These errors are not primarily due to different predictions of race for the same picture, but rather from candidates providing different pictures across election rounds. While not explicitly reported, Black candidates show relatively consistent probabilities over time, whereas mixed-race candidates exhibit the largest inconsistencies in racial classification over time, as expected. I argue that this is not significantly problematic, as the predicted mixed-race category spans a broad range of probabilities (between 20% and 80% of conforming with Black traits). This wide range accommodates the variability in classification, thereby mitigating the impact of these errors on the overall analysis.

Table E.1: Confusion Matrix - All Racial Groups

<i>Vertex AI Prediction:</i>	White	Mixed	Black	Other
<i>Reported Race TSE</i>				
White	79%	3%	18%	0%
Mixed	54%	9%	36%	0%
Black	11%	5%	83%	0%
Other	73%	0%	27%	0%

Note: This table reports the confusion matrix of a multi-label image classification model trained in Vertex AI with candidates from all Census racial groups, as reported by a random subset of candidates running in 2014 Brazilian elections. *Other* includes candidates that self-report “Asian” or “Indigenous”.

Table E.2: Other Racial Classifications from Bueno and Dunning (2017)

	Mean	SD	Min	Max
Afrodescendant	0.737	0.444	0	1
Black (dich.)	0.702	0.462	0	1
Census categories				
- <i>White</i>	0.105	0.310	0	1
- <i>Mixed-race</i>	0.316	0.469	0	1
- <i>Black</i>	0.526	0.504	0	1
- <i>Indigenous</i>	0.035	0.186	0	1
- <i>Asian</i>	0.018	0.132	0	1
Skin-Color Scale	5.965	2.096	1	10

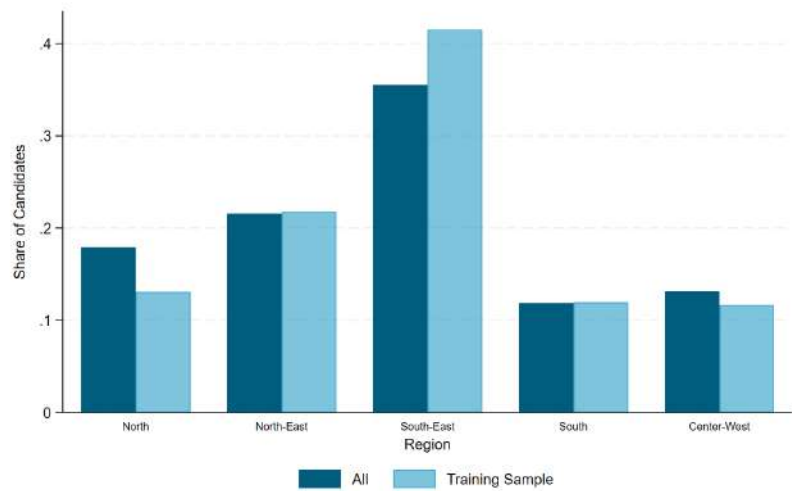
Note: This table displays how candidates that self-report Black to TSE are perceived by survey participants from Bueno and Dunning (2017). Observations: 138.

Table E.3: Representativity of Candidates in the Training Sample

	All Candidates	Training Sample	Diff.	P-value
Age	47.241	47.436	0.194	0.380
Women	0.286	0.303	0.018	0.052
Graduate	0.472	0.459	-0.013	0.179
Married	0.566	0.538	-0.028	0.004
Elected	0.822	0.787	-0.035	0.000
Campaign Spending	2,885,897	2,819,767	-66,131	0.426
Candidates	18,971	2,996	21,967	21,967

Note: This table compares the means of candidates' demographics in the full versus the training sample.

Figure E.1: Geographical Distribution of Candidates in the Training Sample



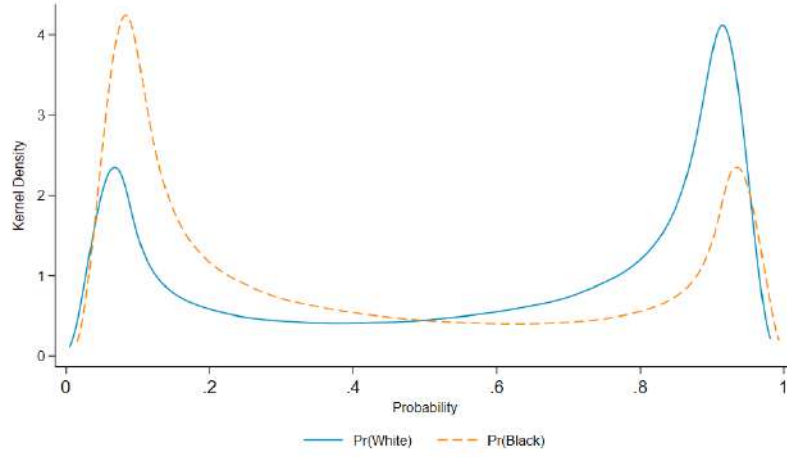
Note: This figure displays the share of candidates running for elections across main geographic regions. Dark columns represent the share of candidates in the full sample, while light-coloured columns represent the share of candidates in the training sample.

Figure E.2: Training Details

Status	Finished
Model ID	2393262979714908160
Version description	—
Created	Mar 6, 2024, 3:38:32 PM
Budget (original)	15 node hours
Budget (actual)	15 node hours
Training time	2 hr 58 min
Region	europa-west4
Encryption type	Google-managed
Dataset	tse_cand_dataset
Dataset ID	7887027803479080960
Annotation set	tse_cand_dataset_icn
Data split	Manually assigned
Total items	3,000
Training items	2,400 (80.0%)
Validation items	300 (10.0%)
Test items	300 (10.0%)
Algorithm	AutoML
Objective	Image classification (Multi-label)
Source	AutoML training

Note: This figure reports the training details input into Vertex AI to train the model.

Figure E.3: Kernel Density of Predicted Probability



Note: This figure displays the Kernel densities of the probability of candidates running in any of the 2006-2022 federal, state, or local elections of being White (blue line) and Black (orange dashed line), according to their phenotypical traits. Kernel function: Epanechnikov. Number of observations: 80,824

Figure E.4: Features Contributing to the Model Prediction



Note: This figure displays 6 pictures from the testing dataset. For each candidate, I report the probability that the candidate conforms to the main racial category that the model predicts. Each picture is overlaid with a pink-green gradient, where greener areas contribute more to the model's prediction. While they are public figures, the faces of candidates were blurred to preserve their anonymity.

Table E.4: Precision and Recall Metrics

	Avg Precision	Precision	Recall	Range
All	0.941	0.930	0.750	0-1
White	0.919	0.933	0.678	0-1
Black	0.955	0.928	0.815	0-1

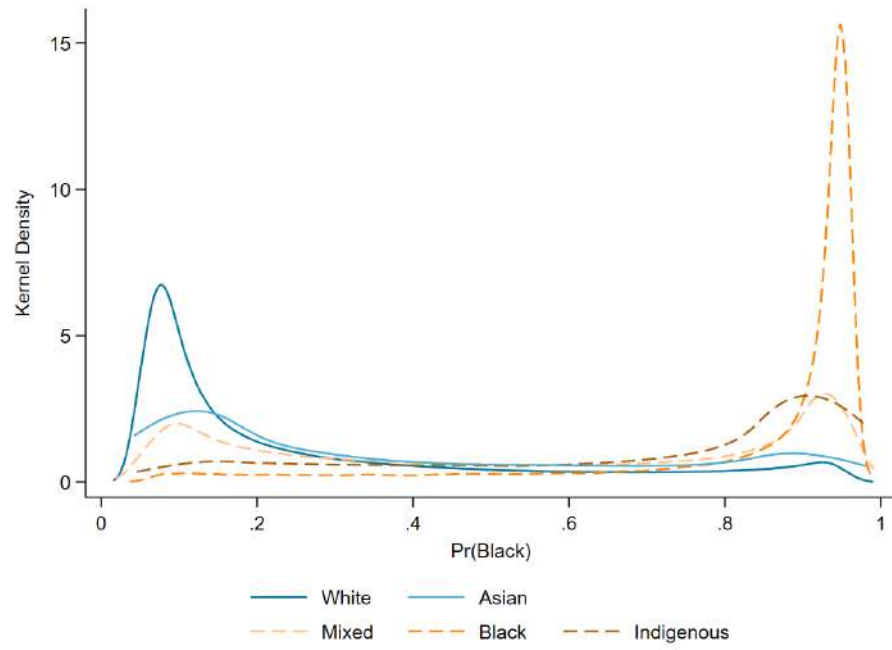
Note: *Average Precision* measures model performance across all confidence thresholds. *Precision* and *Recall* metrics are displayed relative to a 0.8 confidence threshold.

Table E.5: Reported vs Predicted Racial Group

	Confusion Matrix			Probabilities	
<i>Vertex AI Prediction:</i>	White	Mixed	Black	Pr(White)	Pr(Black)
<i>Reported Race TSE</i>					
White	59%	32%	9%	0.722	0.272
Mixed	24%	41%	35%	0.457	0.537
Black	4%	19%	78%	0.172	0.827
Indigenous	11%	38%	51%	0.321	0.675
Asian	41%	41%	18%	0.602	0.391

Note: This table reports the confusion matrix and predicted probabilities for self-reported racial groups vs predicted groups using Vertex AI. The confusion matrix reports the percentages with respect to the total number of candidates that self-report belonging to each racial group in TSE. The last two columns report the average probability of conforming with White or Black phenotypical traits as calculated in Vertex AI. Period: 2014-2022.

Figure E.5: $\Pr(\text{Black})$ by Reported Racial Group



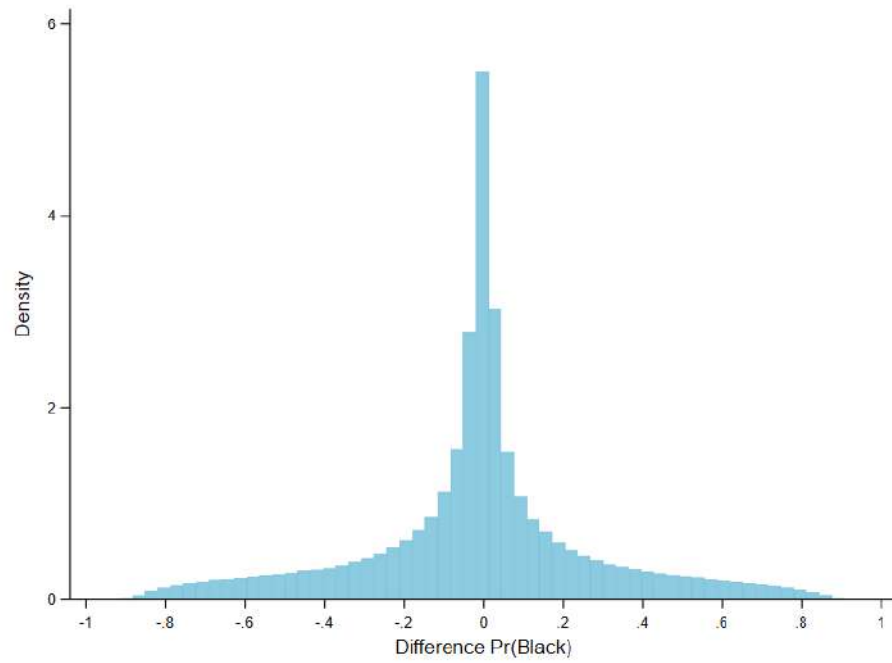
Note: This figure displays the Kernel densities of the predicted probability of complying with Black phenotypical traits, by reported racial group in TSE from 2014-2022.

Table E.6: Within-Candidate Consistency - Summary

	Mean	SD	Min	P25	P75	Max
Difference	-0.01	0.29	-0.92	-0.11	0.09	0.91

Note: This table reports the summary statistics of the difference between the first prediction of $\Pr(\text{Black})$ for a candidate and the subsequent predicted probabilities.

Figure E.6: Within-Candidate Consistency



Note: This figure displays the difference between the predicted probability in each round of election with respect to the probability predicted for the first year that candidates ran for office. I exclude the difference for the first year, as it is 0 for all candidates. Candidates: 374,323.

E.2 Ideological Placement of Political Parties

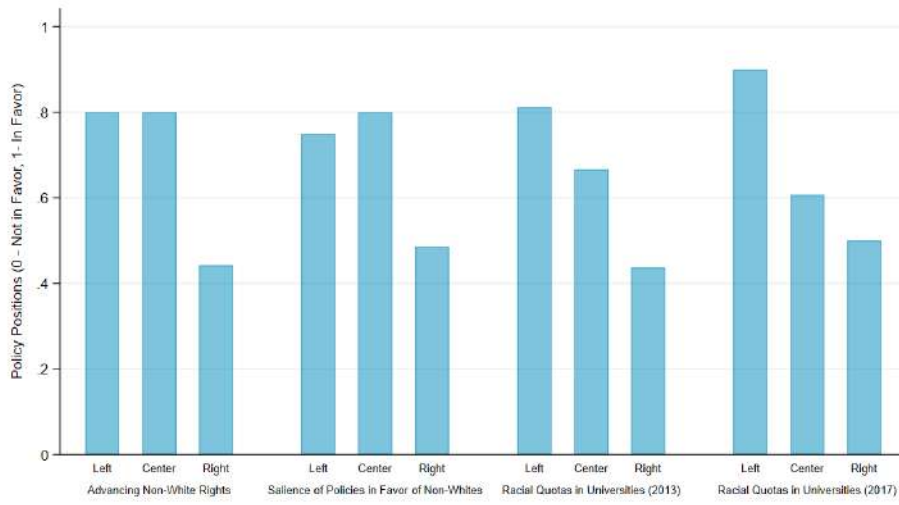
During the period covered by this paper, 2006-2018, 38 parties were officially registered in the Brazilian Supreme Electoral Court (TSE). To simplify my analysis in Section 6, I place parties in the left-right political spectrum, as defined in Zucco and Power (2021). To classify parties, the authors use the Brazilian Legislative Survey (BLS), a rich survey performed to Brazilian federal legislators, and conducted in the years before federal elections in eight waves, from 1990 to 2017. Each of the waves asks about legislators' own-party policy stances and perceptions of other parties' policy stances. Zucco and Power (2021) provide estimates on parties ideological placement calculating Dalton's polarization index for each party, which ranges from -1 to 1. Parties with an index close to -1 are considered to conform to what is understood as the Brazilian left, while parties with an index closer to 1 are considered to belong to the Brazilian right. This measure does not cover all the parties registered in TSE, only the 20 that had access to the national congress. However, the share of votes of the remaining 18 unclassified parties amount to less than 5% in federal elections and 14% in local elections at baseline. I define *Left* as parties with an index below -0.3, *Center* parties with an index between -0.3 and 0.3, and *Right* as parties with an index above 0.3. The rest of Brazilian parties is classified into *Others*.

I complement this data with party positions regarding ethno-racial rights and general redistributive issues from the 2020 Chapel Hill Expert Survey: Latin America (CHES) (Martínez-Gallardo et al., 2020). This dataset contains expert judgements on the policy stances of 12 major political parties in Brazil as of 2020. To extract party positions regarding ethno-racial rights, I focus on two main questions: ETHNIC MINORITIES and ETHNIC MINORITIES SALIENCE. The question ETHNIC MINORITIES asks about the "Position towards advancing rights for indigenous and Afro-descendant peoples" and the question ETHNIC MINORITIES SALIENCE asks about the "Importance/salience of policies in favour of indigenous and Afro-descendant peoples". Regarding redistributive issues, I use the question REDISTRIBUTION, which captures the perception on how parties stand towards redistribution. I also calculate the position of federal legislators regarding their stance towards income and racial quotas in federal universities in years 2013 and 2017 using their responses from the BLS. Provided that they have different scales, I re-scale all variables to a 0-1 scale, where 0 represents "Strongly against/Not salient" and 1 represents "Strongly in favour/Very salient".

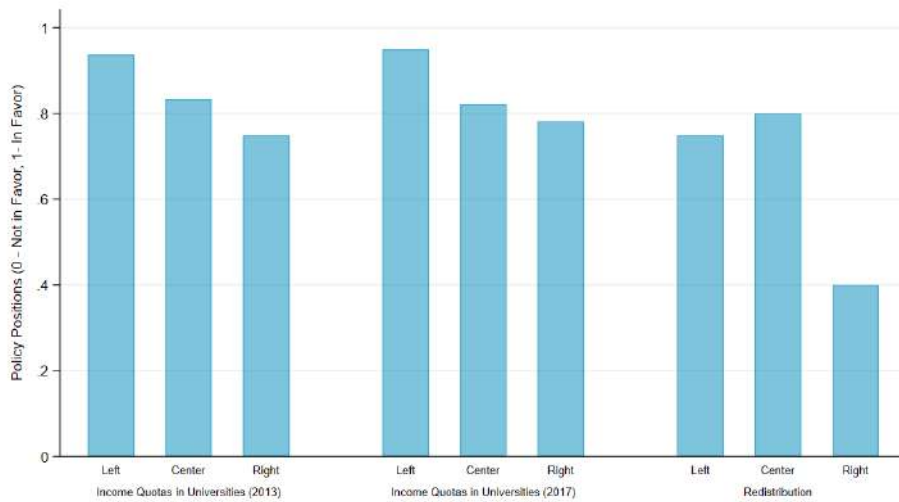
Figure E.7 illustrates the ideological divide among political parties on racial and income-related issues. Both left-leaning and centrist parties are perceived as having a strong agenda for advancing non-White rights in 2020, while right-wing parties show less commitment to these issues. This ideological divide is also evident in the parties' support for racial quotas in public universities during the 2010-2014 and 2014-2018 National Congresses, where support for quotas decreases as parties move further to the right on the ideological spectrum. Although a similar divide exists regarding income-based quotas, the gap in positions is notably smaller. Lastly, left- and center-leaning parties are perceived as being more in favour of redistribution compared to their right-wing counterparts.

Figure E.7: Stances towards Racial and Income Issues, by Ideology

(a) Position towards Racial Issues



(b) Position towards Income Issues



Note: These figures report the stances of political parties along the ideological spectrum regarding racial and income issues. Parties are classified using Zucco and Power (2021)'s ideological score.

E.3 Race-Related Terms for Text Analysis

Table E.7: List of Race-Related Terms

Category	Terms
Race and Ethnicity Terms	raça, etnia, racial, preto, preta, pretos, pretas, afro, africano, africana, africanos, africanas, etnicidade, identidade racial, identidade étnica, negro, negra, negros, negras, afrodescendente, afrodescendentes, quilombola, quilombolas, povo negro, população negra, indígena, indígenas, povos indígenas, pardos, mestiço, mestiça, mestiços, mestiças, população quilombola, minorias étnicas, população indígena, etnias tradicionais, africanos, africanas, afro-brasileiro, afro-brasileira, afro-brasileiros, afro-brasileiras, povos originários, comunidades tradicionais, descendentes africanos
Diversity and Racial Justice	igualdade racial, justiça racial, equidade racial, direitos raciais, inclusão racial, valorização da diversidade, direitos iguais, equidade para todos, reconhecimento de direitos, valorização das minorias, representatividade, justiça social, harmonia racial, inclusão de minorias, pluralidade cultural, respeito à diversidade, diversidade
Systemic Explanations	racismo, discriminação, racista, xenofobia, desigualdade racial, violência racial, etnocídio, branquitude, privilégio branco, etnocentrismo, racismo estrutural, racismo sistêmico, desigualdade estrutural, disparidades raciais, sistemas de opressão, barreiras institucionais, racismo institucional, segregação racial, desigualdades econômicas por raça, injustiça sistêmica, desigualdade sistêmica, exclusão racial, discriminação estrutural, barreiras sistêmicas, minorias, segregação
Group-Based Policies	ações afirmativas, cota racial, cotas raciais, políticas de cotas, políticas inclusivas, políticas públicas de reparação, redistribuição racial, representatividade racial, reparações históricas, programas de igualdade, programas inclusivos, medidas reparatórias, políticas de equidade racial
Social Movements and Activism	movimento negro, movimentos sociais, luta antirracista, antirracismo, antirracista, orgulho negro, mobilização social, reconhecimento histórico, ativismo comunitário, empoderamento negro, resistência cultural, movimento antirracista, ativismo negro, resistência negra, luta pela diversidade
Individualism and Meritocracy	meritocracia, esforço individual, mérito pessoal, responsabilidade individual, competência, igualdade de oportunidades, trabalho duro, superação individual, foco no indivíduo, valorização do esforço, desempenho individual, merecimento, meritocracia plena, autossuficiência, desempenho pessoal
Against Systemic Explanations	narrativa vitimista, culpar o sistema, mito do racismo sistêmico, negação do racismo estrutural, superação individualista, exagero de discriminação, retórica de opressão, desqualificação de políticas raciais, negação de disparidades raciais
Against Group-Based Policies	contra ações afirmativas, contra cota racial, contra cotas raciais, neutralidade racial, igualdade sem distinção, não distinção por raça, color blindness, discriminação reversa, políticas divisivas, foco no mérito, eliminação de cotas, crítica às ações afirmativas, oposição a políticas raciais, oposição a cota racial